Dynamic weather controlling for PSV

RETRACTABLE ROOF DESIGN FOR PHILIPS STADIUM

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1 Introduction

The design of a retractable roof for Philips Stadium involves looking at both the stadium itself and its surroundings. Philips Stadium is located in Eindhoven, home to the football club PSV. It is mainly used for football matches, but also hosts concerts, meetings, and other events. The stadium currently has a large steel roof supported by cantilever trusses and columns, which works well for its current use. However, the addition of a retractable roof will provide more flexibility, offering shelter from bad weather and the ability to adjust the stadium for different types of events.

While there are plans already on the design of a new seating area in the stadium, the roof needs to fit within the new design and avoid any interference with the future extensions to the seating.

This report will go into detail about the design process, looking at the space available, the materials to be used, and the structural needs. The goal is to create a retractable roof that not only meets all technical requirements but also fits well with the existing stadium and enhances its use for many years to come.

2 Analysis of the stadium's environment

2.1 Building plot

In order to create a design for the retractable roof structure, an analysis of the environment is crucial. In this chapter the site of the stadium and its surroundings will be analysed. First with a look on the building plot and the existing stadium, then on the environment in terms of surrounding buildings, infrastructure and transportation roads.

2.1.1 Existing stadium

The Philips Stadium is located on Frederiklaan in Philipsdorp in the Eindhoven district of Strijp, close to the centre of Eindhoven [1]. In Figure 1 a perspective view of the stadium is shown. The stadium is the home stadium of the football club PSV. It is mainly used for PSV to play their home matches, but the stadium is also used for (business) meetings, concerts and other events [2]. The structure of the stadium exists of large steel cantilevering trusses to support the roof, together with large pillars at the corners of the stadium on which the roof is hanging.



Figure 1: Philips stadium

2.1.2 Available building space

In Figure 2, the building space is shown around the stadium. There is 43,014.77 – 30,938.86 = 12,075.91 m^2, so approximately 1.2 hectare of space around the stadium. The most space is at the south and west sides of the stadium. The north and east sides of the stadium is more narrow and close to the busy roads; PSV-laan and Vonderweg.

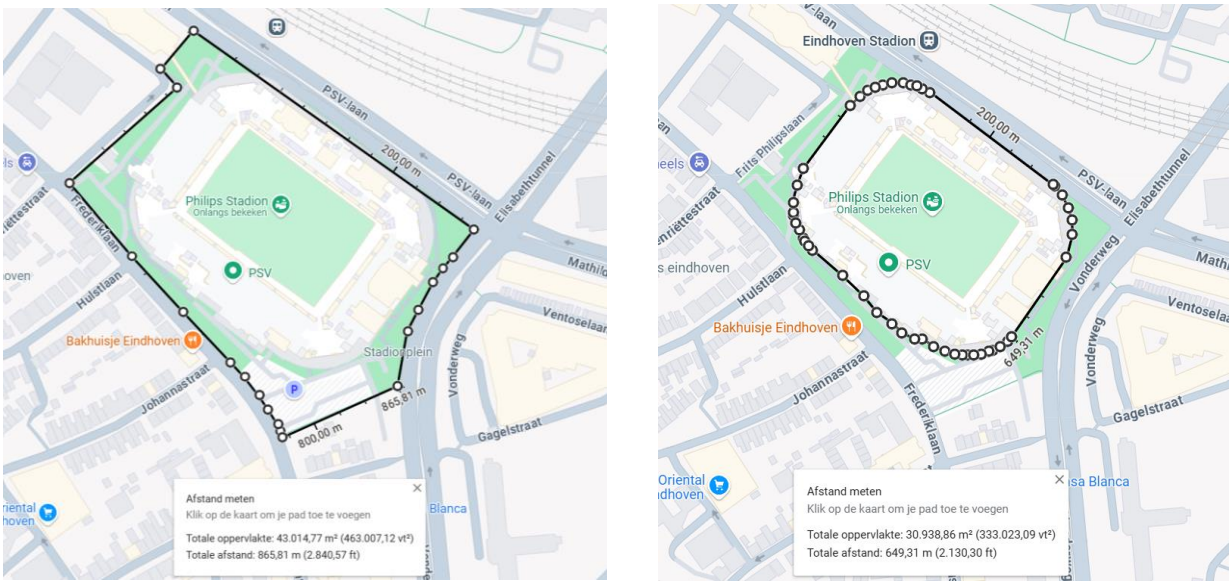


Figure 2: Available space around the Philips stadium

2.1.3 Renovations Philips Stadium

The plans for renovations of the Philips stadium should be kept in mind as well. The upper ring on the north side of the Philips Stadium is getting an upgrade. The outside is being closed, catering points are getting better facilities and more toilets are being added [3]. According to A&E Architecten, the assignment was to increase the stadium capacity to 44,000 seats, or approximately 9,000 extra. In their design, the longitudinal stand on the track side and both head stands are enlarged by adding extra rows of seats. The corners, which are of recent date, remain intact [4].

These upgrades should be kept in mind while designing the new roof structure on top of the stadium. According to the requirements mentioned in the study guide the retractable roof needs to be constructed at a specific height to avoid clashes with future extended tribunes: the bottom face of the roof is 45m above surface level. In Figure 3 an impression of the stadium is given made by A&E Architecten.



Figure 3: Concept for renovations of the Philips stadium

2.2 Environment

The environment around the stadium is important for the accessibility of the stadium for materials to enter the plot as well as for the storage of materials. In this chapter the city of Eindhoven is analysed on different scales according to different aspects.

2.2.1 Surrounding buildings and infrastructure

The buildings around the stadium are relevant to some extent. In Figure 4 a scale is given in for which the surrounding buildings are analysed. In broad terms the north and south-west side of the stadium is stated as residential (yellow parts). The north-west side is Strijp-S and is mainly mixed which means it is used for living, working, public-oriented functions and facilities. At the south-west side of the stadium, the zoning plan is separated in a lot of different functions. According to the colours in the legenda there are residential parts, some greenery, mixed parts, offices and when looking at the far east the centre of the city is stated.



Figure 4: Zoning plan of Eindhoven

2.2.2 Accessibility

Transporting materials from the factory to the plot around the stadium could be done in different ways. One of them being an obvious one which is by trucks driving over the normal roads. In Figure 5 the city of Eindhoven is shown with all the roads indicated in a colour referred to the live traffic information at 15:00h during a weekly day on a scale which indicated the highways and. Figure 5 gives a zoom in to the ring of Eindhoven with more roads indicated by the coloured lines. The maximum clearance height of the roads in the Ring of Eindhoven is 4.1 meter [5]. There are several ways to get to the stadium. Preferably, wide roads without sharp turns are used for transportation of the materials. The elements transported to the plot would presumably be large which means wide straight roads are best for transportation. There are 5 highways which could be used to get in Eindhoven: A58, A2, A50, A270 and the A67. These roads lead to the ring of Eindhoven. When looking at Figure 5 this would mean that roads through the residential areas and Strijp-S should be avoided. Therefore the road called ‘Fellenoord’ is the best option to use for transportation of the materials by trucks. This road is indicated by the red thick line. Another option is using the train for transportation of the materials, while the train tracks are right next to the stadium. However unloading trains in the middle of a train track is not convenient.

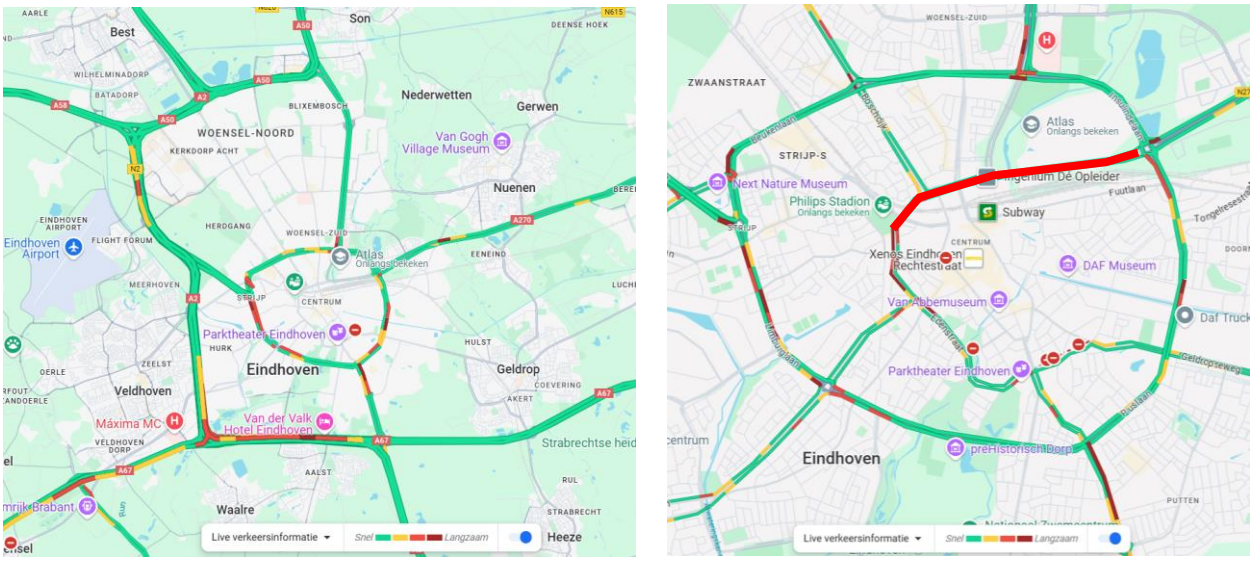


Figure 5: Main roads going to Eindhoven and to the stadium

2.2.3 Stakeholders

When planning a new retractable roof for Philips Stadium, several important stakeholders are involved. First, the club management plays a crucial role, as the leaders of the football club want the roof to enhance the stadium experience for fans and players. They are focused on benefits like increased attendance and the ability to host a wider range of events.

Fans are another key group, as their comfort and enjoyment are essential. They want protection from bad weather and a more exciting atmosphere during games. Local city officials also have a stake in the project because the stadium impacts the surrounding community. They need to consider factors such as traffic, safety, and the effects on local businesses.

Architects and engineers are involved in designing and building the roof, focusing on safety, functionality, and aesthetics. Contractors will manage the actual construction, ensuring that everything is completed correctly and on time. Additionally, sponsors who support the stadium or the team may want to get involved, as they hope to benefit from increased visibility during events.

Finally, environmental groups may raise concerns about the impact of the construction on the local environment and advocate for sustainable practices. Each of these stakeholders has different interests and needs, but they all play a role in making the retractable roof project a success.

3 Design requirements

3.1 Requirements set in the study guide

For the design of the retractable roof, some requirements are set by the study guide. Next to these requirements some additional requirements are set. All the requirements are taken into account during the designing process and will be evaluated at the end of the project. The requirements set by the study go are stated below:

The roof structure...

- …should be retractable.
- …has dimensions 210m x 160m (length x width), covering the south and the extended west, north and east tribunes.
- ...needs to be constructed at a specific height to avoid clashes with future extended tribunes: the bottom face of your roof is 45m above surface level.
- ...the columns should not transfer vertical loads to the tribunes. A total horizontal transfer of 200 kN towards the existing tribunes is allowed in x- and y-directions.

In addition...

- …the positions of the foundations of the columns of the columns are indicated by the red dots in Figure … and Figure ... You cannot change the position of these points or add foundations.
- ...the vertical capacity of each foundation point is limited to 28 MN (28000 kN).
- ...the horizontal capacity of each foundation point is limited to 1 MN (1000 kN).

3.2 Additional requirements

Below the additional requirements are stated:

The retractable roof structure must…

- …use transparent materials for the roof that provide at least 90% light transmission to ensure adequate natural light in the stadium
- …not obstruct the view of the facades between the corners from the outside except for the columns used on the foundation points
- …have a total self-weight that does not exceed 90% of the maximum vertical capacity in the foundation points; this means that the self-weight should not exceed (28000x4x0.9 = 100800kN, 100800x1000/9.81=) 10275 tons
- …not exceed 20 meters above the existing stadium roof
- …have a minimum lifespan of 50 years under normal weather conditions, ensuring long-term performance
- …be designed to fit within standard transport dimensions, allowing for efficient delivery to the site without special permits

The retractable roof structure should…

- …the construction process should be completed within six months, with minimal disruption to ongoing stadium activities, measured by daily disturbance reports
- …be designed for easy disassembly
- …correspond with the existing stadium architecture, maintaining a coherent aesthetic that is approved by design review before construction begins
- …be capable of being opened or closed within 10 minutes, allowing for quick adjustments based on weather conditions or event requirements

4 Design assumptions

4.1 Consequence class

The consequence class of a retractable roof for a stadium is a critical factor in assessing the potential risks associated with its failure and the level of structural safety required. The consequence class is determined based on the potential impact of structural failure in terms of human life, economic loss, and societal disruption.

For a retractable roof in a stadium, the likely consequence class is Class 3, which is the highest class. Class 3 is used for structures where failure would have very serious consequences in terms of human safety, economic impact, and social disruption, all of which apply to a stadium roof.

4.2 Used norms

The design of a retractable roof structure for a stadium involves satisfy to a range of international and national norms and standards that ensure structural safety, reliability, and functionality. These norms cover aspects such as structural integrity, wind and snow loads, material specifications, and safety systems. The following are the key norms and standards typically used in the design of a retractable roof for a stadium:

- NEN-EN 1990: Basis of Structural Design
- NEN-EN 1991: Actions on Structures
- NEN-EN 1992: Design of Concrete Structures
- NEN-EN 1993: Design of Steel Structures

4.3 Loads and load combinations

When designing a retractable roof for a stadium, various loads and load combinations must be considered to ensure the structure's safety, stability, and functionality. All applicable loads will be elaborated and the load combinations will be given for the design. These load combinations will be divided into Ultimate Limit State (ULS) and Serviceability Limit State (SLS). The load combination which is decisive will be used for the dimensioning of the structural elements of the structure.

4.3.1 Loads

The following loads could be taken into consideration for the design of a roof structure over the Philips stadium. Each of these loads will be worked out:

- Self-weight
- Live loads
- Snow loads
- Wind actions
- Thermal actions
- Actions during execution
- Accidental actions
- Traffic loads on bridges
- Actions induced by cranes and machinery

In this report the focus will be on the self-weight, live loads, snow loads, wind actions and thermal loads.

4.3.2 Snow load

A basic way of calculating the snow load is done with the following formula:

$s = \mu_i C_e C_t s$

- $\mu_i = 0.8$
- $C_t = 1.0$
- $C_e = 1.0$
- $s = 0.7\ kN/m^2$
- $s = 0.8 \cdot 1.0 \cdot 1.0 \cdot 0.7 = 0.56\ kN/m^2$

For now this value will be used in the load combinations

4.3.3 Wind load

The wind load of the stadium is calculated now assuming a flat roof.

$F_w = C_s C_d \cdot C_f \cdot q_p(Z_e) \cdot A_{ref}$

- $C_s C_d = 1$
- $q_p = 0.97$
- $C_{f,F} = -1.8$
- $C_{f,G} = -1.2$
- $C_{f,H} = -0.7$

- $q_{w,F} = 1.0 \cdot -1.8 \cdot 0.97 = -1.75 kN/m^2$
- $q_{w,G} = 1.0 \cdot -1.2 \cdot 0.97 = -1.16 kN/m^2$
- $q_{w,H} = 1.0 \cdot -1.8 \cdot 0.97 = -0.68 kN/m^2$

For now these values will be used in the load combinations

4.3.4 Load combinations

The load combinations are based on the safety factors stated in the Eurocode for consequence class CC3.

Ultimate Limit State (ULS)

$$\text{Max}\begin{cases} \sum\gamma_{G,j}G_{k,j} + \gamma_p P + \gamma_{q,1}\psi_{0,1}Q_{k,1} + \sum\gamma_{Q,i}\psi_{0,i}Q_{k,i} & \text{(6.10a)} \\ \sum\zeta_j\gamma_{G,j}G_{k,j} + \gamma_p P + \gamma_{Q,1}Q_{k,1} + \sum\gamma_{Q,i}\psi_{0,i}Q_{k,i} & \text{(6.10b)} \end{cases}$$

6.10a

- $\sum 1.5G_{k,j} + \gamma_p P + 1.65\psi_{0,1}Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j} + \gamma_p P + 1.65\psi_{0,1}Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)

6.10b

- $\sum 1.3G_{k,j} + \gamma_p P + 1.65Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j} + \gamma_p P + 1.65Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)

Serviceability Limit State (SLS)

$\sum G_{k,j} + \gamma_p P + Q_{k,1} + \sum\psi_{0,i}Q_{k,i}$ (Characteristic)

$\sum G_{k,j} + \gamma_p P + \psi_{1,1}Q_{k,1} + \sum\psi_{2,i}Q_{k,i}$ (Frequent)

$\sum G_{k,j} + \gamma_p P + \psi_{2,1}Q_{k,1} + \sum\psi_{2,i}Q_{k,i}$ (Quasi-permanent)

4.4 Main dimensions

The main dimensions of a retractable roof structure over a stadium are influenced by several factors, including the stadium's overall size, the specific design requirements of the roof system, the intended use of the stadium, and environmental considerations. Below are key aspects related to the main dimensions of a retractable roof structure over a stadium:

The roof typically spans the entire length and width of the playing field, with additional coverage for spectator areas and facilities. A football stadium might have dimensions similar to 210 meters (length) by 160 meters (width), similar to the Philips Stadium in Eindhoven. The height of the retractable roof structure must accommodate both the roof mechanisms and any clearance needed for spectators and players. The bottom face of the roof should be positioned above head height to avoid clashes with seating or other structures. The height of the stadium is around 45 meters so the structure should be placed over it.

In the chapter about materialization, the best material option for the structure, based on different criteria, turns out to be steel for the components of the main structure. For the trusses used in the design the largest span would probably be around the same length as the stadium width, which is 160 meter. When looking at Table 1 this span should be divided by 12 or 15 when using steel. When doing so the height of the main trusses should be between 10,6 and 13.3 meter.

The dimensions of the columns are based on the material, which is concrete or steel, and coming from the materialization chapter as well. For a height of 45 meter a concrete column should be approximately be between 3.75 and 4.5 meter. This would mean that gigantic massive concrete reinforced columns should be used. For steel the difference in cross section differs a lot. However, when using steel for the columns, trusses would be the best option to avoid massive sections.

Table 1: Rules of thumb

Dakbalkhoogten (mede afhankelijk van belasting!)			
staal	stalen warmgewalste I-profielen voor daken (tot 15 à 16 m) ($h.o.h. \leq {}^{l}/_{3}$)	enkel veld:	1/30 x lengte
		doorgaand:	1/40 x lengte
	stalen vakwerkliggers met evenwijdige randen	enkel veld:	1/12 à 1/15 x lengte
Kolommen (l = lengte per verdieping)			
Beton	1 bouwlaag	$l \leq 8\,m$	Breedte = ${}^{l}/_{12}$ à ${}^{l}/_{15}$
	Meerdere bouwlagen	$l \leq 4\,m$	Breedte = ${}^{l}/_{10}$ à ${}^{l}/_{12}$
staal	1 bouwlaag	$3 \leq l \leq 8\,m$	Breedte = ${}^{l}/_{20}$ à ${}^{l}/_{25}$
	Meerdere bouwlagen	$3 \leq l \leq 4\,m$	Breedte = ${}^{l}/_{7}$ à ${}^{l}/_{18}$

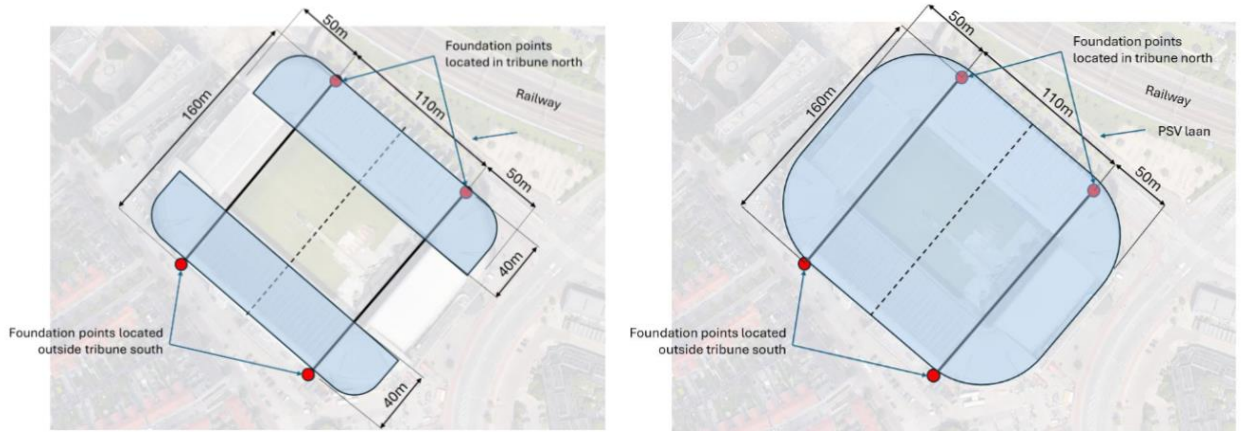


Figure 6: Dimensions of the stadium roof

5 Variant study and selection

5.1 Variants

With inspiration coming from the references stated in Appendix B together with the requirements set in chapter 3 and the design assumptions in chapter 4, some variants could be designed. The variants will be elaborated and discussed. With the use of a Multiple Criteria Analysis (MCA) the variants will be rated and one variant will be chosen for further elaboration.

5.1.1 Variant 1

The first variant is based on the Johan Cruyff Arena and is shown in Figure 1 The structure exists of two large portal trusses with two perpendicular trusses placed in between for stability. At both ends of the stadium the structure has cantilevering trusses shaped around the corners at sides of the stadium. The roof is placed above all tribunes and the retractable roof slides between the two main trusses, the same as for the Johan Cruyff Arena. This roof mechanism is shown in Appendix B1, Figure …. The main dimensions of the structure are given in Appendix B1.

The vertical load distribution is shown in the Appendix B1. Loads coming from the roof are distributed into the two main trusses and the cantilevering trusses at the sides. All the horizontal trusses are supported by the vertical trusses placed on the foundation points. The horizontal stability is shown in Appendix B1. Wind coming from the direction of the red arrow is distributed by the portal frame trusses. Lateral loads coming from the direction of the blue arrow are distributed by the truss between the two portal frames.

The design does have some points of improvement. The cantilevering part gives relatively high moments in the column truss compared to the main trusses placed between the main trusses. These high moments result in a stiff moment connection between the column and the cantilevering part, which makes the structure complex and introduces large cross sections. Furthermore, the design gives a basic impression without spectacular architectural elements. The third point of improvement could be the stability of the structure. Using braces for example would give the structure a better stability and could also help with the complexity of the connections by making them hinged for example.

All the trusses, except for the main ones, are the same principle; a square hollow section. The three dimensional trusses are all the same and not based on the forces acting on them. The last point of improvement of the design is

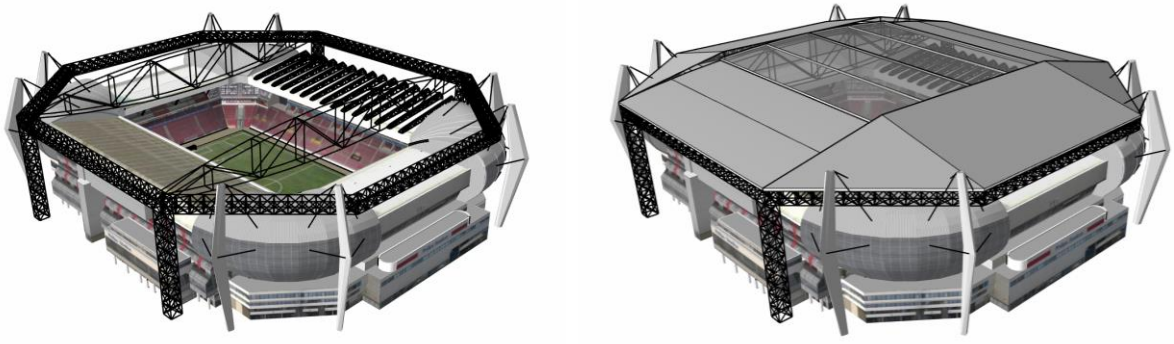


Figure 7: Variant 1

5.1.2 Variant 2

The second variant is based on a bridge principle and is shown in Figure 8. The columns on the foundation points are higher than the ones of variant 1 to hang the cables of the main hanging truss. The cantilevering truss in the middle of each column is there to connect the cables which are used to prestress the main cable truss structure. The roof above the tribunes works the same way as variant 1. However the retractable roof in this case is based on the roof structure of the Bernabeu stadium and is folding to open and close. In Appendix C2 the mechanism of the roof is shown.

The vertical load distribution is shown in Appendix B2. Loads from the roof are going into the cables and are then going to columns. The prestressed cables are used to reduce the cross section of the cables. When a load is applied on the top cable, the bottom cable reduces in tension. However, still all elements remain in tension. Reducing moments and base the elements on axial load in the structure helps with the material efficiency and reduces cross sections. Compression has the second order effect of buckling which is a disadvantage. Therefore tension elements are the most efficient. The cantilevering parts at the sides are the same as in variant 1.

The horizontal load transfer is shown in Appendix B2. Lateral loads go through the main truss into the cantilevering trusses on which all the cables are attached. These cables will bring the forces to the foundation points. This is shown by the red arrows. The blue arrows show the wind distribution in the other direction. The stability here is maintained by the horizontal truss perpendicular to the main cable trusses.

The second variants has some points of improvement as well. First of all the structure would not work in terms of load distribution. When the top and bottom cable are both in tension, there are large tension forces at the connections with the columns. The tension forces of the top cable are tensioned by the diagonal cables going to the middle of the columns and then by the diagonal cables going from the middle to bottom of the columns. When the bottom cable is in tension, the two columns will be pulled towards each other. Therefore the bottom cable should be replaced with a compression element. The cantilever in the middle of the column will be pulled towards the stadium and therefore a compression element should be placed between the two columns. Next to that, prestressing all cables would mean that the foundation should resist the tension coming from the diagonal cables at the bottom. This would give a complex connection at the bottom which needs a larger stiff foundation. Another point of improvement is the cantilevering part which is the same as in variant 1. This would give complex moment connections in the columns. The stability is a part to be approved in the direction which was showed by the blue arrows in Appendix B2. This is the same as in variant 1 as well.

Furthermore, the structural height of this variant is a critical point. More height means more material use, which should be avoided. The last point of improvement would the use of all the same three dimensional trusses which are based on squared hollow sections. To improve the design all elements should be based on the forces acting on them. The main trusses are different from the rest, but the other trusses are all the same.

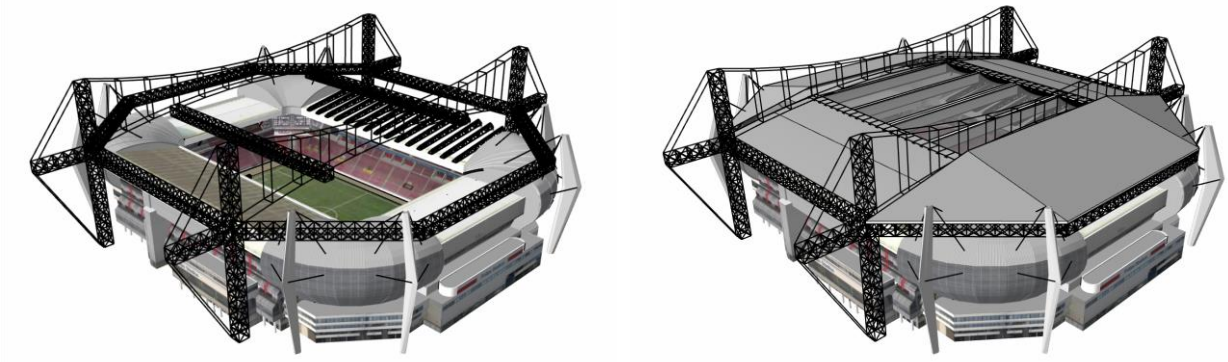


Figure 8: Variant 2

5.1.3 Variant 3

Variant 3 is based on the Arena Națională and has the same roof structure. There is a compression ring at the outside and a tension ring at the inside which are connected by tension cables. However, the Arena Națională has supports all around the compression ring which is not possible for this design. Therefore are large table truss structure is created on which the roof structure is hanging. The opening and closing of the roof is shown in Appendix C3 and exists of a box in the middle of the tension ring from which the roof is sliding open.

The ring hanging below is a stiff element which is hanging from the truss structure. All loads applied on top are going from the ring into the cables into the horizontal trusses and then into the columns. This is shown in Appendix B3. The lateral forces will replace the ring to different sides. To avoid this, not only vertical cables are used, but also diagonal cables going in all directions. The table structure is stabilized in both direction as portal frames with clamped connections at the foundation points.

Variant 3 has some points of improvement. The first being the stability of the whole structure. Using braces in both directions would give the structure more stability and bring more axial forces into the structure instead of moments which will result in smaller cross section and easier connections. Next to that the trusses spanning over the stadium have large points loads in the middle. Shaping this truss based on the moment line would mean that the truss will be higher at the points where the cables are attached. This would mean that smaller cross sections could be used, which reduces the amount of material.

The overall architecture of the structure does not match the stadium. There is a table placed over the stadium with 90 degree angels which are not coming back in the stadium design. Also the hanging circle does not cover all of the area which should be covered. The last point of improvement is the height of the structure which is high because the ring is hanging under it. The structural height increases the amount of material, which should be avoided.

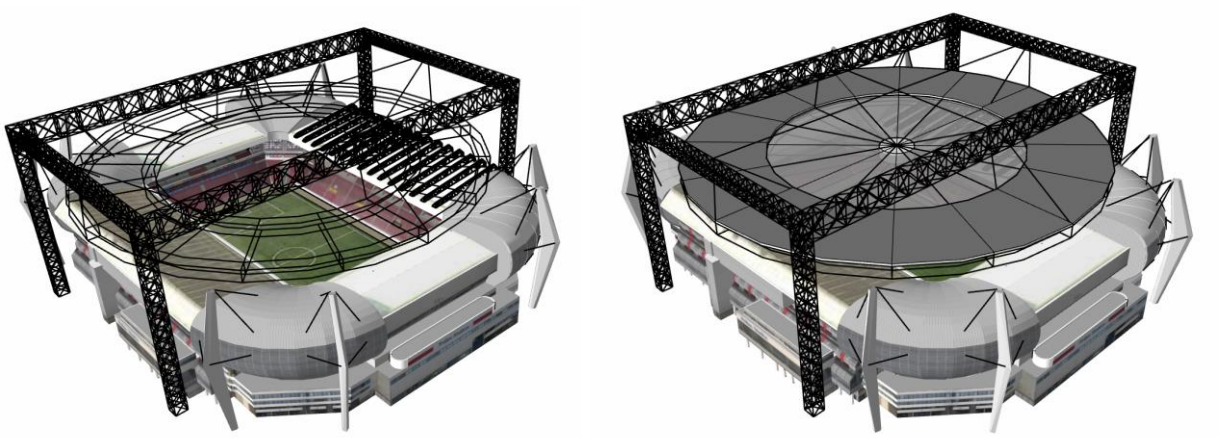


Figure 9: Variant 3

5.2 Materialisation

The materialization of a retractable roof structure for a stadium is a crucial aspect that influences the roof's functionality, durability, aesthetics, and overall performance. Here are key points regarding the materials commonly used in the construction of such roofs:

5.2.1 Main structure

For the main structure the most used material in the references is steel. Steel is one of the most common materials for the structural framework of retractable roofs due to its high strength-to-weight ratio, durability, and flexibility in design. Steel allows for large spans with minimal support, essential for creating open spaces in stadiums. Furthermore steel can withstand dynamic loads, such as wind and seismic forces, and can be prefabricated, facilitating quicker assembly on-site.

However, there is another option for the structure which is aluminium. One significant benefit is its lightweight nature. Aluminium is much lighter than steel, which can reduce the overall load on the supporting structure and foundation. This can lead to cost savings in construction and may allow for more innovative architectural designs. Aluminium also has excellent corrosion resistance, which is particularly important for outdoor structures exposed to the elements. This durability means that the roof can maintain its appearance and functionality over time, often requiring less maintenance than steel.

When comparing steel to aluminium, there are some advantages for steel. One key factor is its strength and load-bearing capacity; steel has a higher tensile strength than aluminium, which allows it to support heavier loads. This is particularly important for a retractable roof that must withstand environmental forces like wind and snow. Cost-effectiveness is another significant consideration. Generally, steel is less expensive than aluminium, especially in large quantities, which can lead to considerable savings during construction—particularly for large structures like a stadium roof.

Welding and fabrication are easier with steel, which simplifies the construction process and improves the integrity of the joints. This use of fabrication is beneficial for the complex designs typically required in retractable roofs. Fire resistance is another advantage of steel; it performs better under high temperatures compared to aluminium, making it a safer option for structures. Finally, maintenance and repair can be simpler with steel, as damaged areas can often be welded back together, whereas aluminium may require different techniques for repairs.

While aluminium has its own benefits, such as being lighter and more resistant to corrosion, the strength, cost, and fabrication advantages of steel often make it the preferred choice for the structural components of a retractable roof.

5.2.2 Roof material

The roof material should be lightweight. This is because the span of the trusses over the stadium is 160 meters. With a large span the dead load is preferred to be low. Transparency of the roof is important as well. There are some different option for the material for the roof which will be explained. One of them is chosen for the final design.

Polycarbonate panels is the first option and stands out due to their exceptional strength-to-weight ratio. They are lightweight and durable, allowing for large spans without the need for extensive support structures. Their high transparency allows natural light to come into the stadium while offering a clear view of the sky. Additionally, polycarbonate panels are UV-resistant, protecting spectators and interior spaces from sun exposure.

Acrylic, or PMMA, is another option that offers excellent clarity and is slightly lighter than polycarbonate. It is also UV-resistant, making it suitable for outdoor applications. However, acrylic is less impact-resistant compared to polycarbonate, meaning it can be more prone to cracking under stress. While it can provide a visually appealing option, its lower durability in comparison to polycarbonate makes it less ideal for large roof structures.

ETFE foil or PTFE foil are other options often used in membrane structures. They are extremely lightweight and flexible, often used in cushion systems for roofing. Both of them are transparent and durable. Its ability to allow light while maintaining a lightweight profile makes it appealing for modern architectural designs.

While all these materials have their advantages, polycarbonate panels combine strength, lightweight nature, transparency, and resistance to weather, making them a particularly suitable choice for a retractable roof over a stadium. Their durability under extreme conditions and cost-effectiveness further enhance their appeal, ensuring a reliable and efficient roofing solution.

5.3 Multiple Criteria Analysis (MCA) of the variants

The variants discussed before will be rated based on the Multiple Criteria Analysis (MCA) seen in Table 2. The criteria are based on personal preferences and are corresponded to a certain weight of importance. Each criterium will be discussed for all variants. Then the variant with the best rating will be chosen as the variant to use for the further process.

Focusing on architecture the first criterium is transparency. All three variants are quite transparent due to the truss structures which are quite open. Most of the roof is closed for all variants and the retractable part is transparent. Therefore all variants are rated the same. In terms of architecture the first variant is most integrated with the design of the stadium. The trusses are hanging on the high columns of variant 2 is the same idea as the large columns holding the already existing roof. Variant 1, however is better matching with the stadium due to the shape of the roof which fits tightly on top of the existing roof. The last variant has a structure which looks really rough in comparison to the stadium. Related to the transparency is the visibility of the existing stadium. All structures only have the same columns in front of the facades. There are no other elements obstructing the façade of the stadium at the outside

The structure is divided into four criteria, the first one being the self-weight of each structure. The self-weight of variant 2 seems to be the lowest due to the use of the prestressed cables. However, the structural height of this variant is higher compared to the first variant. Therefore they would come out to be approximately the same. The third variant has the lowest roof weight due to its ring structure and has the lowest weight. The structural height is the lowest for variant 1. Variant 2 and 3 are of the same height. The environmental impact of all variants is rated low, because of the use of steel in all variants and not focusing enough on the loads applied on the structure which causes larger cross-sections. The third variant however, makes use of structural tension and compression ring which are based on tension and compression and reduce material use. The durability of the materials is the same for all variant because steel is used in all of them for most of the structure. Steel has a long lifespan if it has the right maintenance.

In terms of construction there are three main criteria. Variant 1 and 2 have a truss structure existing of steel elements which are transportable separately. The third variant has the rings which are separated into parts as well. These parts are curved and therefore more difficult to transport. The construction complexity of the first variant is the lowest. The second variant has the prestressed cables which are difficult to install and create large forces in the foundation, which results in complex connections. Variant three has the compressions and tension ring which should connected before lifting the whole structure. This process is complex as well. The demount-ability of the variant 1 and 2 is easier than for the third variant because of the tension and compression ring which have to be disconnected after above the stadium somehow before the structure is able to be disconnected.

Table 2: Multiple Criteria Analysis (MCA) of the variants

Multiple Criteria Analysis (MCA) of the variants

CATEGORY	CRITERIA	WEIGHT	VARIANT		
			1	2	3
Architecture	Transparency	5%	7	7	7
	Matching architecture to the existing stadium	5%	8	7	5
	Visibility of the existing stadium	10%	8	8	8
Structure	Self-weight	20%	6	6	7
	Structural height	5%	8	6	4
	Environmental impact	10%	5	5	6
	Durability of materials	15%	6	6	6
Construction	Transportability to site	5%	7	7	5
	Construction complexity	10%	8	5	5
	Demountable structure	15%	7	7	6
Total		**100%**	6.7	6.4	5.9

5.4 Chosen variant

After evaluating the different design options for the retractable roof structure at Philips Stadium, Variant 1 is the best choice for further development. This variant, inspired by the Johan Cruyff Arena, offers a good balance of structural efficiency, architectural fit, and ease of construction.

In terms of architecture, Variant 1 integrates well with the existing stadium design. The roof shape fits tightly above the tribunes, creating a implemented look. The roof is also transparent, allowing natural light into the stadium and offering a good view of the sky. Unlike Variant 3, which has a more bulky and disconnected appearance, Variant 1 is more smoothly with the stadium's existing structure.

Structurally, Variant 1 is the simplest and most efficient. The roof is supported by two large portal trusses and additional cantilevering trusses, distributing the load evenly. The self-weight of the structure is relatively low, and the height of the structure is kept to a minimum, which helps reduce material use and costs. While the cantilevering trusses create higher moments at the column connections, this issue can be addressed by refining the design, such as adding braces for improved stability. These adjustments can simplify the connections and reduce the need for large cross-sections.

For the materials, the use of steel for the structure ensures durability and strength, while polycarbonate panels for the roof are lightweight and transparent, providing both functionality and aesthetics. Although the design could benefit from optimizing the truss design (for example by using trusses based on the forces they carry), the overall simplicity of the structure makes Variant 1 a practical choice.

In summary, Variant 1 is the most suitable option for the retractable roof at Philips Stadium. It offers a good balance of structural performance, architectural integration, and ease of construction. With some adjustments to improve stability and connection complexity, this variant can meet the technical and visual requirements of the project.



6 Conceptual design

6.1 Flowchart of the steps taken in the conceptual design

For the conceptual design process a particular design approach is used. Below in Figure 10 this design approach is illustrated by a flowchart.

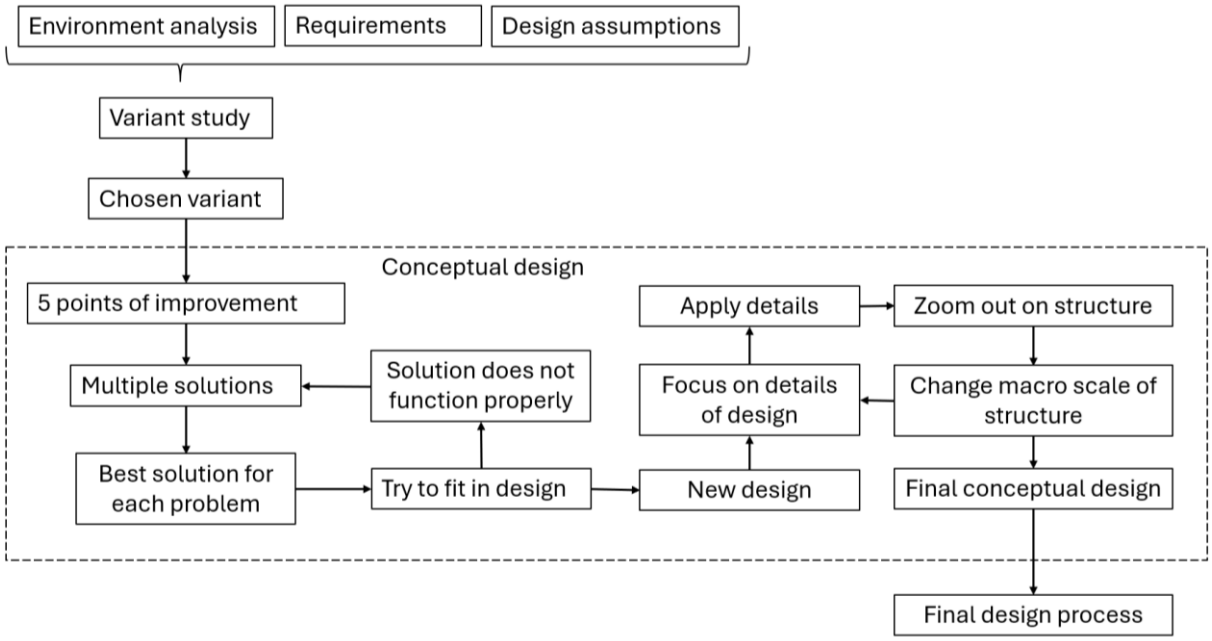


Figure 10: Flowchart of conceptual design process

With the use of the flowchart the chosen variant has changed. Below in Figure 11 the improved design is shown. The main modifications are the added braces which are included for stability and as an extra support for the cantilevering trusses. While the braces are not capable of going through the structure, the angle of the cantilevering part of the structure has changed. These angels are now 90° angles. In the chosen variant the angels were following the roof structure in the corners of the stadium.

The other modification is the change of the main truss. The truss is now a three-dimensional truss which is horizontal instead of the arch truss in the chosen variant. For the first stage hand calculations these trusses are simplified to make quick calculations. In the further design process these main trusses will be elaborated in depth and the best truss structure will be chosen after analysing different options. In Appendix C3 some views of the new design are shown.

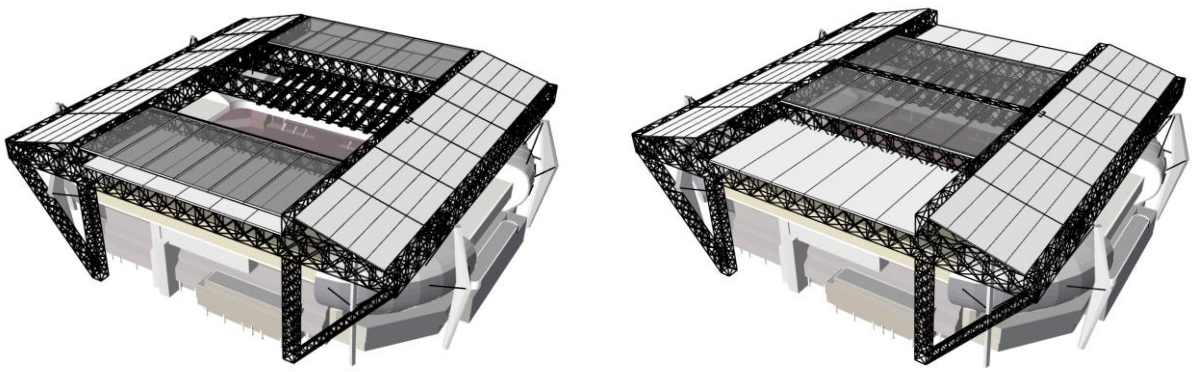


Figure 11: New conceptual design after modifications

6.2 Materialisation

The materialisation of the structure of the retractable roof system is already discussed in chapter 5.2. For the conceptual design the same materials will be used. In general there are 5 components in the structure, Table 3 below shows the division and materials per component.

Table 3: Materials of each component

Component	Material
Roof material	Polycarbonate panels
Truss structure directly attached to roof material	Aluminium
Main truss structure	Steel
Truss columns	Steel
Foundation	Concrete

6.3 Main dimensions

The main dimensions of the roof structure is shown in Figure 13. The roof exists of 6 parts which are supported by trusses underneath. All trusses are now three-dimensional trusses of 5 meter high and 2.5 meter in width except for the truss spanning between point A and B. This truss has a height of 15 meters and a width of 5 meter (Figure 12). There are two different types of roof panels. One at both sides indicated by 5 and 6 in Figure 13. and the others in the middle indicated by 1, 2, 3 and 4. The panels 5 and 6 are 160 by 50 meter and panels 1 till 4 are 40 by 110 meter each.

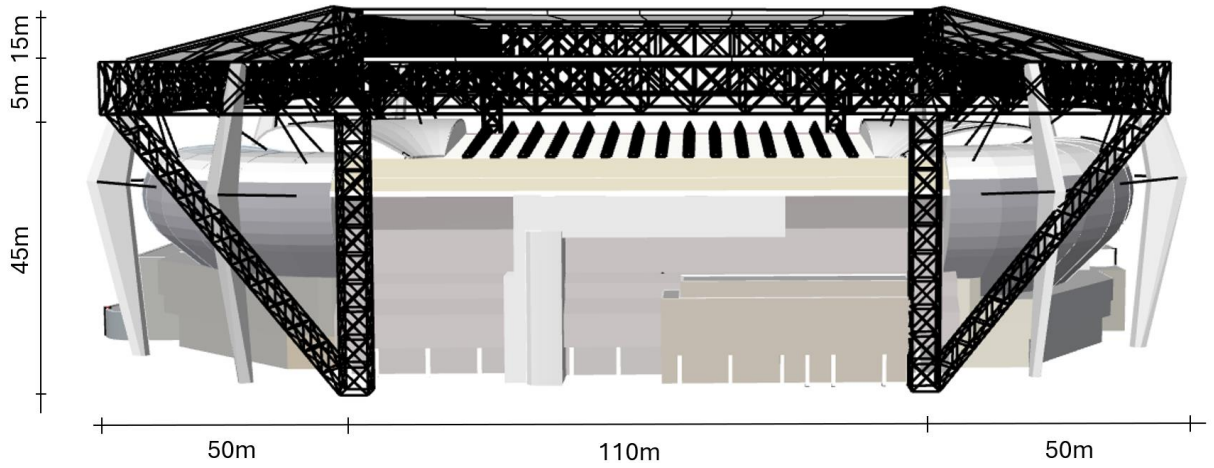


Figure 12: Side view of the stadium

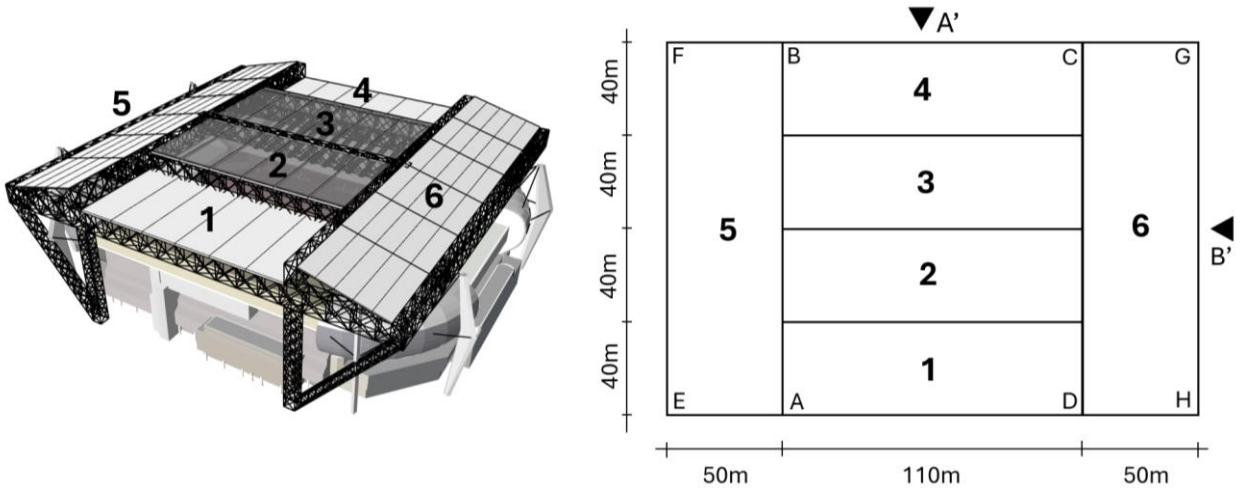


Figure 13: Roof components

6.4 Mechanical schemes

The structure is divided into different parts. All lines in Figure 14 and 15. are trusses. However, for simplicity, only lines are used, to create a clear overview of the connections between the elements.

In Figure 13 two elevations of the structure are shown; A' and B'. In Figure 14. elevation A' is given. The largest truss in this elevation is the one of 210 meter supported by the columns and overhanging on both sides. The diagonals are supporting this truss as well at both ends. The truss is made out of one element to reduce the maximum moment in the middle of the truss.

The column is connected to the foundation with a fixed support for extra stability. While the columns exist of three-dimensional trusses as well, connecting them at all outer sides would be possible. However, moments in the foundation should then be taken into account. The diagonal is meant for stability as well and is hinged at both sides. When lateral forces are applied to the structure, one of the diagonals will work in tension while the other works in compression.

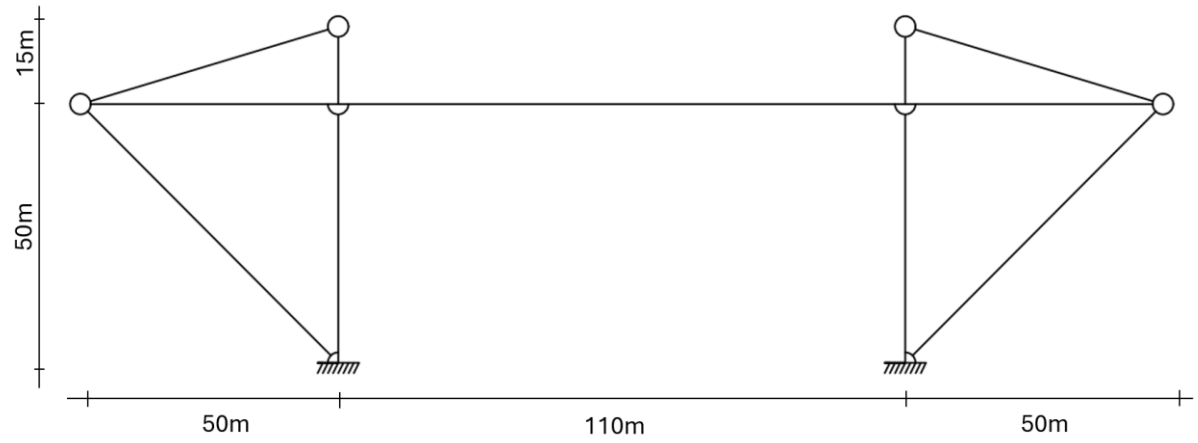


Figure 14: Mechanical scheme elevation A'

Elevation B' in Figure 13 is given in Figure 15. The main truss spanning 160 meter is seen in this elevation. This truss is placed on top of the 210 meter spanning truss which is to be seen in Figure …. directly above the columns of the structure. When the 160 meter truss would be placed between the 210 meter trusses, moments in the 160 meter truss would cause torsional moments in the 210 meter trusses. To prevent this the trusses are placed on top.

Stability in the direction of the 160 meter truss is realized by a portal frame with fixed connections at the foundation points as mentioned before. The 160 meter truss is then connected by hinges on both ends.

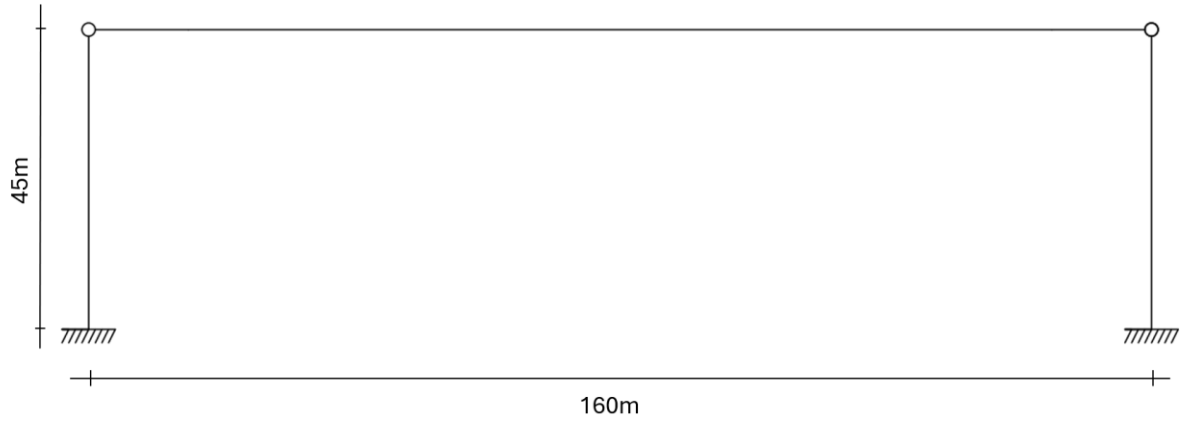


Figure 15: Mechanical scheme elevation B'

6.5 Cross-sectional dimensions of the main structural elements

To find the cross-sectional dimensions of the main structural elements, some hand calculations will be used. In order to find all acting loads on each element, the structural elements should be calculated from top to bottom. In general this means that first the roof dimensions will be calculated. With the use of the self-weight of the roof and the variable acting loads, the dimensions of the trusses could be calculated. After this, the total loads coming from the roof and the trusses will be used to calculate the dimensions of the columns.

6.5.1 Roof

For the first calculations of the trusses the weight of the roof should be calculated. In this early stage of the calculations the weight of the roof will be taken from a reference project. As mentioned before, the roof will exist of 6 parts. The first calculated truss is the one which has the largest span and the highest loads. Assuming all roof parts have approximately the same weight per square meter, the trusses spanning between A and B and between C and D (Figure 13) have the highest loads and the largest span. For the calculation of the main truss spanning between point A and B half of the roof parts 1 till 4 are used for the loads distributed on the truss. Assumed is that the truss between A and B is divided into two trusses. One of them is supporting half of roof parts 1 till 4 and the other one is supporting half of roof part 5.

The roof parts 1 till 4 are based on the polycarbonate panels of the Johan Cruyff Arena. The Johan Cruyff Arena has two sliding roof panels with each having dimensions of 37 x 120 meter [6]. These two roof panels together have a weight of 520 tonnes, which means each roof panel weighs 260 tonnes.

As to be seen in Figure 16 the sliding roof panels have the whole system included. This means not only the weight of the polycarbonate panels is included in the total weight, but also the trusses on which these panels are attached directly. The roof panels used in the design for the retractable roof structure over the Philips stadium have almost the same dimensions, which is why the reference has realistic values to use for this design. This counts for roof panels 1 till 4.

Assumed is that the trusses spanning between A and B used in in the design for the retractable roof structure over the Philips stadium have larger dimensions then the main trusses seen in Figure 16. This is because these trusses only support the two sliding panels while the other roof panels are supported by other elements as well, such as the overhanging elements. The truss between A and B in Figure 13 supports four roof panels on its own.

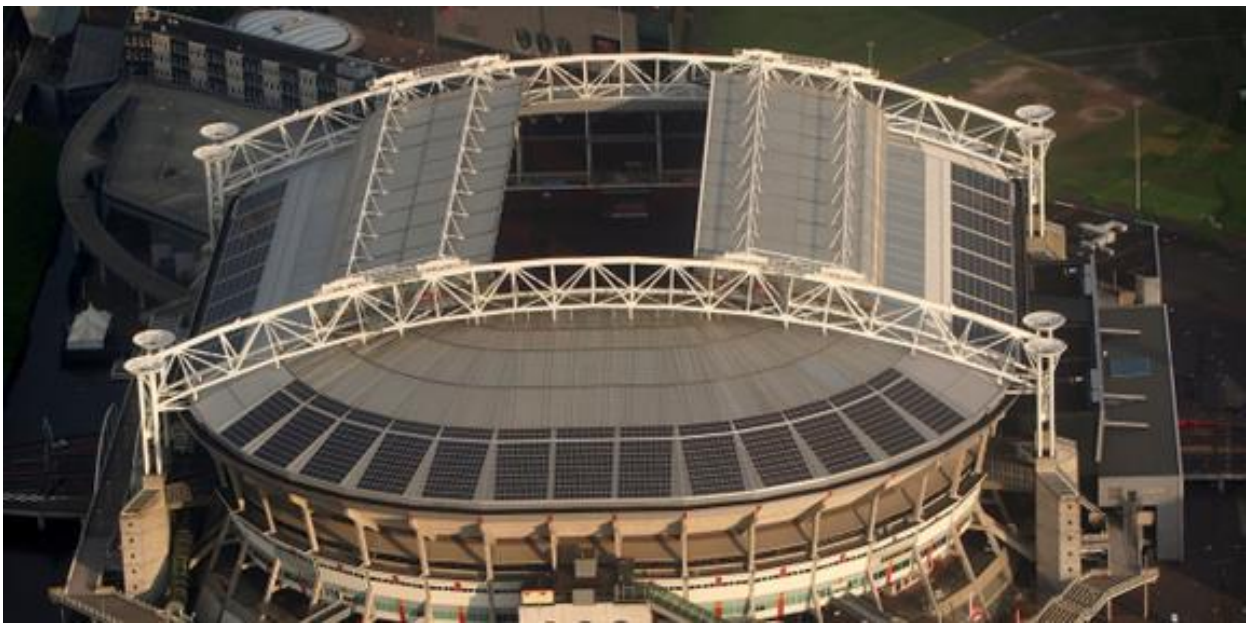


Figure 16: Roof Johan Cruyff Arena

6.5.2 Trusses

As discussed, the first step is to calculate the main truss spanning from A to B in Figure 13. This truss is shown in Figure 17. The span is 160 meter and the height is still to be determined. With the use of an Excell sheet different options are given for different parameters for the truss design. In this chapter these different parameters will be discussed to come to the most efficient structure while fulfilling the requirements.

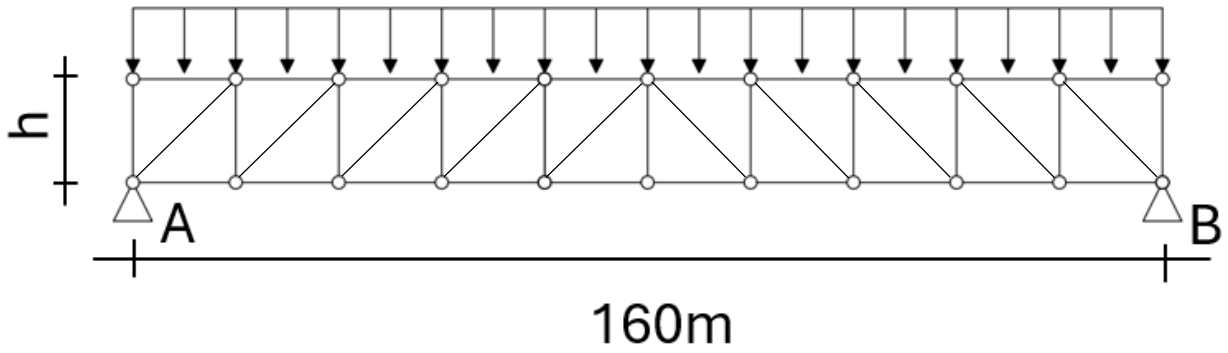


Figure 17: Mechanical scheme truss spanning between A and B

The first calculation of the truss is based on a closed roof with and an unfavourable load combination with safety factor 1.3 for the dead load and safety factor 1.65 for the snow load. The snow load is assumed to be the decisive live load and is taken as the dominant variable load. The other variable loads are multiplied by factor 0. This unfavourable load combination is assumed to be the decisive one. Results are shown in Appendix D1:

6.10b

- $\sum 1.3G_{k,j} + \gamma_p P + 1.65Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)

In this calculation steel grade s235 is used and the height of the truss is 16 meter. All diagonals are placed at 45° of the bottom chord. Different cross sections are filled in until the smallest cross section has enough strength. For the top chord the maximum CHS is used, which is 720x65mm. The top chord is assumed to be in compression and as the maximum compressive force half of the yield strength is used.

As visible in Figure 30 in Appendix E1 the minimum allowed cross sectional area is 2.4e+05 mm^4. The maximum cross sectional area is 1.6e+5 mm^4, which is too low. The bottom chord is assumed to be loaded in tension with an allowable strength equal to the yield strength. CHS 790x55mm satisfies the cross sectional area needed for the bottom chord. The diagonals are assumed to be in tension as well, which means the strength is equal to the yield strength. CHS 620x40 meets the minimal cross sectional area.

The SLS calculation meets the requirements of the maximum deflection of the truss.

While the top chord is 50% too low to resist the compression force, some parameters should be changed in order to find a standard dimension, which is able to handle the compressive force.

In the second calculation the following parameters have changed:

- Steel grade s235 -> s355
- Requirements set: roof should be open while snowing, this means the surface for snow is now half of the surface compared to the first situation
- The load combination changes into the other unfavourable load combination, because the snow load has reduced. The new used load combination is:

6.10a

- $\sum 1.5G_{k,j} + \gamma_p P + 1.65\psi_{0,1}Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)

The calculation is repeated several times until the cross sectional area of all elements is higher than the minimum calculated from the yield strength of s355. The calculation is shown in Figure 31 in Appendix E1. The following cross sections are used for the truss:

- Top chord CHS 780x50mm
- Bottom chord CHS 612x36mm
- Verticals/diagonals CHS 600x30mm

6.5.3 Columns

The columns are made up of steel three-dimensional trusses. These trusses should be capable of resisting the reaction forces which are calculated in the supports of the trusses. For now only the reaction forces of the main trusses are calculated. While the columns are three-dimensional trusses, only one two-dimensional truss is used a support for the main trusses.

While the loads on the columns are compressive loads on top of the end of the truss, all forces are distributed only through the side chords as this is the shortest way to get to the foundation. The diagonals and horizontal parts of the column trusses are only there for stability and to prevent buckling. When calculating the cross sections of the side chords, the reaction force should be divided by two and then the compressive resistance of the steel CHS should be calculated.

In Appendix E2 this calculation is given. It turns out that the cross sections of the side chords are CHS 400x20mm. However, this is only based on the compressive strength of steel. To make sure the cross section could withstand all forces, the second order effect buckling should be taken into account.

6.6 Working principles of the retractable roof sliding system

The retractable roof system has the same principle as the one from the Johan Cruyff Arena. The roof sliding system is shown in Figure 18. For further calculations the loads of the sliding panels could be referred to as traffic loads. The panels are sliding over the trusses with a certain velocity which means loads are moving over the trusses. This process is complex to calculate by hand. By the use of computer software such as SCIA Engineer, these loads could be calculated.

Another approach to calculate the moving loads is to certain points in time where the roof panels are standing still on a certain position. For example the truss could be calculated when the roof is open entirely. This means more force is distributed closer to the supports. Another critical point is when the roof is entirely closed. Then the loads are distributed equally over the whole truss. To make a final check, the truss could be calculated when the roof panels are halfway between closed and open. At the end the calculations with the largest resulting dimensions are decisive.

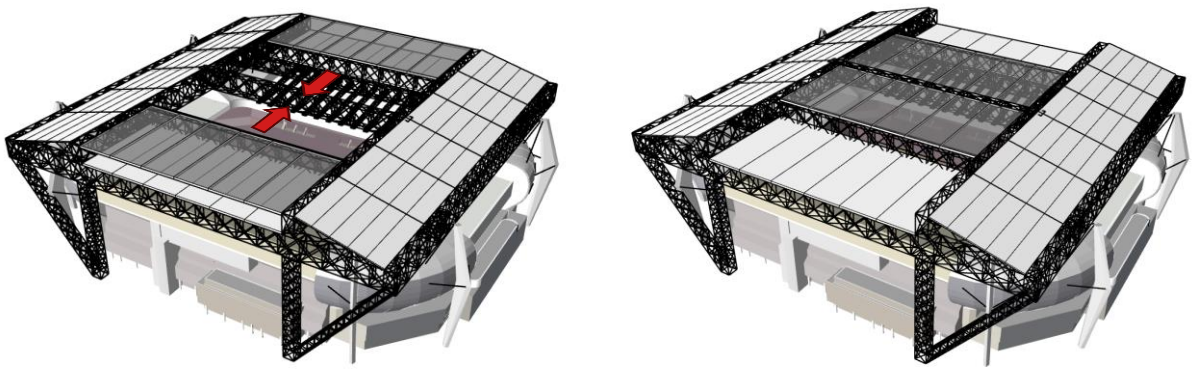


Figure 18: Retractable roof sliding system

6.7 Conclusion

The conceptual design for the retractable roof at Philips Stadium has been carefully developed, with several important changes made to improve stability and support. As shown in the updated design, braces have been added to the cantilever trusses to make the structure more stable. Additionally, the angles of the cantilevering trusses have been changed to 90°, which makes it possible to use the added braces. Another key change is the replacement of the original arch trusses with three-dimensional trusses, which will be further refined during the detailed design process.

The materials selected for the roof structure remain the same as previously discussed. Polycarbonate panels will be used for the roof, while the trusses that support the roof will be made of aluminium, and the main trusses and columns will be made from steel. The foundation will be concrete. This combination of materials is both strong and durable, ensuring the roof will perform well for at least 50 years.

The main dimensions of the roof structure are now clearer. The roof will be made up of six sections, each supported by three-dimensional trusses. The trusses will vary in size, with the largest ones being 160 meters and 210 meters long. The roof panels are designed in two different sizes to ensure proper support and load distribution.

Initial calculations have been made to determine the loads on the trusses, including the weight of the roof panels and snow loads. These calculations will be updated in later stages to ensure the trusses can handle all the expected loads. The dimensions of the trusses and columns have been chosen to meet the required strength and stability. Steel grades S355 is used for the trusses and columns.

The retractable roof system will work similarly to the one at the Johan Cruyff Arena, with panels that slide over the trusses. These moving panels create dynamic loads, which will be analyzed using specialized software to ensure the design can handle all the different forces as the roof opens and closes.

In summary, the conceptual design for the retractable roof at Philips Stadium has been developed with a focus on stability, material choice, and load distribution. While the initial calculations are based on estimated values, further detailed analysis will be carried out to refine the design and ensure it meets all the necessary safety and performance standards.

7 Final design

7.1 Flowchart of the steps taken in the final design

After the conceptual design was finished, the final design could be elaborated. However, some new ideas came up during the process. These concepts will be worked out and after that the final design will be finished. Below the flowchart is given of the process during the final design phase (Figure 19).

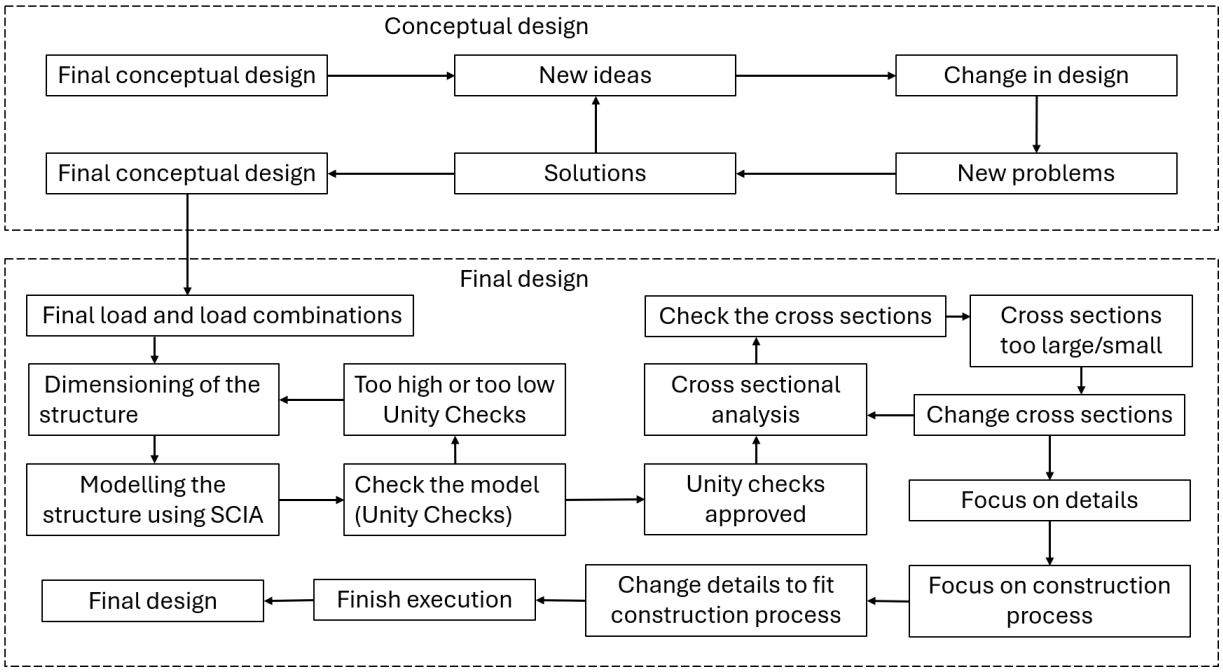


Figure 19: Flowchart final design

During the process, first the conceptual design had some changes. Below in Figure 20 the modifications are shown. The trusses with the largest span are were first changed into lenticular trusses and after changed into arches with tension elements. The first modification (truss to arch) was implemented to reduce moment in the elements and have a better water accumulation and more aesthetics. After calculating the new lenticular arch it turned out that the height in the middle did not met the requirement. When using a horizontal tension element instead of an arch the total height of the whole truss in the middle of the stadium decreases, which leads to a lower support point at the side.

After the changes in the whole design, all parts of the structure could be elaborated. Starting with the loads and load combinations, the dimensioning of the structural elements is calculated. Then with the use of a FEM program; SCIA, the whole structure is checked as working principle of all elements together. When all unity checks have fulfilled the requirements, the cross sections of the elements will be checked. The details are determined implementing different aspects, such as the design of the whole structure, mechanical efficiency and the execution. At last this execution process will be explained



Figure 20: Modifications in design

7.2 Loads and load combinations

7.2.1 Snow load

The snow load on the roof of the stadium is dependent on different factors. Below all factors are explained and applied on the roof of the stadium. All factors are used in the following formula:

$s = \mu_i C_e C_t s_k$

- $\mu_i = snow\ load\ shape\ coefficient\ depending\ on\ the\ shape\ of\ the\ roof$
- $C_e = exposure\ coefficient$
- $C_t = thermal\ coefficient$
- $s_k = characteristic\ value\ for\ snow\ load\ on\ the\ ground$

First of all for the characteristic value of the snow load on the ground (s_k) in the Netherlands, s_k = 0.7 kN/m2 must be assumed for each location.

The exposure coefficient C_e should be used to determine the snow load on the roof. When choosing C_e, future developments around the location should be taken into account. C_e is equal to 1.0, unless otherwise specified for different environments.

The heat coefficient C_t should be used to take into account reduction of snow load on roofs with a high heat transfer (> 1 W/m2K), such as glass roofs, due to melting snow through heat losses. For all other cases: C_t = 1.0. Although the roof is made of transparent polycarbonate panels, there is no heat transfer through the panels because the stadium itself is not warmed up. The heat coefficient for every building in the Netherlands is: C_t = 1.0.

The snow load shape coefficient is based on the shape of the roof. For all parts of the roof (1 till 6) in Figure 21 the roof has a cylindrical form. The stadium is not surrounded by taller building structures. For the cylindrical roof the following formulas are used:

$$\beta > 60°, \mu_4 = 0$$
$$\beta \leq 60°, \mu_4 = 0.2 + 10 \cdot \frac{h}{b}$$

The maximum angle of the arched roof is 28° (Figure …), which indicates that every slope on the arch has an angle smaller than 60°. In Figure … in Appendix … the snow load shape coefficient is illustrated for cylindrical roofs. Case (i) is the undrifted snow load arrangement which is $\mu_1 = 0.8$. Case (ii) is the drifted snow load arrangement which is calculated with the formula given above. The height h of the arch is 20 meter and the width b of the arch is 160 meter which is also equal to I_s because the angle is below 60° for the whole arch.

$$\beta \leq 60°, \mu_4 = 0.2 + 10 \cdot \frac{20}{160} = 1.45$$

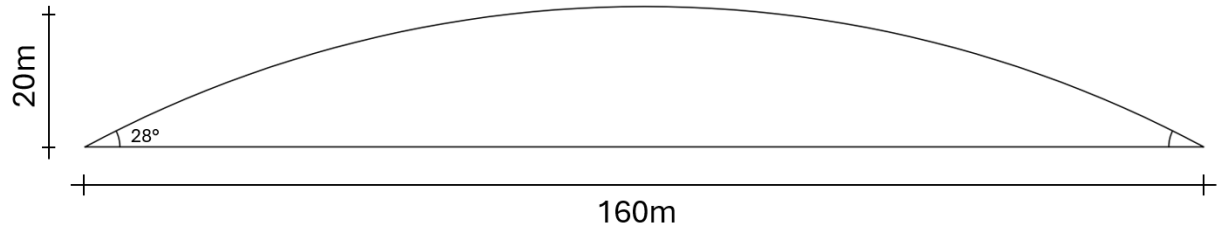


Figure 21: The slope of the arches

$s = \mu_i C_e C_t s_k$

- $\mu_i = 0.8$
- $C_t = 1.0$
- $C_e = 1.0$
- $s = 0.7\ kN/m^2$
- $s = 0.8 \cdot 1.0 \cdot 1.0 \cdot 0.7 = 0.56\ kN/m^2$

7.2.2 Wind load

The wind load on the roof of the stadium is divided into two parts as well as for the wind load. Again there are different factors playing a role in the determination of the wind load. Below the general formula is given for the wind load.

$$F_w = C_s C_d \cdot C_f \cdot q_p(Z_e) \cdot A_{ref}$$

- $C_s C_d = 1$
- C_f
- $q_p = 0.97$
- $C_{f,F} = -1.8$
- $C_{f,G} = -1.2$
- $C_{f,H} = -0.7$

- $q_{w,F} = 1.0 \cdot -1.8 \cdot 0.97 = -1.75 kN/m^2$
- $q_{w,G} = 1.0 \cdot -1.2 \cdot 0.97 = -1.16 kN/m^2$
- $q_{w,H} = 1.0 \cdot -1.8 \cdot 0.97 = -0.68 kN/m^2$

7.2.3 Thermal load

The thermal load has a significant influence on the structure of the roof, due to its large span and the use of steel for the load bearing structure.

Assumed is a clear light colour at the outside of the roof. Therefore the significant factor 0.5 is used for the formula of the thermal load. As seen in Table … the temperature T_{out} is therefore is chosen to be $T_{max} + T_3$.

T_{max} = 30°C and T_3 = 20°C so T_{out} = 30 + 20 = 50°C

T_{min} = -25°C

The total temperature difference is then: 50 – (-25) = 75°C

Table 2: Indicative temperature difference

Tabel 5.2 — Indicatieve temperaturen T_{out} voor gebouwen boven het grondniveau

Seizoen	Significante factor		Temperatuur T_{out} in °C
Zomer	Relatief absorberend vermogen afhankelijk van de kleur van de buitenzijde	0,5 helder licht	$T_{max} + T_3$
		0,7 licht gekleurd	$T_{max} + T_4$
		0,9 donker	$T_{max} + T_5$
Winter			T_{min}

The values of T are all given in Table … below.

Table 3: Temperature values

Waarin:

T_{max} = 30 °C

T_{min} = −25 °C

T_3 = 20 °C

T_4 = 30 °C

T_5 = 45 °C

7.2.4 Load combinations

To find the decisive load combination all possible load combinations are shown below. The following load cases are checked:

LC1: Dead load
LC2: Dead load + wind load
LC3: Dead load + snow load

Max $\sum\gamma_{G,j}G_{k,j} + \gamma_p P + \gamma_{q,1}\psi_{0,1}Q_{k,1} + \sum\gamma_{Q,i}\psi_{0,i}Q_{k,i}$ (6.10a)
$\sum\zeta_j\gamma_{G,j}G_{k,j} + \gamma_p P + \gamma_{Q,1}Q_{k,1} + \sum\gamma_{Q,i}\psi_{0,i}Q_{k,i}$ (6.10b)

6.10a

- $\sum 1.5G_{k,j} + \gamma_p P + 1.65\psi_{0,1}Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j} + \gamma_p P + 1.65\psi_{0,1}Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)

6.10b

- $\sum 1.3G_{k,j} + \gamma_p P + 1.65Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j} + \gamma_p P + 1.65Q_{k,1} + \sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)
-

LC1:
$\sum 1.3 \cdot 2.9 + 0 + 0 + \sum 0 = 3.77\ kN/m^2$

LC3
$\sum 1.3 \cdot 2.9 + 0 - 1.65 \cdot 1.75 + \sum 0 = 0.88\ kN/m^2$

LC2:
$\sum 1.3 \cdot 2.9 + 0 + 1.65 \cdot 0.56 + \sum 0 = 4.7\ kN/m^2$

The snow load is governing in the ultimate limit state.

7.2.5 Dimensioning of the structure

With the use of the calculated loads and load combinations the dimensioning of the structural elements could be performed. With the use of hand calculations an Excel sheet is made. This Excel sheet is made in a way that makes it possible to change certain factors and compare different outcomes. This is used to minimize cross sectional areas and therefore the weight of the structure.

First the mechanical schemes of the new structure are given in Figure … and Figure …. The structure is divided into different parts. For the hand calculations each element is calculated separately. Starting at the top of the structure and ending at the foundation, all elements are determined. Each time an element is calculated, the weight of this element is used in the calculation for the next element. The order of the elements is shown below:

1. Truss roof 1-4
2. Truss roof 2-3
3. Truss roof 5-6
4. Main arch
5. Side arch
6. Stability truss
7. Column
8. Brace

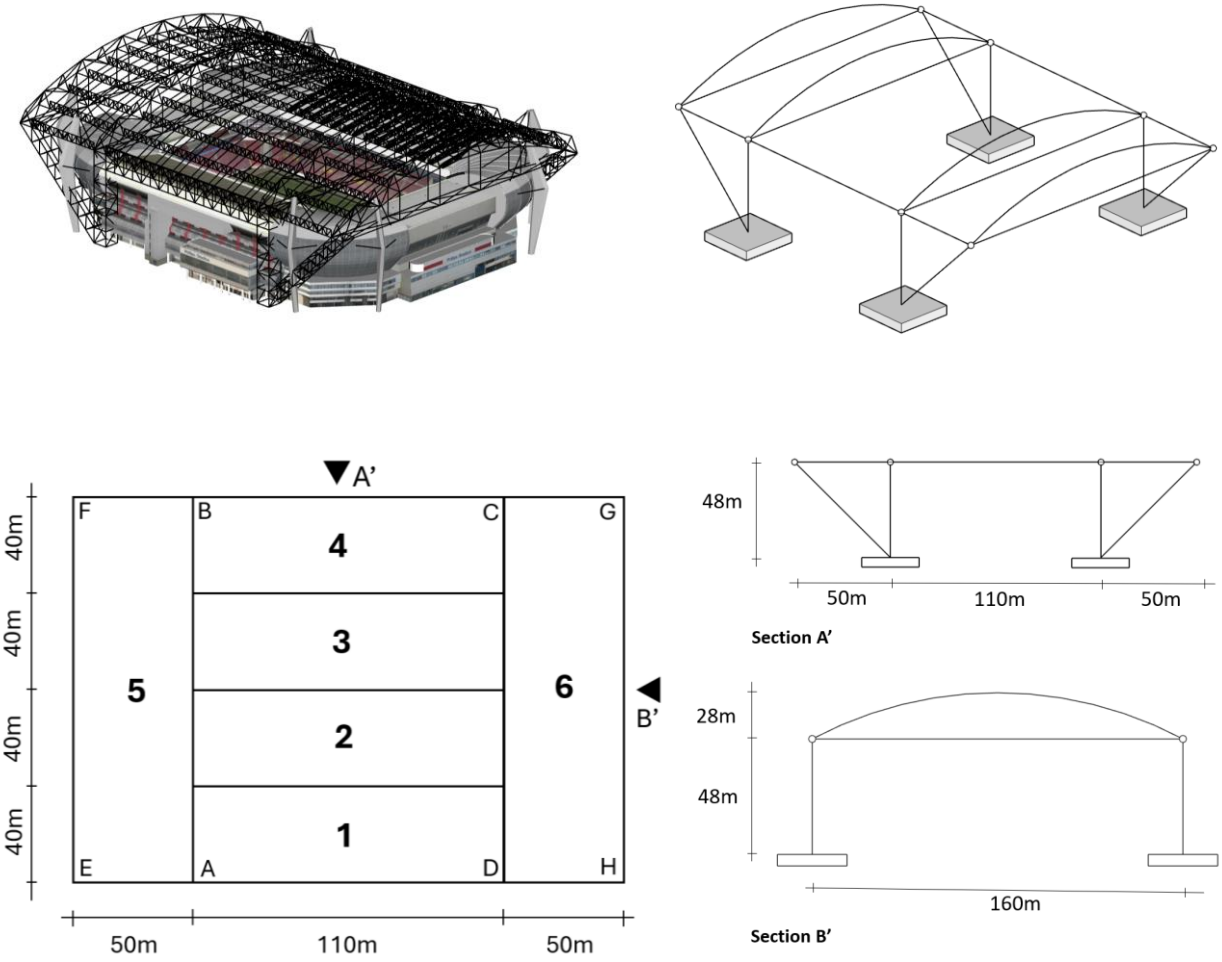


Figure 22: Structure and mechanical schemes

1. Truss roof 1-4

The first truss that will be calculated is the truss which supports the roof panels. They are supporting the part of the roof between the main arches which does not move. When looking at Figure … these trusses are only supporting part 1 and part 4. The height of these trusses is 5 meter and the support points are placed on the bottom chords of the two main arches. The height of the trusses is based on the height of the main arch. Preferably the height of the trusses is as high as possible to reduce cross sectional areas. However, the retractable roof should be able to be placed on top of these roof parts 1 and 4. The retractable roof is attached to the top chords of the main arches. The main arch is 8 meters in height and the retractable roof trusses span the same distance as the trusses that support the roof parts 1 and 4. This means that the height of both kind of trusses should be close to each other. When the support points of the trusses of the retractable roof are in the middle of the of the height of the truss, the height of these trusses could be 5 meter as well. In short this means that 2.5 meter is placed below the top chord of the main arch and 2.5 meter is above it. When the trusses of roof part 1 and 4 are 5 meter, there is 8-5-2.5 = 0.5 meter left between the retractable roof and the roof trusses of parts 1 and 4.

With the use of hand calculations the dimensions of the cross sections of the elements in the roof trusses are calculated. The decisive load combination is used which includes snow. In Appendix … the hand calculation is shown. With an Excel sheet different cross sections are practiced to see which one has the smallest cross section with a unity check below 1.0. This is done to reduce the amount of steel used in the structure. Below in Figure … the mechanical scheme is presented. The maximum moment of the truss is calculated due to the load. This moment is divided by the height to find the axial forces in the top and bottom chord. Assumed is that the top chord is only loaded in compression and the bottom chord is loaded in tension. Steel grade S355 is used to find the minimum cross sectional area needed in the ultimate limit state. The serviceability limit state is checked as well for the cross sections which are used.

It turns out that the dimensions of the elements are most efficient when the following cross sections are used:

Truss 1-4
Top chord:
Bottom chord:
Verticals/diagonal:

With the use of the finite element program SCIA Engineer, The truss is modelled and checked individually. The results are shown in Figure … in Appendix …. It turns out that additional supports in the middle should be used on some points to avoid lateral torsional buckling. In the individual model these support will be placed on some points on the truss. In the model of the total structure, these supports are replaced by elements perpendicular to the trusses on the top and bottom chords.

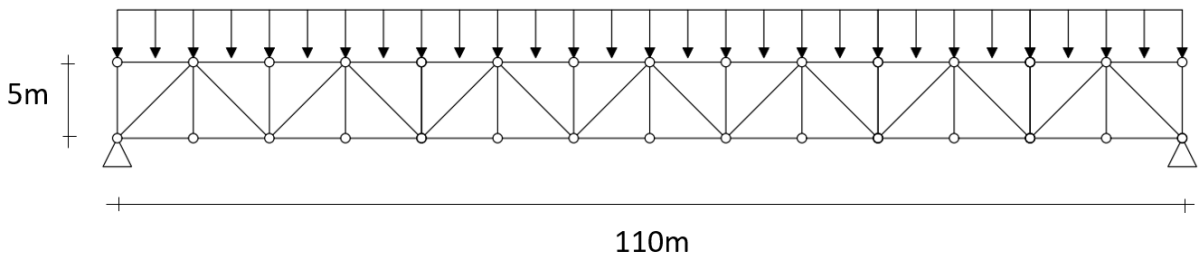


Figure 23: Truss roof 1-4

2. Truss roof 2-3 (retractable roof)

The retractable roof has the same span as roof parts 1 and 4. However, a different setup is made for this roof. To make the sliding principle work, only two support points are desired. This principle is the same as for the reference project of the Johan Cruyff Arena (Figure …). When using only two support points per retractable roof per side, the roof panel is able to slide easily. The more supports used, the more stiff the support. This makes it more difficult to rotate over the arch.

To avoid snow from accumulating on the roof, most of the roof structure is below the poly carbonate panels. In Figure … the setup of the roof is shown with two sections. The roof exists of a framework with poly carbonate panels placed on top. Most of the trusses have the same height and cross sectional dimensions as the truss used in the roof parts 1-4. However, due to the fact that only two support points are used per side of the roof panel, the truss which is attached to the arch is 2.5 meter higher. This truss distributes all loads on the roof to the support points, this means more load is distributed then the other trusses. When the roof panel was checked in SCIA Engineer the stresses and displacements in the roof where higher then required. Therefore extra elements are implemented in the panel. These elements are the diagonals spanned between the top of the small trusses and the top of the higher trusses as to be seen in Figure ….

The following dimensions are used in the trusses of the retractable roof parts:

Side truss
Top chord:
Bottom chord:
Verticals/diagonal:

Supported middle trusses
Top chord:
Bottom chord:
Verticals/diagonal:

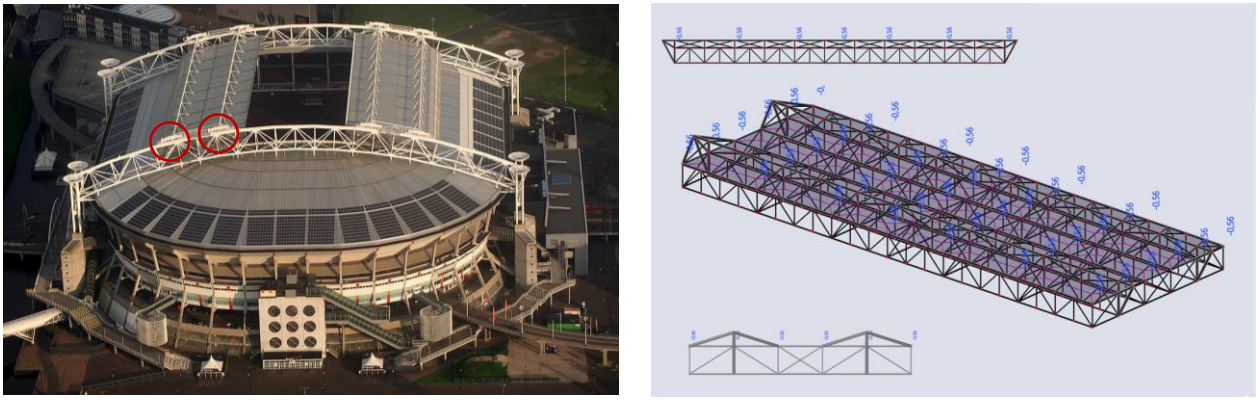


Figure 24: Sliding roof principle

3. Truss roof 5-6

The trusses which are placed between the main arch and the side arch, have a span of 50 meter. They support the poly carbonate panels of the roof in part 5 and 6. With the same calculation as the calculation of the trusses for roof parts 1 and 4, these trusses are calculated. With an Excel sheet shown in Appendix … Figure … the cross sections of all elements are calculated. The following dimensions are used for the truss of roof parts 5 and 6:

Truss roof 5-6
Top chord:
Bottom chord:
Verticals/diagonals:

4. Main arch

The largest elements in the structure are the main arches. These arches span 160 meter and have a height of 20 meter. According to the requirements, the height of the structure should not exceed 20 meter above the height of the stadium. Again an Excel sheet is used to calculate the cross sectional dimensions. In Appendix … the Excel sheet is shown. With the use of the hand calculations shown in Figure … in Appendix … the dimensions of the top and bottom chord are calculated. The reference project of the Johan Cruyff Arena is used to predict the dimensions of the chords. When the total height of the truss is limited to 20 meter and the distance between the top and bottom chord is 8 meter (based on the Johan Cruyff Arena), the height of each truss is limited till 12 meter. It turns out that the dimensions of the chords in this case are significantly higher than the cross sectional dimensions of the Johan Cruyff Arena. Therefore the decision is made to exceed the height requirement. The total height is now 28 meter.

In Figure … the truss and its dimensions are shown. The cross sectional dimensions are shown as well. The structure is checked in SCIA Engineer, with the loads from the roofs. The arch is checked on different load combinations. Even though the snow load is the decisive load combination, the thermal load combination and wind combination is checked for the main arch as well. The arch is modelled together with the columns. The connection between the arch and the column is hinged but not a roller. The columns are moment fixed in the foundation. When the main arch elongates/shrinks due to thermal loads, the columns will be pushed outwards or pulled inwards. Therefore the stiffness of the columns should be low enough for the top part to move along with the arch. When the stiffness is too high, the internal stresses will exceed the stress limit of the steel. In Figure … in Appendix … the SCIA Engineer results are shown for the thermal load combination. The wind is checked in SCIA Engineer as well as shown in Figure … in Appendix …. This is relevant as well, because wind is coming from the sides as well. Stability should be checked based on the wind load.

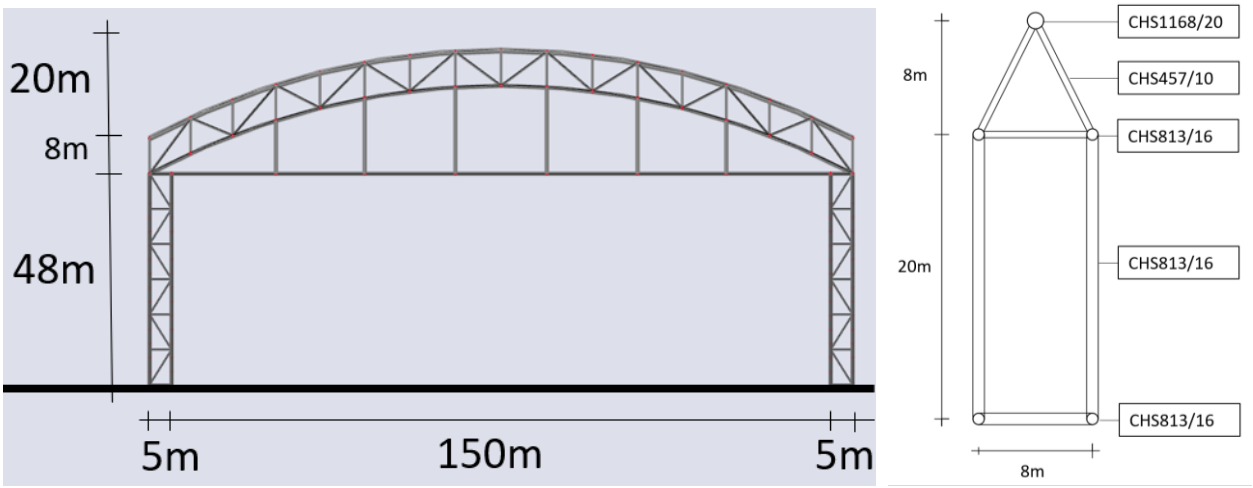


Figure 25: Main arch dimensions

5. Side arch

The side arch is based on the same principle as the main arch The side arch is placed on the braces instead of the columns. In Figure … the dimensions and cross sections are shown for the side arch. The loads on this arch are lower which means a slender structure is used for the side arch. In Appendix … the Excel sheet is shown for the side arch, and the arch is checked in SCIA Engineer as well. These results are found in Figure … in Appendix ….

The height of the side arch is the same as the height of the main arch. The reason for this is that the roof truss could be placed horizontally. When the side arch is lower then the main arch, diagonal trusses should be used for the roof. Diagonal trusses introduce horizontal forces in the main arch as well which should preferably be prevented.

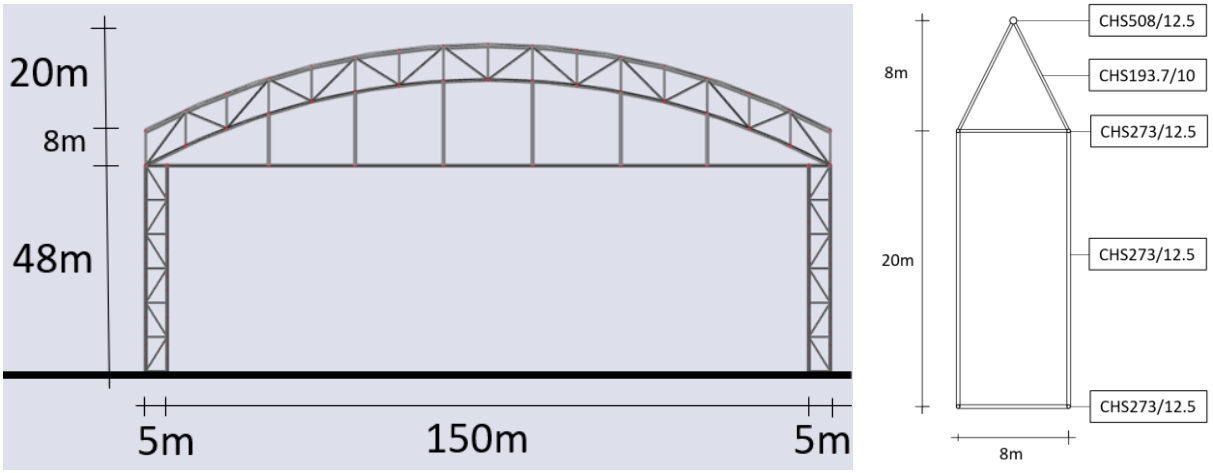


Figure 26: Side arch dimensions

6. Stability truss

To stabilize the two arches, there is a truss placed between them. This truss is shown in Figure … below. In Appendix … an Excel sheet is used to determine the cross sectional dimensions. The truss is only meant for stability and does not carry any vertical loads except for its self-weight. Therefore relatively small cross sections are used for the elements.

The stability truss is a three-dimensional trusses instead of two-dimensional trusses. For all roof trusses the trusses are two-dimensional combined with perpendicular elements to resist lateral torsional buckling. The stability truss is an individual member which will not be attached to the roof members. Therefore, over a span of 110 meter, the truss needs some thickness to prevent lateral torsional buckling.

In SCIA, the stability truss is not calculated separately, but only in the full structure. With the use of a the wind load combination the stability for the whole structure could be checked. The structure is also checked with the snow load combination.

Between the top of the diagonal and the column seen in Figure … there is a tension element. This element is used to prevent the side truss from falling. Again this truss is only there for the stability of the structure and therefore only distributing a vertical load coming its self-weight. This truss is spanning 50 meter and is designed as a two-dimensional truss placed horizontally. The height of the truss is not needed vertically because the tension is significantly higher than the moment in the truss. The tension elements exists of two chords with diagonal and horizontal elements in between with the following dimensions:

Stability truss

Side chords:

Verticals/diagonals:

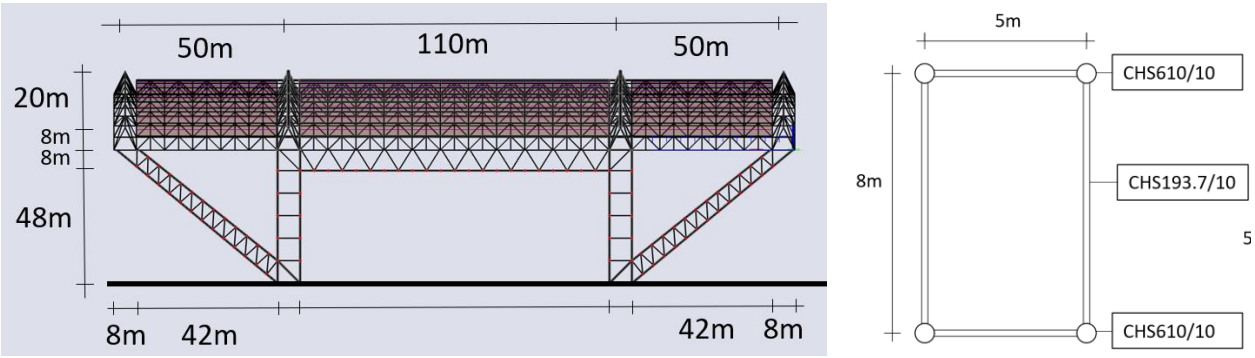


Figure 27: Stability truss dimensions

7. Column

The columns of the structure are three-dimensional trusses to divide the stresses from the structure into more columns with smaller cross sectional areas. To reduce the buckling length of the columns, horizontal and diagonal elements are implemented. The column and its dimensions is shown in Figure …. The column is moment fixed at the bottom. With the use of SCIA Engineer the column is checked with all loads.

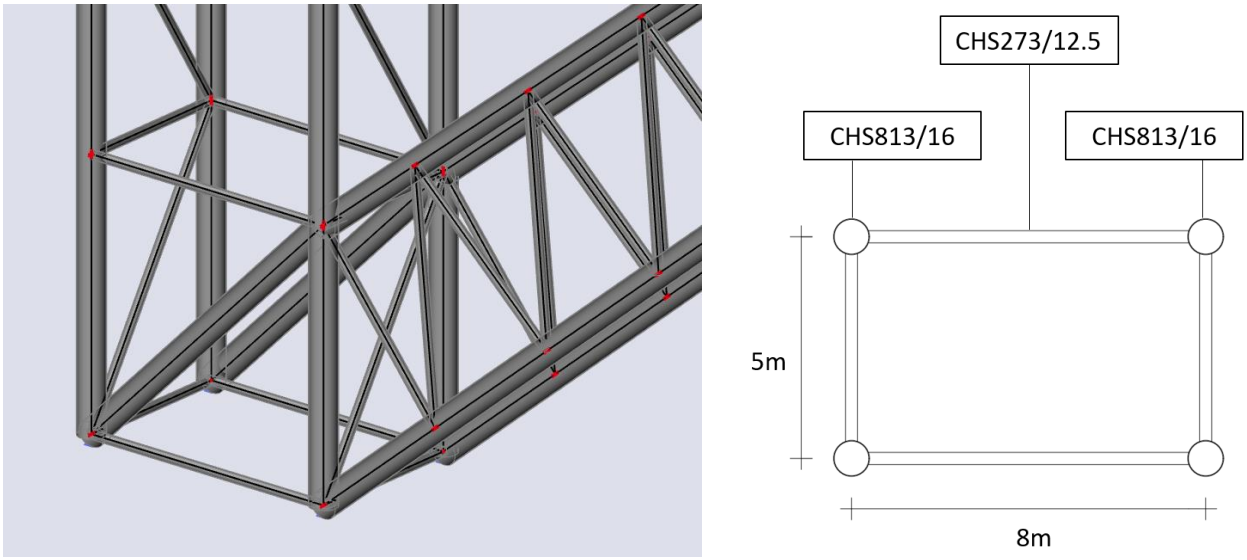


Figure 28: Column dimensions

8. Brace

The diagonal brace in the structure is implemented to transfer the loads coming from the sides directly into the foundation. The cross sectional areas of the elements in the brace are the same as for the columns. While the total weight of the side arches is significantly lower, the brace is still designed this way. Due to the diagonal direction the forces are not only vertical, but horizontal as well. In Figure … below the brace is shown attached to the side arch and the tension element.

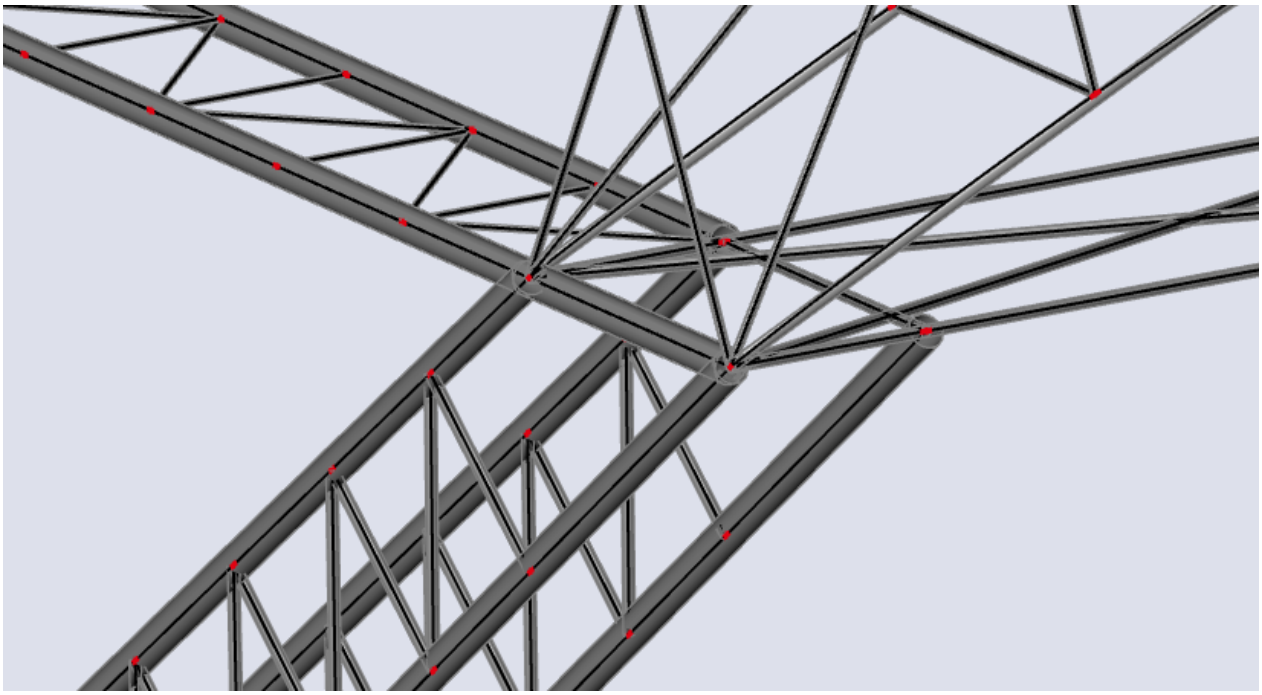


Figure 29: Diagonal brace

7.3 Force distribution

Design decisions are partly based on the force distribution. In this chapter, the force distribution will be explained. With the use of the finite element program SCIA Engineer the structure will be checked as a whole to see how elements of the structure work together and to check if all element are able to resist the stresses coming from the load combinations.

In Figure … the general force distribution is shown for vertical forces. Forces distributed on the roof are transferred to the roof trusses. The roof trusses distribute those forces into the arches. The arches are placed on the columns or diagonals and distribute the forces via them to the foundation. The horizontal load distribution is shown in Figure …. The diagonals in all trusses pay a role in the distribution of the horizontal forces. Also the large brace is implemented for horizontal load distribution.

With the use of SCIA Engineer the structure is calculated as a whole. The structure is divided into two parts: one part exists of all elements in the roof structure (open and closed) and is shown in Figure … and the other part exists of the columns stability trusses and arches, without the roof trusses (Figure …). The retractable roof is modelled separately as well (Figure …). The reason for this are the problems occurring in SCIA Engineer. When implementing the whole structure into one model, SCIA Engineer crashed.

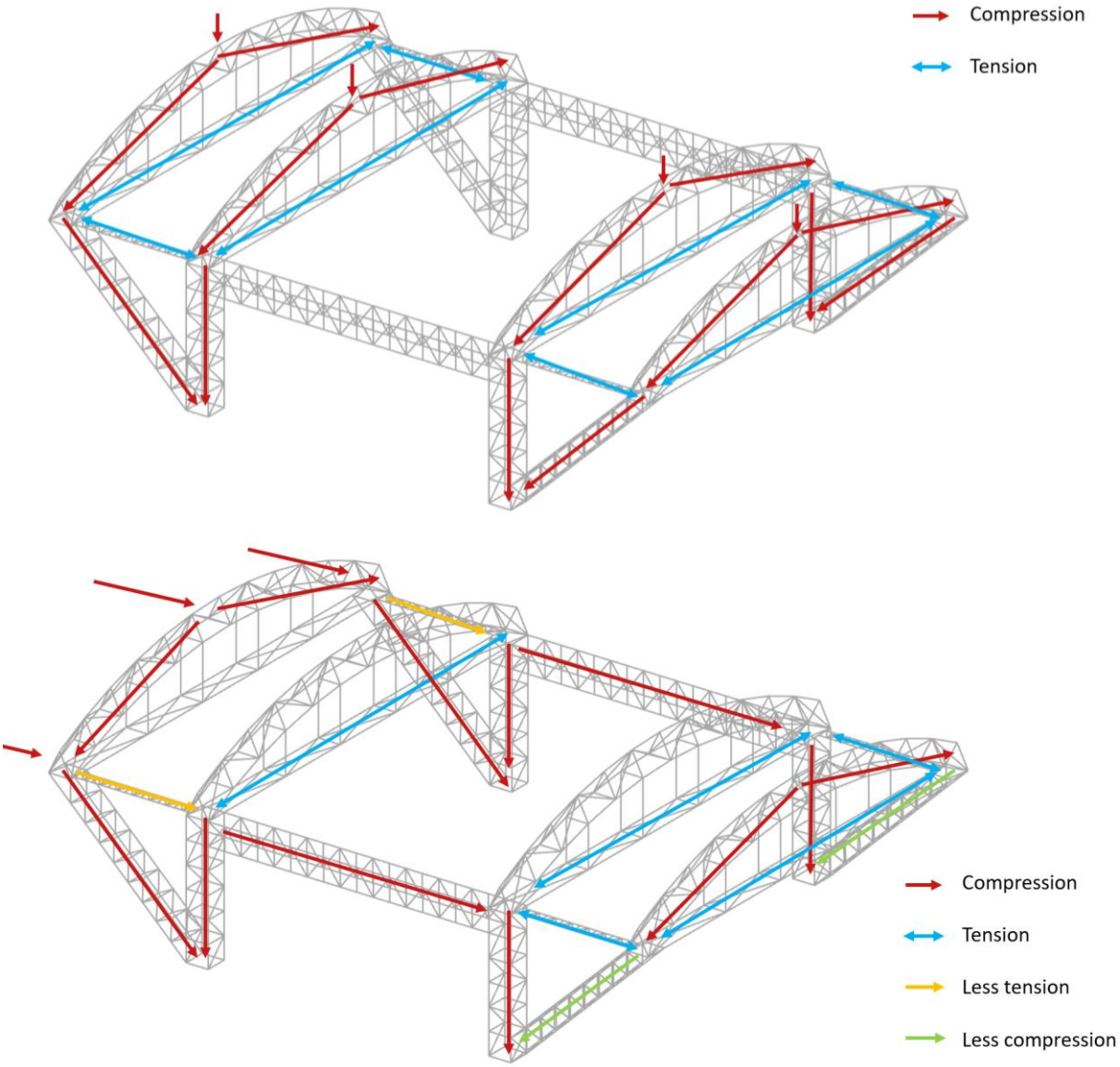


Figure 30: Vertical and horizontal load distribution

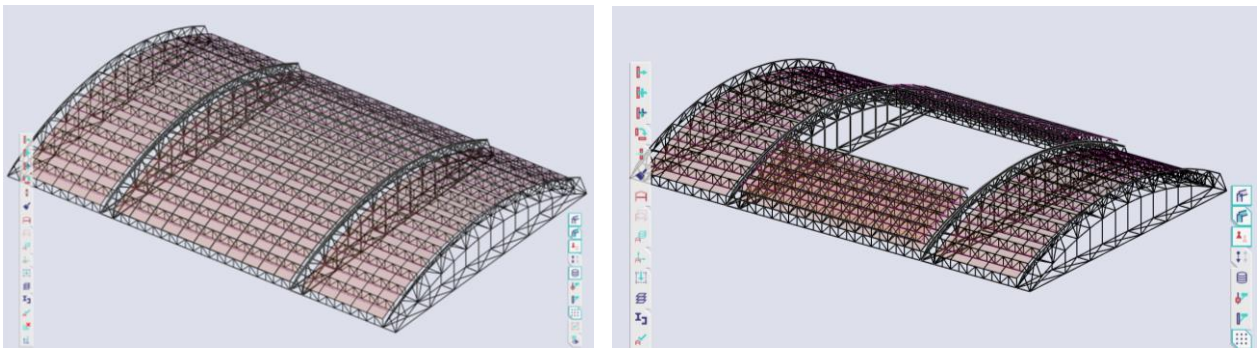


Figure 31: Closed and open roof structure

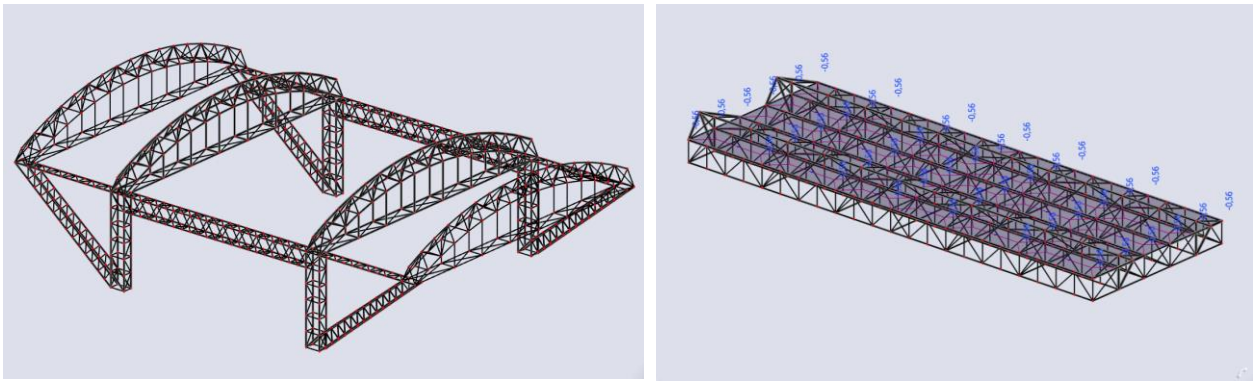


Figure 32: General structure and retractable roof

Different zoom-ins are performed to check the stresses in certain elements. In Appendix … the results from SCIA Engineer are given. All elements are checked with use of the different load combinations. An important part of the structure is the total load distributed on the foundation. The requirements state that the vertical capacity of each foundation point is limited to 28 MN (28000 kN). The additional requirements stated that only 90% of this vertical capacity should be used. In Figure … the reaction forces in the supports at the foundation are shown. 5263+1575+4391+8642 = 19871 kN < 25200 kN. It turns out that the vertical reaction force is smaller than the required force and so the requirement has been met.

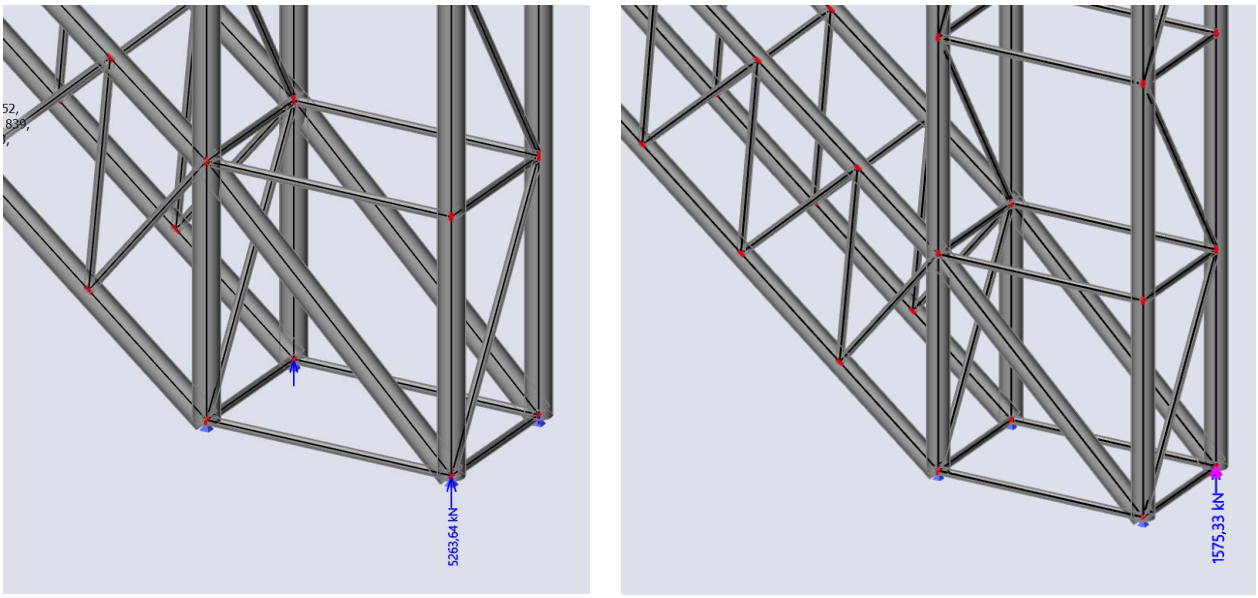


Figure 33: Vertical support reactions

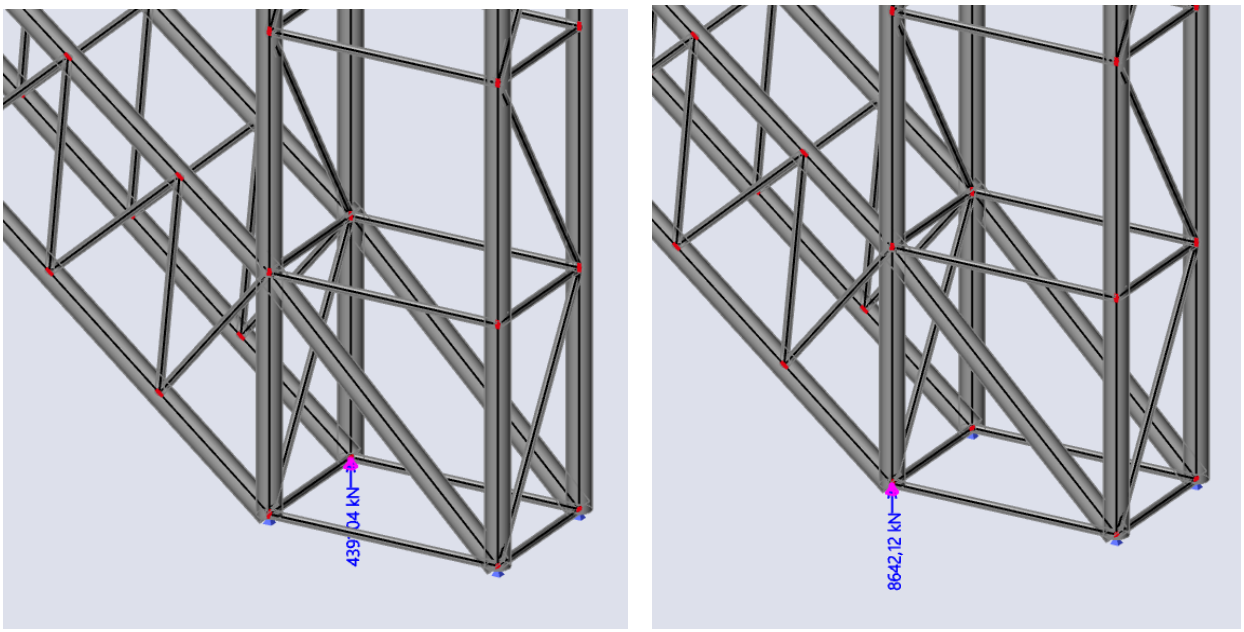


Figure 34: Vertical support reactions

The main reason for the low weight of the whole structure is partly realised by the diagonal bracings. The bracings distribute the loads from the sides directly into the foundation. Therefore no large cantilevering elements are used. When these large cantilevering elements are used, the moment in the trusses would be larger and therefore larger cross sections should be used.

The low vertical reaction forces come with a price. Due to the brace, horizontal forces are introduced in the foundation. The requirements state that the horizontal capacity of each foundation point is limited to 1 MN (1000 kN). The total horizontal reaction force is shown in Figure …. and is 3560 kN. The requirements is not met: 3560kN > 1000 kN.

In Figure … the stresses in the columns are shown. Visible is that most of the stresses are in the blue columns. These are the columns on which the arch is placed directly. The other (green columns) have a lower stress. For the diagonals the same principle counts. The compression on the blue diagonal is the highest. This is one of the diagonals on which the arch is placed directly.

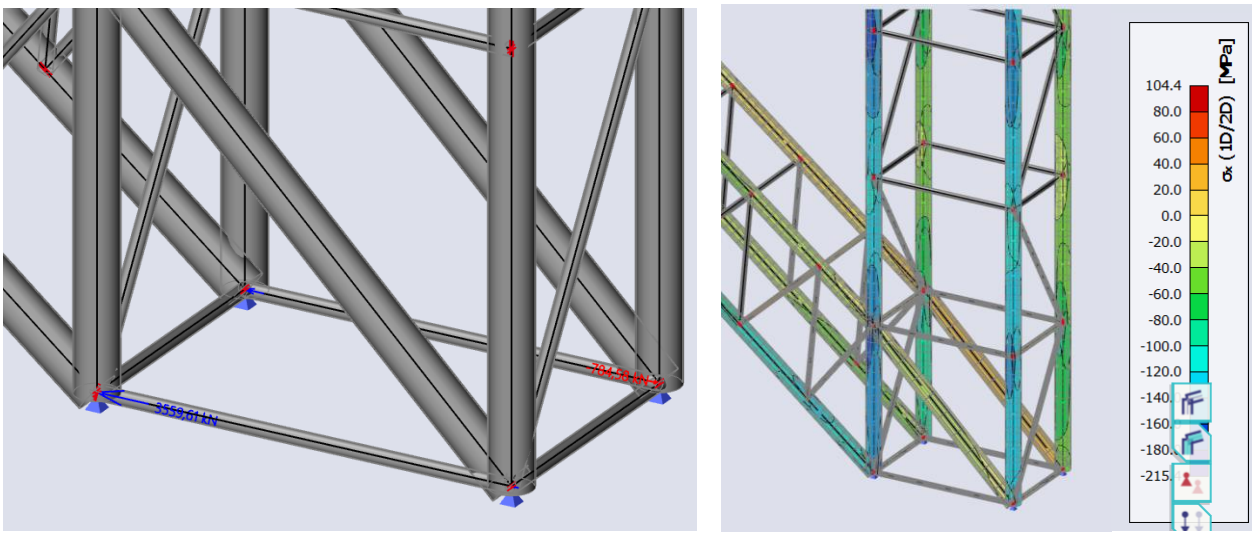


Figure 35: Horizontal support reactions and stresses

7.4 Details

7.4.1 Detail 1 (including calculation)

The first and most important details are the connections of the elements in the arches. These arches are distributing the highest loads and have the largest span. The forces in the elements are mainly axial loads in tension or compression. With the use of end plates on the elements the connections are bolted. For these details the welds and bolts will be calculated. In Figure … and Figure … some details are shown. In Appendix … Figure … the images of the details are presented more clearly.

With the use of the calculations found in Appendix … the thickness of the welds and the amount and dimensions of the bolts could be found. It turns out that the welds should have a thickness of 5 mm and for the bolted connection 6 bolts are used of M20

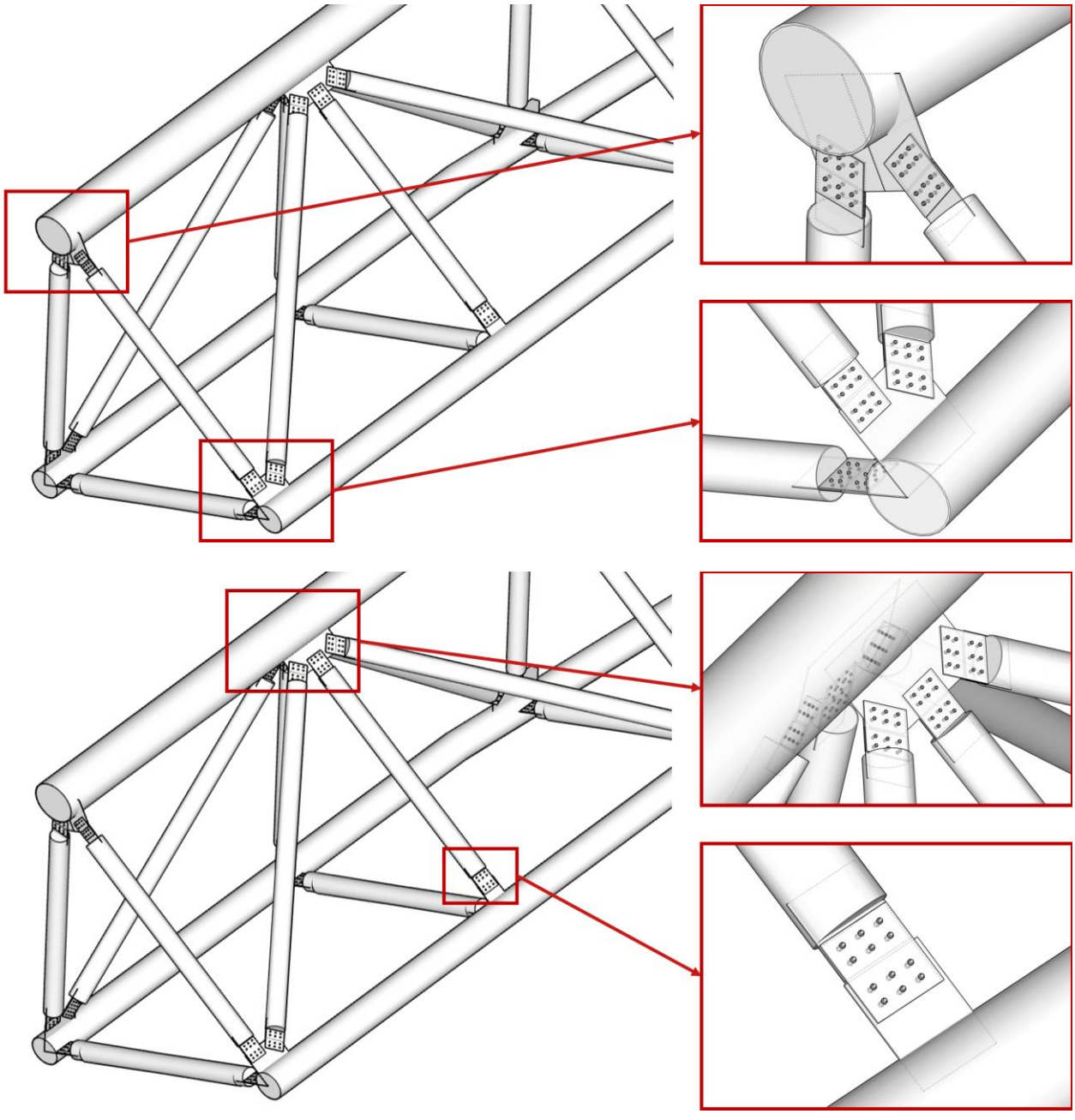


Figure 36: Details of the arch

7.4.2 Detail 2

The second detail is focused on the foundation. Due to the high loads from the structure, the foundation has to resist large forces. The steel circular hollow section is connected to a concrete foundation with a moment fixed connection as shown in Figure …. With the use of large bolts the steel column is connected to a steel end plate and the foundation. An extra ring is used in the concrete for the bolts to distribute their forces to the concrete.

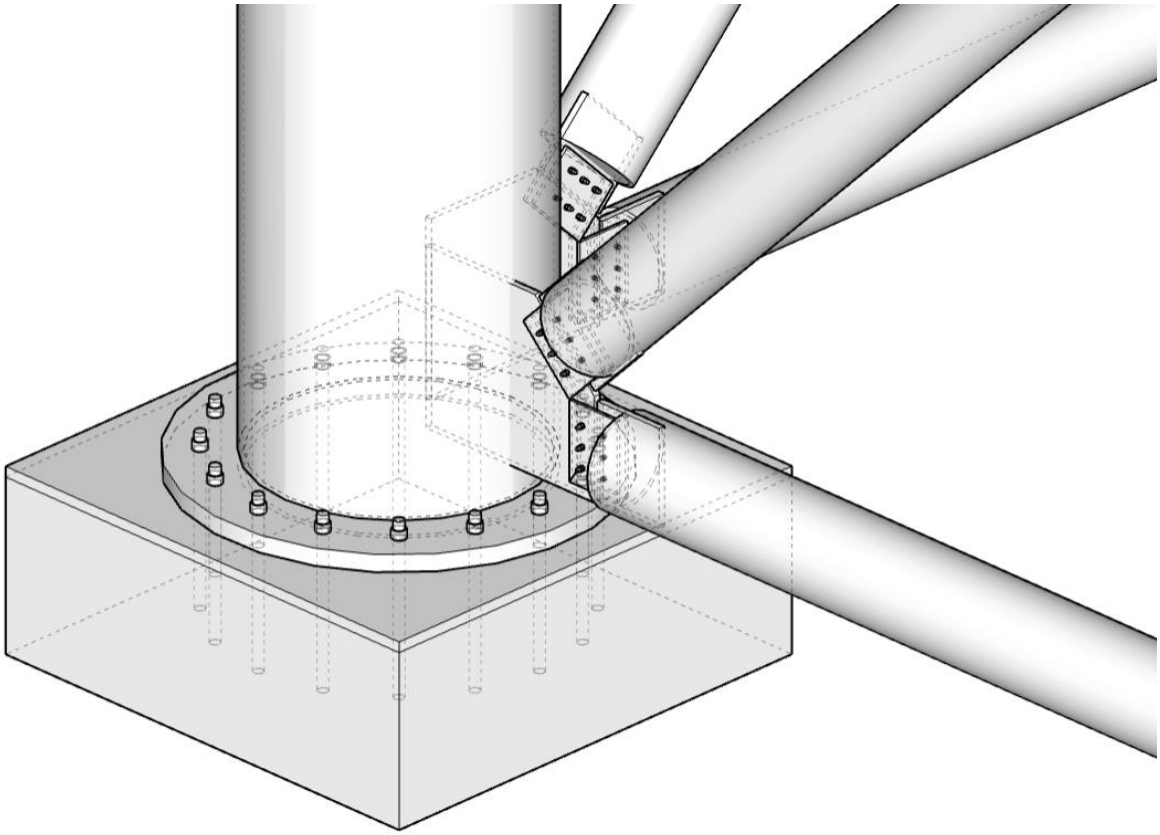


Figure 37: Detail of the foundation

7.4.3 Detail 3

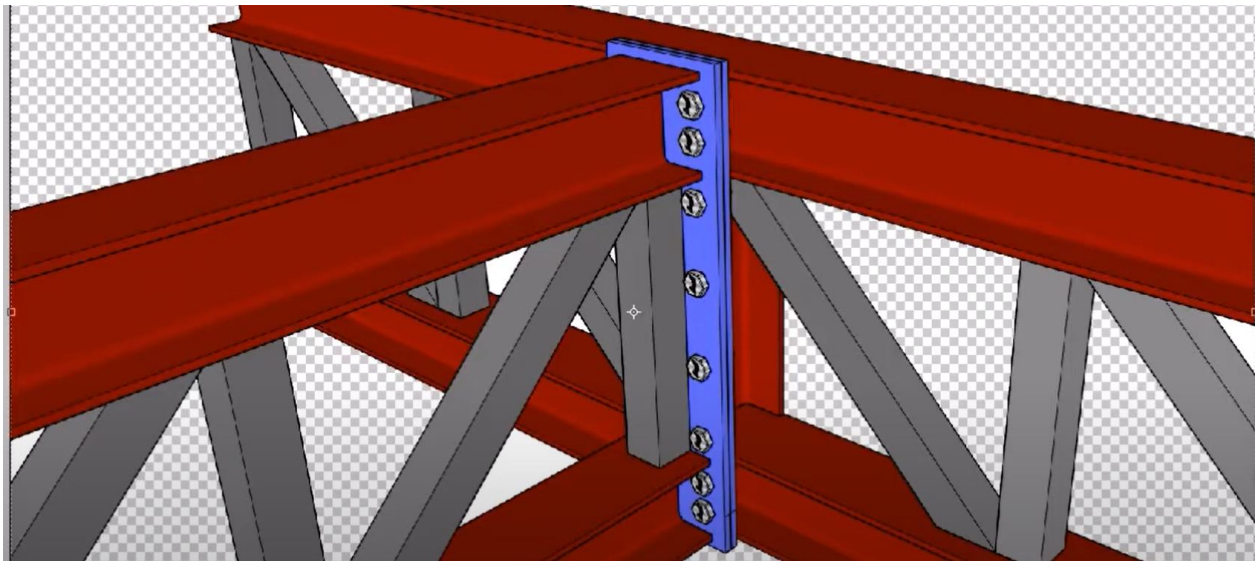


Figure 38: Reference project of truss to truss detail

8 Execution

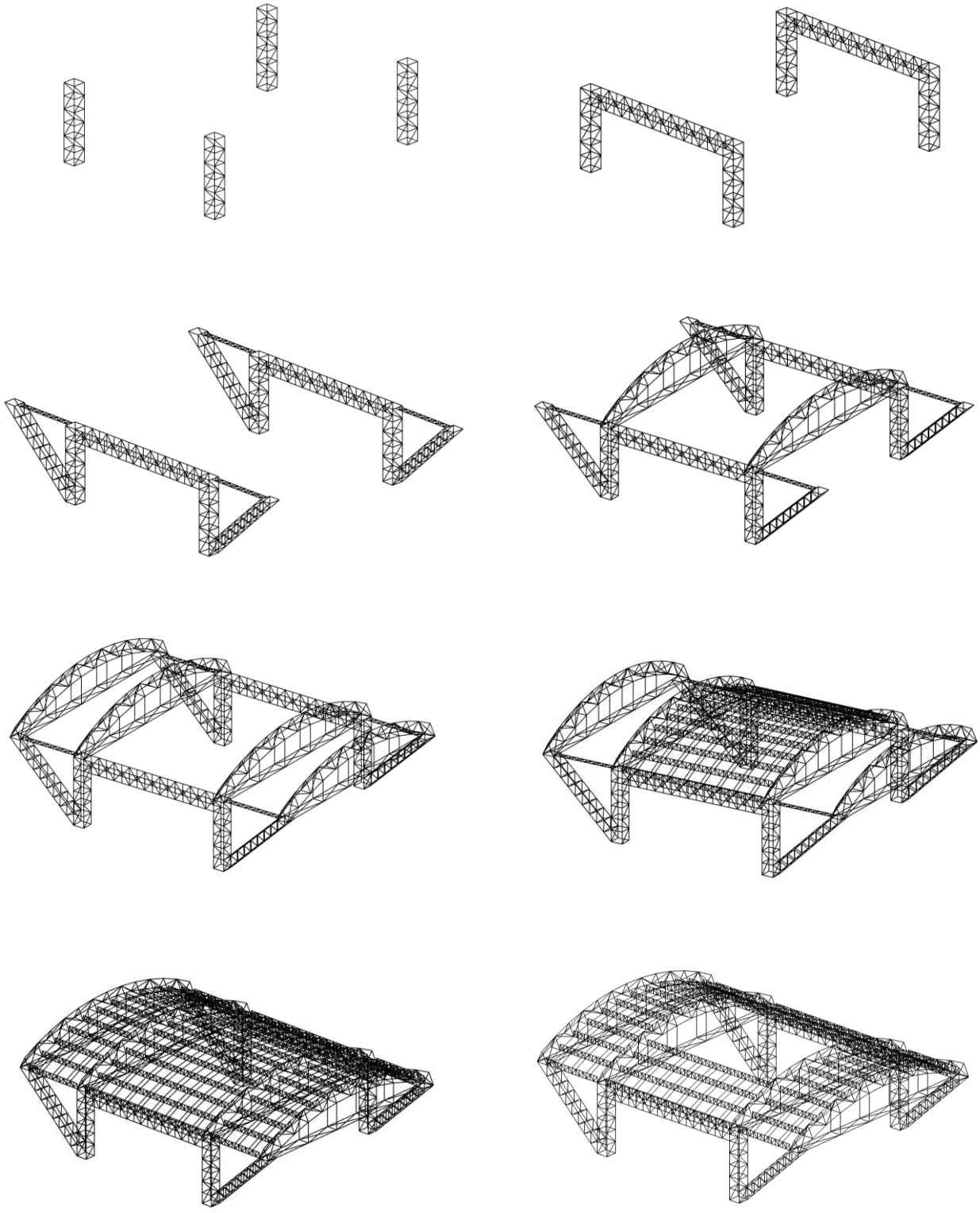


Figure 39: Construction method

9 Conclusions

The final design of the stadium's retractable roof structure successfully integrates all specified requirements while maintaining structural efficiency, aesthetics, and functional adaptability. The total roof spans 210m in length and 160m in width, covering the south, extended west, north, and east tribunes, ensuring complete protection.

The retractable mechanism is designed to enable rapid opening and closing within 10 minutes, accommodating changing weather conditions and event requirements. The selected transparent polycarbonate panels ensure a minimum of 90% light transmission, while maintaining visibility from the exterior, as required.

The roof did not met the requirement of the height of the stadium above surface level. The roof is too high because the cross sectional areas where to be minimized. Additionally, the design guarantees that vertical loads are not transferred to the tribunes.

Structural support is provided exclusively at predetermined foundation points, as indicated in the referenced figures. The column foundations are strategically positioned without modification, with each point maintaining a vertical load capacity of 19871 kN, which is lower than the required 25200 kN (28000x0.9). The horizontal reaction force requirement is not met. The horizontal reaction force is 3560kN which is higher than the required 1000 kN.

The structural framework incorporates optimized arches, braces and tension elements to enhance moment distribution and minimize material usage, thereby ensuring a lightweight yet robust system. Stability trusses and bracing elements are incorporated to manage dynamic loads, including wind, snow, and thermal expansion. All load calculations were performed using a combination of hand calculations and SCIA Engineer simulations, ensuring that the structure meets all safety and performance criteria.

To maintain architectural coherence with the existing stadium, the roof design was reviewed and approved through a design assessment process. The structure does not obstruct the facades between the corners from an external viewpoint, except for the necessary supporting columns, preserving the visual integrity of the stadium.

Practical considerations such as transport and construction timelines were also addressed. The roof components are designed to fit within standard transport dimensions, facilitating efficient delivery without requiring special permits. The construction process is planned to be completed within six months, with minimal disruption to ongoing stadium activities, as monitored through daily disturbance reports.

Additionally, the roof system is engineered for easy disassembly, allowing for future modifications or dismantling if required.

The final design ensures a minimum lifespan of 50 years under normal weather conditions, guaranteeing long-term durability and performance. The lightweight, high-strength materials used in construction further contribute to the structure’s resilience and efficiency.

In conclusion, the stadium's retractable roof meets all structural, functional, and aesthetic requirements while optimizing material usage, load distribution, and construction feasibility. The design achieves a balance between innovation and practicality, ensuring seamless integration with the existing stadium and delivering a high-performance, adaptable roofing solution.

10 Discussion and recommendations

During the process of the project some problems came up. In the early stages of the project, these problems could be solved. However at later phases of the project these problems were accepted. This chapter elaborates some points of discussion and recommendations for future projects.

When problem that occurred during the modelling process in SCIA was the way the trusses were connected to the bottom chords of the arches. The trusses were connected at the bottom chords, which resulted in different deflections and stresses in both of the bottom chord. This would not be a problem if both trusses had exactly the same weight, because then the arch would still be balanced. However, The weights on both bottom chords were different. This created torsion in the arch which would result in different internal stresses which would then change the stresses in the columns as well. An idea to prevent this is to be seen in Figure … below.

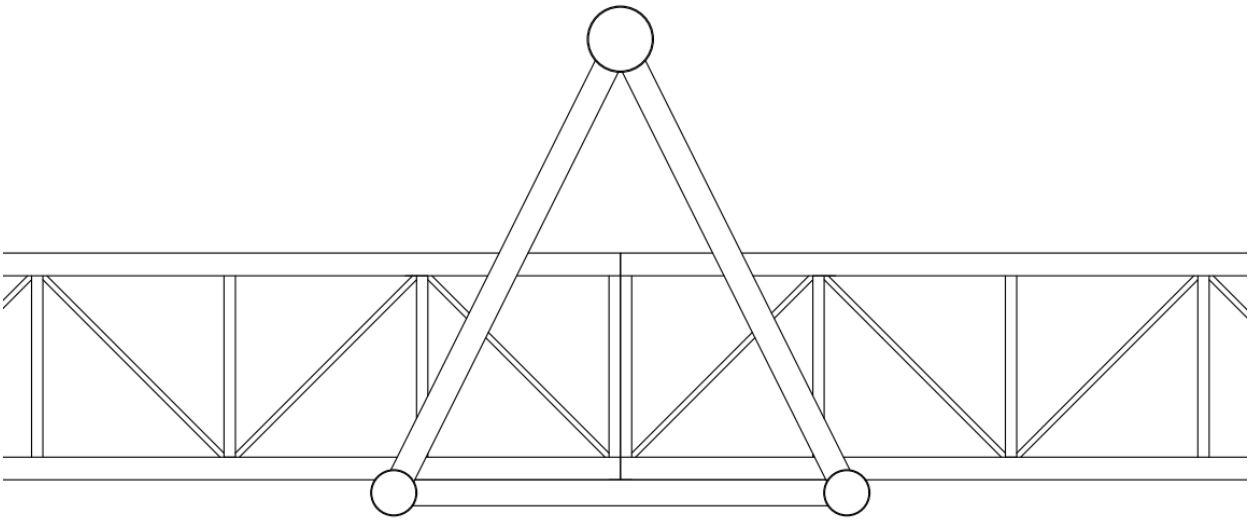


Figure 40: New design of the truss connection to the arch

Figure … shows the cross section of the arch. In the arch, the roof trusses are connected in the middle. When both trusses have a different load, the loads would be distributed in the centre of the arch. This would result in a vertical force in the centre of the arch which is desired. This adaptation was not elaborated during the project.

Another point of discussion would be the design. The added brace included early in the design was a choice which turned out to be efficient in terms of total weight of the structure. One of the requirements was based on the total weight and this requirement was kept. However, the brace did result in higher horizontal forces in the foundation then required.

One last point of discussion would be the overall design. During the project, a lot of design decisions were based on the reference project of the Johan Cruyff Arena. Most of the dimensions for example were compared to the Johan Cruyff Arena to see if the outcomes were realistic. Also, the roof principle was based on the Johan Cruyff Arena as well as the arches. For the next project, some more in depth study could be performed based on more references. This way the design would become more original.

Overall the project was sufficient; most of the requirements were kept and design decisions were based on different aspects. The process had some trial and error, but this resulted in a learning process which helps a lot in the future.

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[7] “Kls,” [Online]. Available: https://nl.wikipedia.org/wiki/Kls. [Accessed 24 September 2024].

Declaration of generative AI and AI-assisted technologies in the writing process

While preparing this work, the author used ChatGPT to improve some sentences in the report. After using ChatGPT, the authors again reviewed and edited the content as needed and take full responsibility for the content of this publication

Appendices

Appendix A. Views of the plot and environment



Figure 41: Views of the stadium with foundation points



Figure 42: Stadium division

Appendix B. Reference projects

B1. Johan Cruyff Arena (Amsterdam, Netherlands)
The Johan Cruyff Arena in Amsterdam is the home stadium of AFC Ajax and one of the best sports venues in the Netherlands. It opened in 1996 as the Amsterdam ArenA and was renamed in 2018 to honour the famous Dutch footballer Johan Cruyff. The arena is designed to host many events and features modern technology and architecture.

One of the standout features of the Johan Cruyff Arena is its retractable roof, which allows it to be used for events all year round. The roof has two large sections that can slide open or closed depending on the weather or the type of event. This means the stadium can host outdoor football matches as well as indoor concerts and other events. The roof is supported by a strong steel truss system that ensures it can handle loads like snow and wind. The sections are made of a lightweight, see-through material that lets in natural light when the roof is closed, creating a bright atmosphere and reducing the need for artificial lighting.

The roof extends out over the seating areas, providing shelter for most spectators from rain and bad weather. Large steel columns around the stadium support the roof, reducing the need for internal columns that could block views. This design allows fans to see the pitch clearly while staying dry. The unique shape of the roof also gives the arena a modern look.

The main structure of the stadium is built with reinforced concrete, which forms the foundation, lower stands, and core walls. This concrete provides the strength needed for the large seating areas and other parts of the stadium. Steel beams and trusses are used in the upper tiers and roof to support their weight, allowing for wide open spaces inside.

The stadium rests on deep pile foundations that can support the heavy loads from the roof and upper stands. These foundations ensure stability, even with large crowds during events. The loads from the roof and upper sections are spread evenly to the foundations using steel columns and concrete walls. Care is taken to manage the dynamic loads of the retractable roof, preventing stress on the structure.

The Johan Cruyff Arena is where AFC Ajax plays its home games, but it also hosts concerts and other events. The stadium's structure is easy to see from the outside and features two large trusses. The retractable roof spans between these trusses and is connected to a sliding mechanism.

A key part of the structure is how the foundation connects to the main trusses. Each truss column splits into two, connecting to two concrete columns that help distribute the weight to nearby foundation points. This design helps keep the structure stable, and the forces acting on it are managed by the frame of the curved truss.



Figure 43: Outside and inside view of the Johan Cruyff Arena



Figure 44: Self taken photos of the outside of the Johan Cruyff Arena

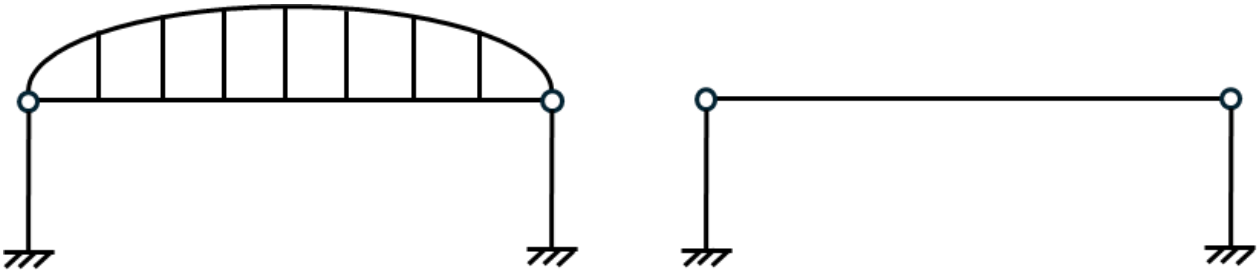


Figure 45: Mechanical scheme of the Johan Cruyff Arena

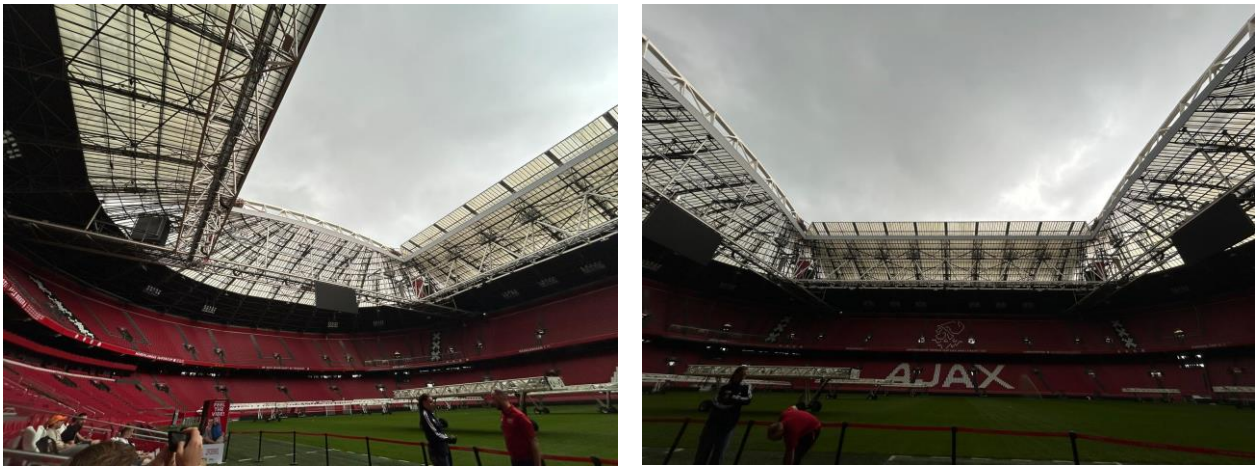


Figure 46: Self taken photos of the inside of the Johan Cruyff Arena

B2. Bernabeu (Madrid, Spain)
Currently, the Bernabeu Stadium, home to Real Madrid, is being redeveloped. Like the Johan Cruyff Arena, this stadium is mainly used for Real Madrid's home matches. One of the most noticeable changes is the addition of a fully retractable roof. This roof allows the stadium to be completely covered, making it usable all year round for various events, including concerts and other activities.

The new roof is a lightweight structure made of steel and a see-through membrane that lets natural light in while keeping out the weather. It is supported by large trusses that stretch across the stadium, allowing the roof to open and close without blocking views or affecting the seating area. The foundations have been strengthened to support the extra weight of the new roof and structures. Deep foundations and load-bearing columns help transfer the roof's weight without impacting the pitch or seating.

The foundation system uses deep piles to keep the overall structure stable, especially with the added weight from the roof and new upper levels. Reinforced concrete columns carefully distribute vertical loads to avoid excessive stress on any part of the foundation.

The Bernabeu features steel mega-trusses, which are crucial for the retractable roof system. These large trusses span the stadium and support the roof when it is closed. Steel is used for these trusses because it is strong yet lightweight. To ensure clear views from the upper seating tiers, they are supported by cantilevered sections, reinforced with steel beams and cables for stability. The stadium also includes advanced systems to manage vibrations from crowds, especially during exciting football matches or big events.

The force distribution is shown in Figure 26. The large main trusses are spanned over the stadium and supporting the roof. Between these two main trusses the secondary trusses are spanned on which the roof are sliding. The rest of the roof is supported by the cantilevering trusses. The roof works as different sliding membrane panels which are folding open.



Figure 47: outside view and retractable roof principle of Bernabeu

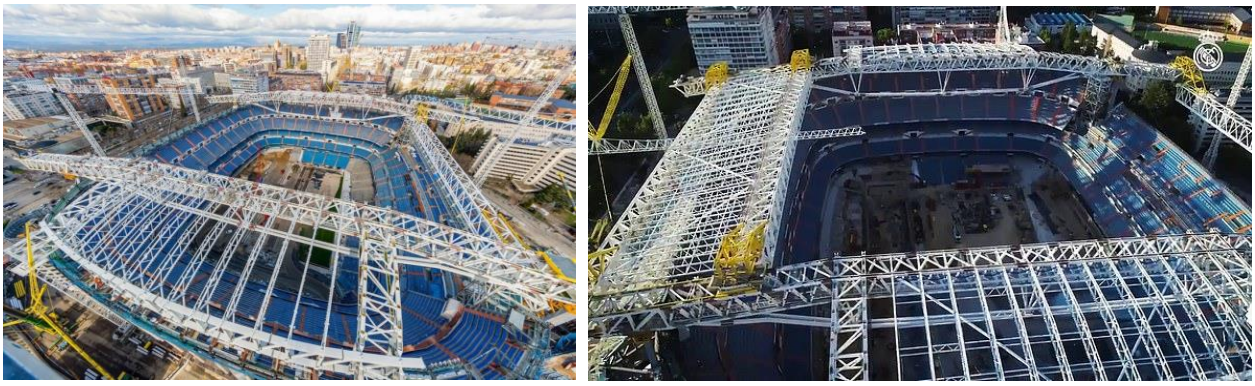


Figure 48: Construction of Bernabeu

B3. Arena Națională (Bucharest, Romania)
The Arena Națională in Bucharest, Romania, is the largest stadium in the country and an important place for football matches and big events. Like many modern stadiums, it has a bowl-shaped design, with seats that wrap around the field. This layout creates an exciting atmosphere for fans and ensures everyone has a good view of the action. The main structure of the stadium is made of reinforced concrete, which provides strong support for the lower and upper seating areas. This type of concrete can hold a lot of weight, making it ideal for large crowds. The symmetrical design of the stadium helps distribute weight evenly and gives it a balanced look from all sides.

The roof is made from a PTFE (Polytetrafluoroethylene) membrane, which is lightweight, strong, and weather-resistant. This membrane allows some natural light to come in, even when the roof is closed. The roof can open and close smoothly using a system of steel cables that support the retractable sections. It covers the entire field and seating area, keeping players and fans safe from the weather. The roof can open or close in about 20 minutes, thanks to motorized systems that move the large sliding sections.

The roof is supported by steel trusses and beams that help spread its weight across the outer edges of the stadium. This design not only protects the seating areas but also helps reduce the weight on the retractable roof. To support the size and weight of the structure, the Arena Națională is built on deep pile foundations made of reinforced concrete. These foundations keep the stadium stable, even with the movements from crowds, wind, and the retractable roof.

The weight of the roof and seating areas is carefully distributed through concrete pillars and beams, which helps avoid putting too much stress on any one part of the structure. The stadium's design takes into account the wind conditions in Bucharest, and the roof and outer walls are built to handle strong winds. The roof's membrane and truss system are specifically designed to reduce wind uplift.



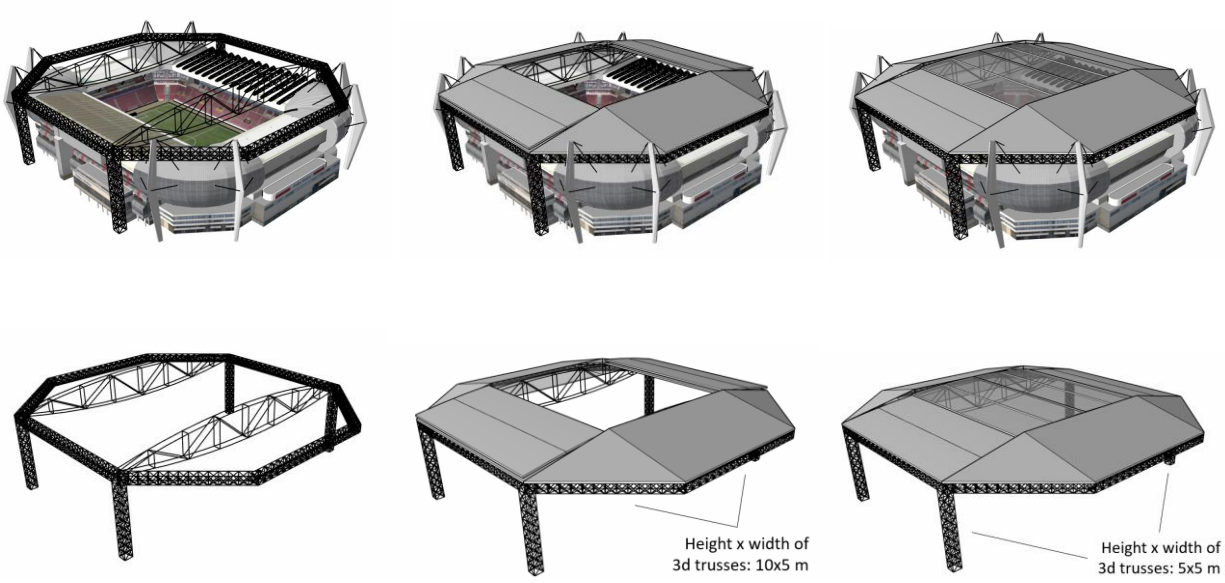
Figure 49: Outside view of Arena Națională

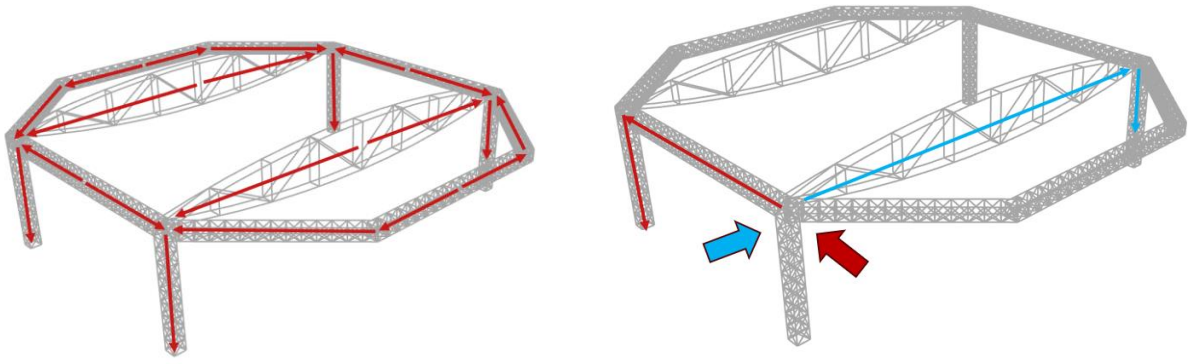


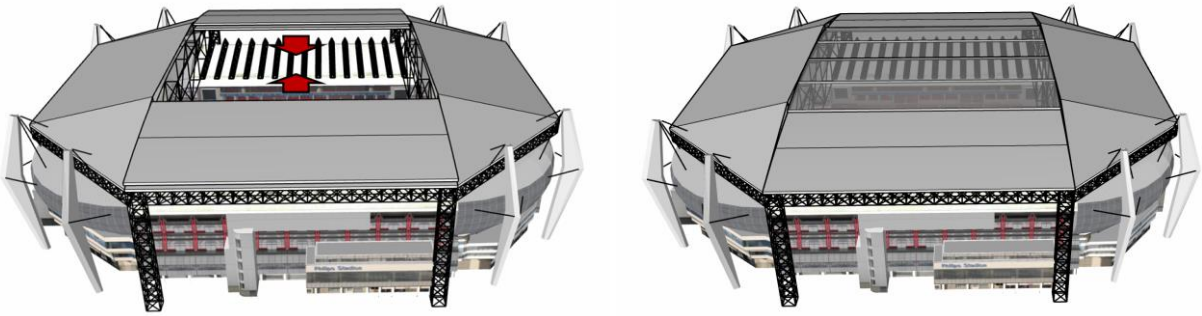
Figure 50: Inside view of Arena Națională

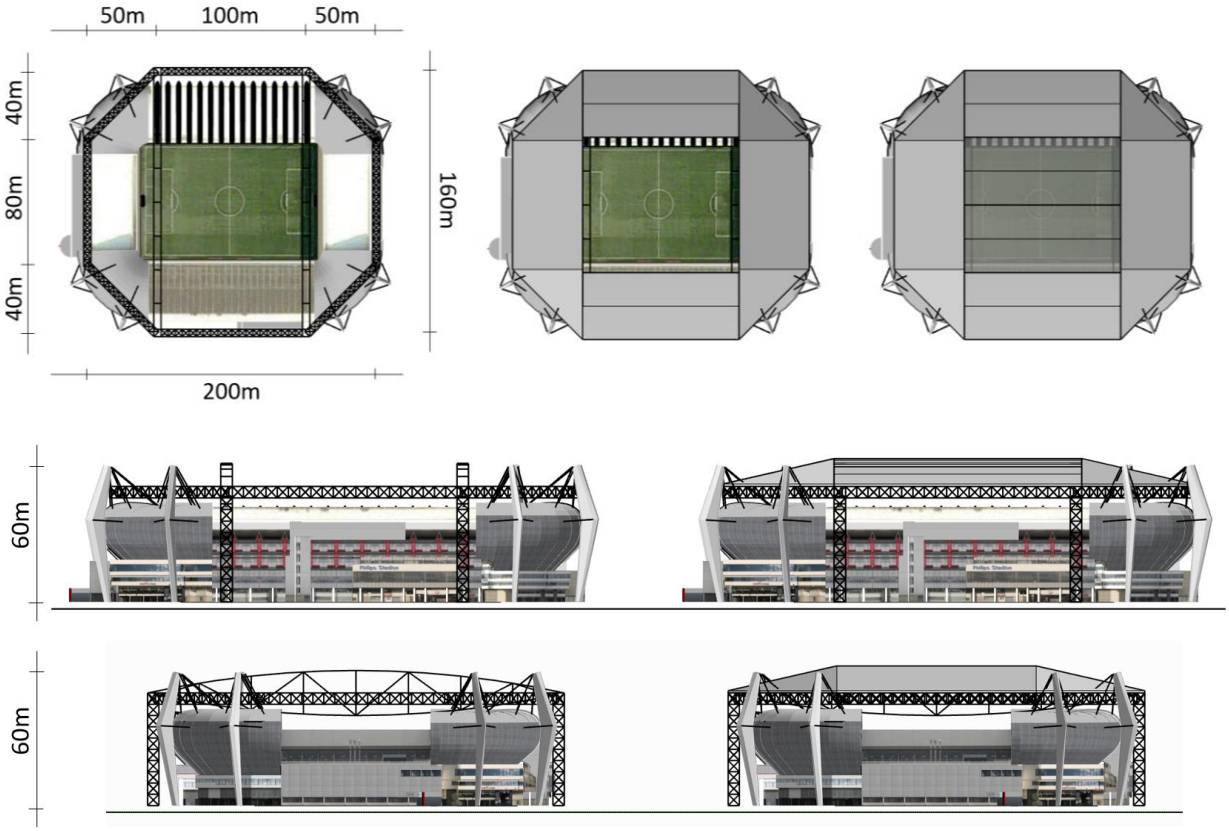
Appendix C Drawings of the variants

C1. Variant 1

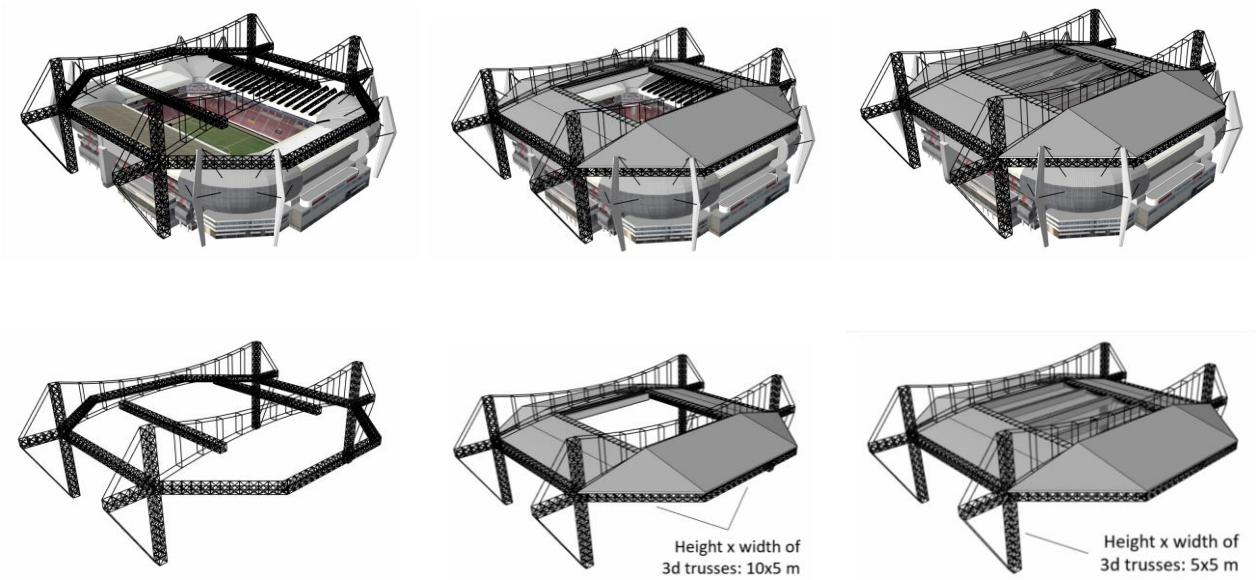


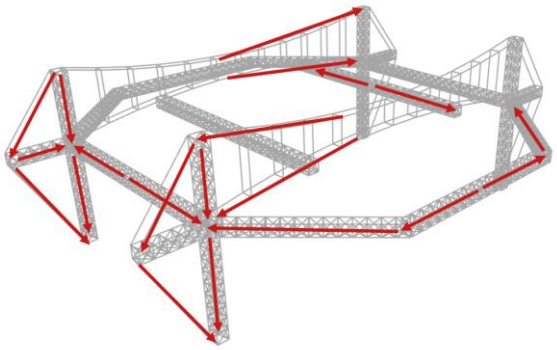


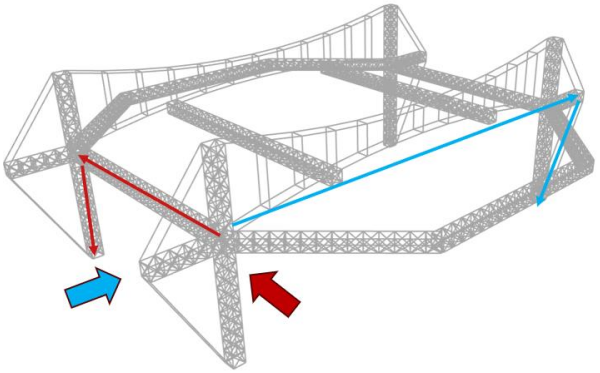


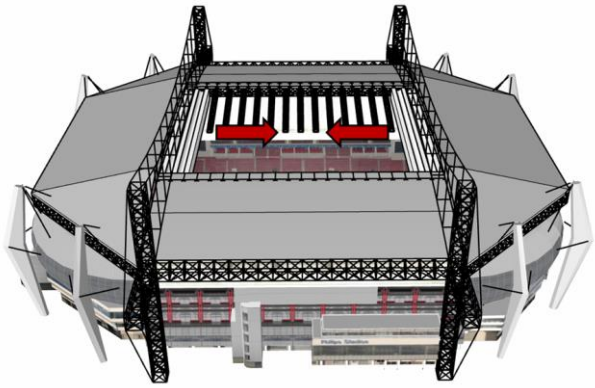

50m
100m
50m
40m
80m
40m
160m
200m
60m
60m

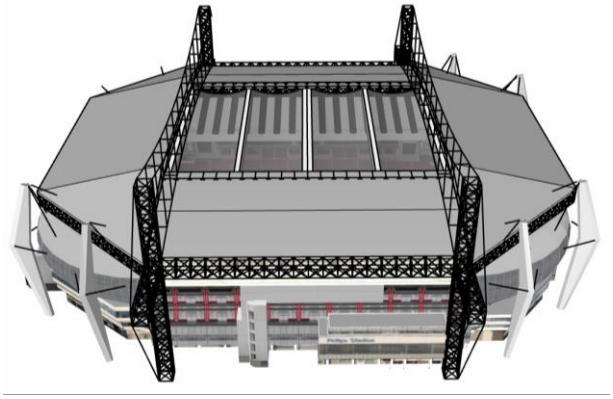
C2. Variant 2

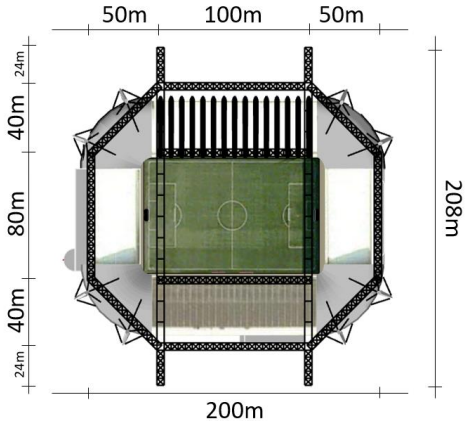

Height x width of
3d trusses: 10x5 m
Height x width of
3d trusses: 5x5 m

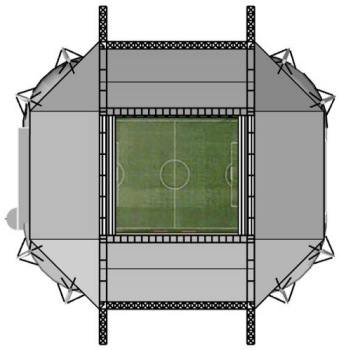


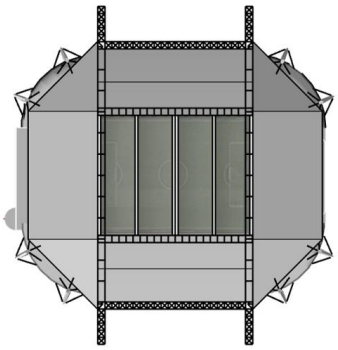


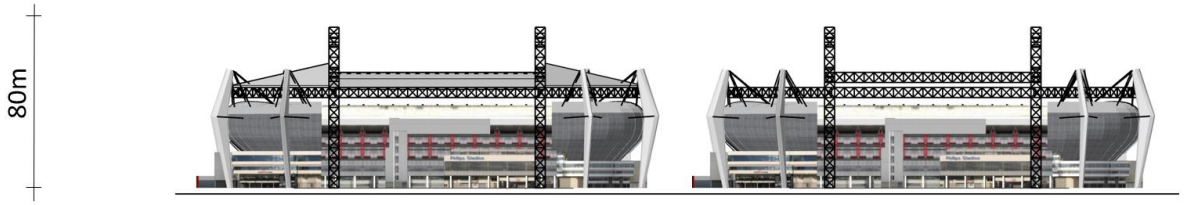


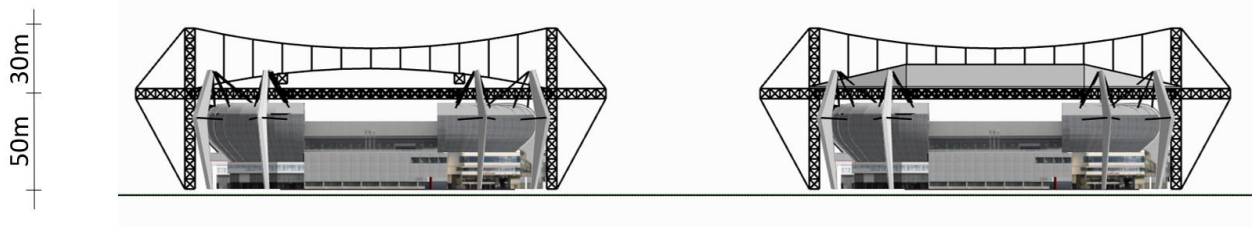



50m
100m
50m
24m
40m
80m
40m
24m
208m
200m

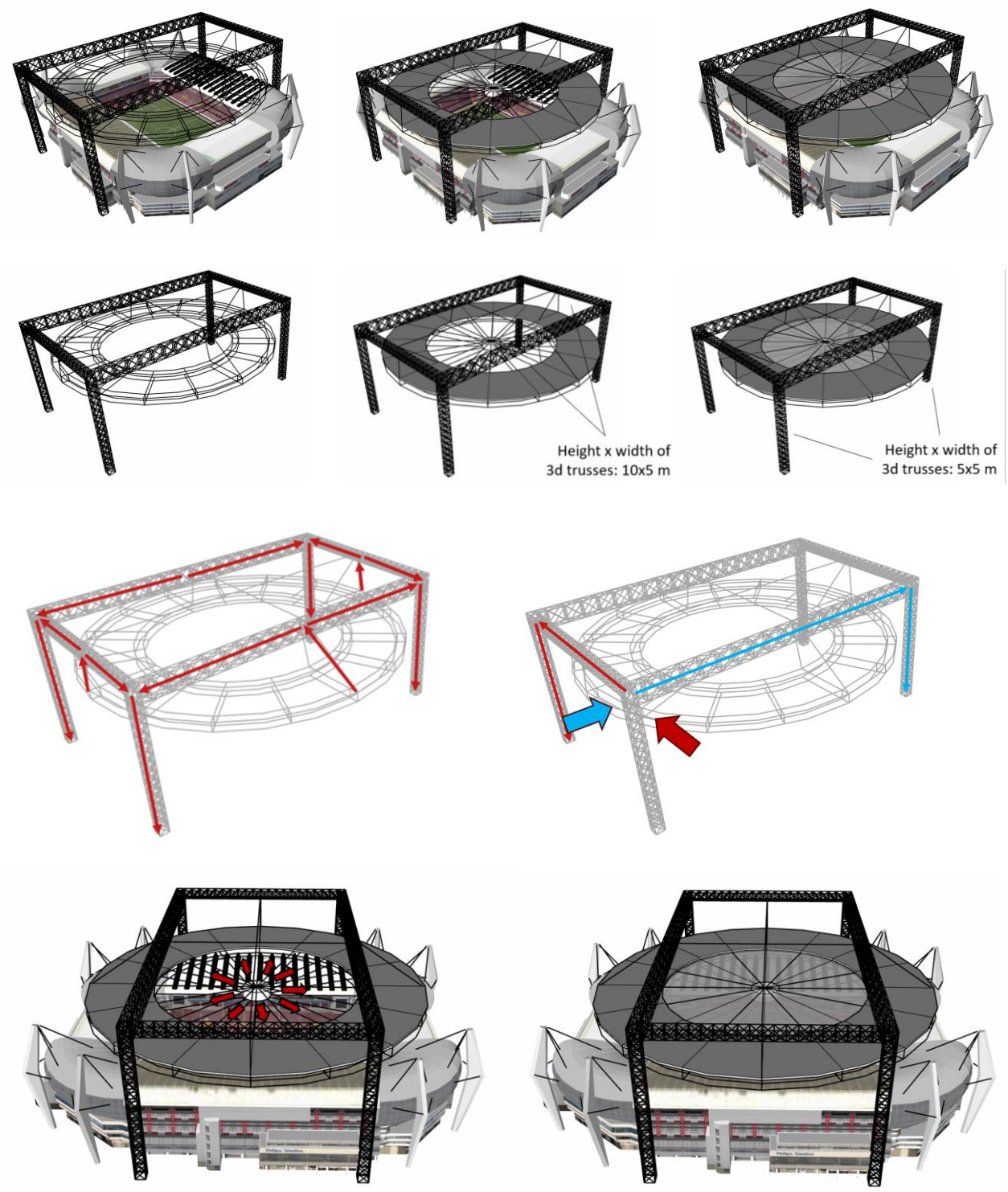


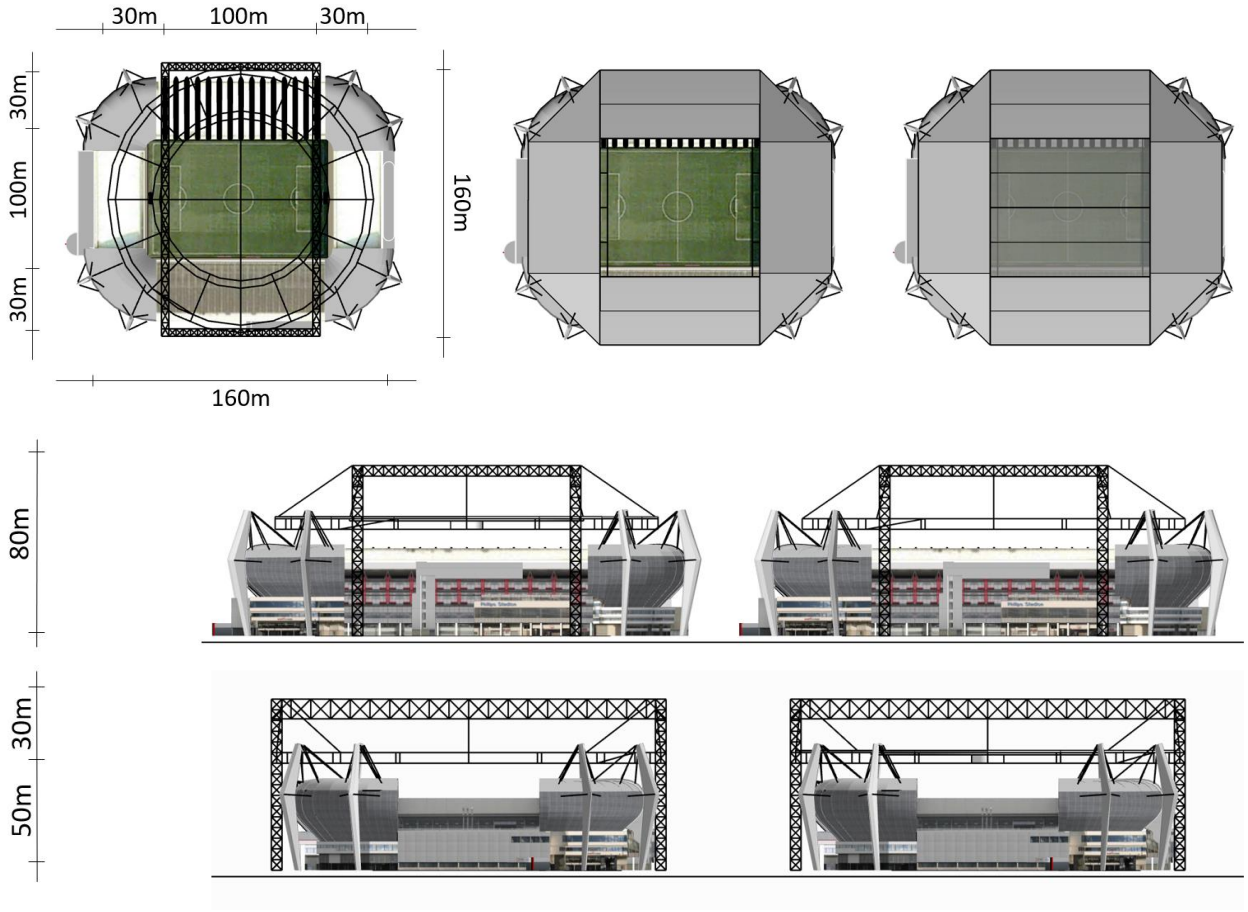



80m


30m
50m

C3. Variant 3


Height x width of
3d trusses: 10x5 m
Height x width of
3d trusses: 5x5 m



C4. Modification variant 1



Appendix D Loads and load combinations

D1. Snow load

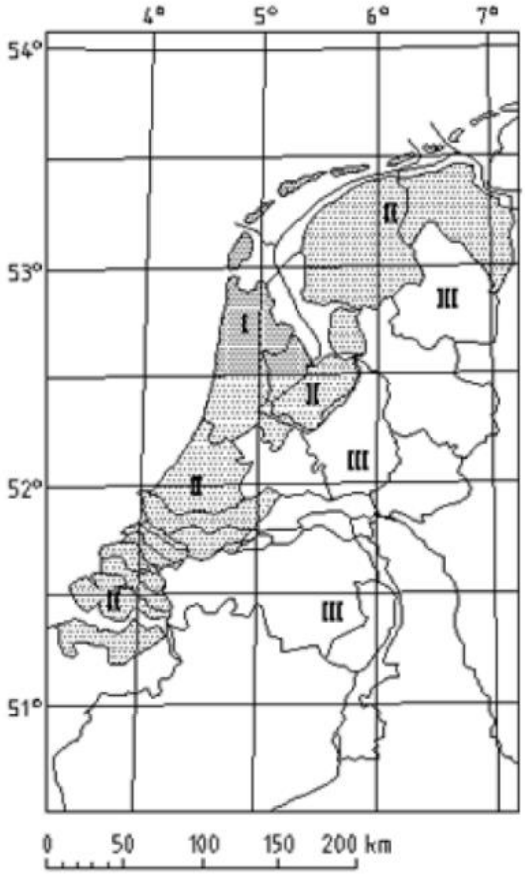
Table 4: Snow load

Hoek van het dak α	$0° \leq \alpha \leq 30°$	$30° < \alpha < 60°$	$\alpha \geq 60°$
$\mu_1(\alpha)$	$\mu_1(0°) \geq 0{,}8$		0,0
$\mu_2(\alpha)$	0,8		0,0
$\mu_3(\alpha)$	0,8 + 0,8 α/30	1,6	--

$s = \mu_i C_e C_t s$

- $\mu_i = 0.8$
- $C_t = 1.0$
- $C_e = 1.0$
- $s = 0.7\ kN/m^2$
- $s = 0.8 \cdot 1.0 \cdot 1.0 \cdot 0.7 = 0.56\ kN/m^2$

D2. Wind load



Hoogte m	Gebied I			Gebied II			Gebied III	
	kust	onbebouwd	bebouwd	kust	onbebouwd	bebouwd	onbebouwd	bebouwd
1	0,93	0,71	0,69	0,78	0,60	0,58	0,49	0,48
2	1,11	0,71	0,69	0,93	0,60	0,58	0,49	0,48
3	1,22	0,71	0,69	1,02	0,60	0,58	0,49	0,48
4	1,30	0,71	0,69	1,09	0,60	0,58	0,49	0,48
5	1,37	0,78	0,69	1,14	0,66	0,58	0,54	0,48
6	1,42	0,84	0,69	1,19	0,71	0,58	0,58	0,48
7	1,47	0,89	0,69	1,23	0,75	0,58	0,62	0,48
8	1,51	0,94	0,73	1,26	0,79	0,62	0,65	0,51
9	1,55	0,98	0,77	1,29	0,82	0,65	0,68	0,53
10	1,58	1,02	0,81	1,32	0,85	0,68	0,70	0,56
15	1,71	1,16	0,96	1,43	0,98	0,80	0,80	0,66
20	1,80	1,27	1,07	1,51	1,07	0,90	0,88	0,74
25	1,88	1,36	1,16	1,57	1,14	0,97	0,94	0,80
30	1,94	1,43	1,23	1,63	1,20	1,03	0,99	0,85
35	2,00	1,50	1,30	1,67	1,25	1,09	1,03	0,89
40	2,04	1,55	1,35	1,71	1,30	1,13	1,07	0,93
45	2,09	1,60	1,40	1,75	1,34	1,17	1,11	0,97
50	2,12	1,65	1,45	1,78	1,38	1,21	1,14	1,00
55	2,16	1,69	1,49	1,81	1,42	1,25	1,17	1,03
60	2,19	1,73	1,53	1,83	1,45	1,28	1,19	1,05
65	2,22	1,76	1,57	1,86	1,48	1,31	1,22	1,08
70	2,25	1,80	1,60	1,88	1,50	1,34	1,24	1,10
75	2,27	1,83	1,63	1,90	1,53	1,37	1,26	1,13
80	2,30	1,86	1,66	1,92	1,55	1,39	1,28	1,15
85	2,32	1,88	1,69	1,94	1,58	1,42	1,30	1,17
90	2,34	1,91	1,72	1,96	1,60	1,44	1,32	1,18
95	2,36	1,93	1,74	1,98	1,62	1,46	1,33	1,20
100	2,38	1,96	1,77	1,99	1,64	1,48	1,35	1,22
110	2,42	2,00	1,81	2,03	1,68	1,52	1,38	1,25
120	2,45	2,04	1,85	2,05	1,71	1,55	1,41	1,28

Figure 51: Wind areas and Extreme thrust in kN/m2 as a function of height

$$F_w = C_s C_d \cdot C_f \cdot q_p(Z_e) \cdot A_{ref}$$

- $C_s C_d = 1$
- C_f
- $q_p = 0.97$
- $C_{f,F} = -1.8$
- $C_{f,G} = -1.2$
- $C_{f,H} = -0.7$

- $q_{w,F} = 1.0 \cdot -1.8 \cdot 0.97 = -1.75 kN/m^2$
- $q_{w,G} = 1.0 \cdot -1.2 \cdot 0.97 = -1.16 kN/m^2$
- $q_{w,H} = 1.0 \cdot -1.8 \cdot 0.97 = -0.68 kN/m^2$

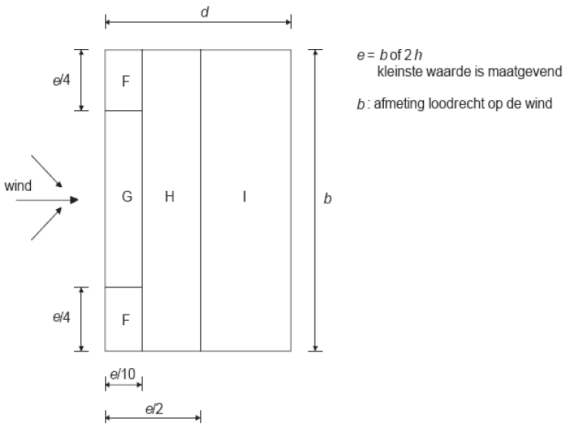


Table 5: External pressure coefficients for flat roofs

Daktype	Zone							
	F		G		H		I	
	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$
Scherpe dakranden	-1,8	-2,5	-1,2	-2,0	-0,7	-1,2	+0,2	
							-0,2	-0,5

D3. Load combinations

Table 6: Calculation values of loads (EQU) (group A)

Blijvende en tijdelijke ontwerpsituaties	Blijvende belastingen		Overheersende veranderlijke belasting	Veranderlijke belastingen gelijktijdig met de overheersende	
	Ongunstig	Gunstig		Belangrijkste (indien aanwezig)	Andere
(Vgl. 6.10)	1,1 $G_{k,j,sup}$	0,9 $G_{k,j,inf}$	1,5 $Q_{k,1}$		1,5 $\Psi_{0,i} Q_{k,i}$ (i >1)

Table 7: Calculation values of loads (STR/GEO) (group C)

Blijvende en tijdelijke ontwerpsituaties	Blijvende belastingen		Overheersende veranderlijke belasting	Veranderlijke belastingen gelijktijdig met de overheersende	
	Ongunstig	Gunstig		Belangrijkste (indien aanwezig)	Andere
(Vgl. 6.10)	1,0 $G_{k,j,sup}$	1,0 $G_{k,j,inf}$	1,3 $Q_{k,1}$		1,3 $\Psi_{0,i} Q_{k,i}$ (i >1)

Table 8: Calculation values of loads (STR/GEO) (group B)

CC	Blijvende en tijdelijke ontwerpsituatie	Blijvende belastingen		Overheersende veranderlijke belasting	Veranderlijke belastingen gelijktijdig met de overheersende	
		Ongunstig	Gunstig		Belangrijkste (indien aanwezig)	Andere
3	(Vgl. 6.10a)	1,5 $G_{k,j,sup}$ [a]	0,9 $G_{k,j,inf}$		1,65 $\psi_{0,1}Q_{k,1}$	1,65 $\psi_{0,i}Q_{k,i}$ (i >1)
	(Vgl. 6.10b)	1,3 $G_{k,j,sup}$ [b]	0,9 $G_{k,j,inf}$	1,65 $Q_{k,1}$		1,65 $\psi_{0,i}Q_{k,i}$ (i >1)

a	Bij vloeistofdrukken met een fysiek beperkte waarde mag zijn volstaan met 1,1 $G_{k,j,sup}$ bij CC1 en met 1,35 $G_{k,j,sup}$ bij CC3.
b	Deze waarde is berekend met ξ = 0,89.

$$\text{Max}\begin{cases}\sum\gamma_{G,j}G_{k,j}+\gamma_p P+\gamma_{q,1}\psi_{0,1}Q_{k,1}+\sum\gamma_{Q,i}\psi_{0,i}Q_{k,i} & \text{(6.10a)}\\ \sum\zeta_j\gamma_{G,j}G_{k,j}+\gamma_p P+\gamma_{Q,1}Q_{k,1}+\sum\gamma_{Q,i}\psi_{0,i}Q_{k,i} & \text{(6.10b)}\end{cases}$$

6.10a

- $\sum 1.5G_{k,j}+\gamma_p P+1.65\psi_{0,1}Q_{k,1}+\sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j}+\gamma_p P+1.65\psi_{0,1}Q_{k,1}+\sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)

6.10b

- $\sum 1.3G_{k,j}+\gamma_p P+1.65Q_{k,1}+\sum 1.65\psi_{0,i}Q_{k,i}$ (unfavourable)
- $\sum 0.9G_{k,j}+\gamma_p P+1.65Q_{k,1}+\sum 1.65\psi_{0,i}Q_{k,i}$ (favourable)

Table 9: Calculation values of loads for load combinations

Combinatie	Blijvende belastingen G_d		Veranderlijke belastingen Q_d	
	Ongunstig	Gunstig	Overheersende	Andere
Karakteristiek	$G_{kj,sup}$	$G_{kj,inf}$	$Q_{k,1}$	$\psi_{0,i}Q_{k,i}$
Frequent	$G_{kj,sup}$	$G_{kj,inf}$	$\psi_{1,1}Q_{k,1}$	$\psi_{2,i}Q_{k,i}$
Quasi-blijvend	$G_{kj,sup}$	$G_{kj,inf}$	$\psi_{2,1}Q_{k,1}$	$\psi_{2,i}Q_{k,i}$

$$\sum G_{k,j} + \gamma_p P + Q_{k,1} + \sum \psi_{0,i} Q_{k,i} \quad \text{(Characteristic)}$$
$$\sum G_{k,j} + \gamma_p P + \psi_{1,1} Q_{k,1} + \sum \psi_{2,i} Q_{k,i} \quad \text{(Frequent)}$$
$$\sum G_{k,j} + \gamma_p P + \psi_{2,1} Q_{k,1} + \sum \psi_{2,i} Q_{k,i} \quad \text{(Quasi-permanent)}$$

Table 10: ψ-factors for buildings

Belasting	Ψ_0	Ψ_1	Ψ_2
Voorgeschreven belastingen in gebouwen, categorie			
Categorie A: woon- en verblijfsruimtes	0,4	0,5	0,3
Categorie B: kantoorruimtes	0,5	0,5	0,3
Categorie C: bijeenkomstruimtes	0,6/0,4 [a]	0,7	0,6
Categorie D: winkelruimtes	0,4	0,7	0,6
Categorie E: opslagruimtes	1,0	0,9	0,8
Categorie F: verkeersruimte, voertuiggewicht ≤ 25 kN	0,7	0,7	0,6
Categorie G: verkeersruimte [b], 25 kN < voertuiggewicht ≤ 160 kN	0,7	0,5	0,3
Categorie H: daken	0	0	0
Industrieel gebruik waarbij de veranderlijke belasting:			
— niet langdurig aanwezig is	0,5	0,5	0,3
— langdurig aanwezig is	1,0	0,9	0,8
Sneeuwbelasting	0	0,2	0
Belasting door regenwater	0	0	0
Windbelasting	0	0,2	0
Temperatuur (geen brand)	0	0,5	0

[a] De waarde 0,6 geldt voor delen van het gebouw die in geval van een calamiteit zwaar kunnen worden belast door een mensenmenigte (vluchtroutes, trappen enz.); de waarde 0,4 geldt in overige gevallen.

[b] Met verkeersruimte wordt in dit geval een ruimte bedoeld waar voertuigen kunnen rijden, bijvoorbeeld parkeergarages

Appendix E Structural calculations

E1. Calculations main truss

Steel truss calculation

Steel grade	s235	Unit
Yield strength	235	N/mm²
Tensile strength	360	N/mm²
Weight	7850	kg/m³
Modulus of Elasticity	210000	N/mm²

Cross section	Top chord	Bottom chord	Diagonals/verticals	Unit
Diameter outer circle	850	790	620	mm
Thickness	65	55	40	mm
Diameter inner circle	720	680	540	mm
A	160299,765	126998,883	72884,94956	mm²
W_y	29252349,8	21832956,71	9933513,287	mm³
I_y	1,2432E+10	8624017900	3079389119	mm⁴
Weight	1258,35316	996,9412317	572,1468541	kg/m

Dead load self-weight	Amount	Weight per unit	Total weight	Load over lenght of trus	Unit
Top chord	1			12,34444446	kN/m
Bottom chord	1			9,779993483	kN/m
Verticals	11	89,80417022	987,8458724	6,174036702	kN/m
Diagonals	10	127,0022754	1270,022754	7,937642215	kN/m

Dimensions truss	Amount	Unit
Heigth (h)	16	m
Length	160	m

Dead load roof Johan Cruyff Arena	Amount	Unit	Amount	Unit
Dimensions (Wikipedia)	2x37x120	m	8880	m²
Weight (Wikipedia)	520	ton	5101,2	kN
Dead load per square meter			0,574459459	kN/m²
Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN
Dead load roof per truss per meter			31,59527027	kN/m
			Unit	**Amount**
Total dead load			67,83138714	kN/m
Total variable load (snow)	0,56	kN/m²	30,8	kN/m

Ultimate Limit State (ULS)	Amount	Unit
Safety factor (G)	1,3	
Safety factor (Q)	1,65	
Moment	444802,5705	kNm
Axial force bottom/top chord	27800,16066	kN
Top chord		
A	236597,112	mm²
Bottom chord		
A	118298,556	mm²
Verticals/diagonals		
N	15726,14569	kN
A	66919,76889	mm²

Servicibility Limit State (SLS)	Amount	Unit
Safety factor	1	
Moment	315620,4388	kNm
Moment of Inertia		
Top chord	1,02716E+13	mm⁴
Bottom chord	8,13655E+12	mm⁴
Verticals/diagonals		mm⁴
Total	1,84082E+13	mm⁴
Maximum deflection	640	mm
Deflection	217,7228178	mm

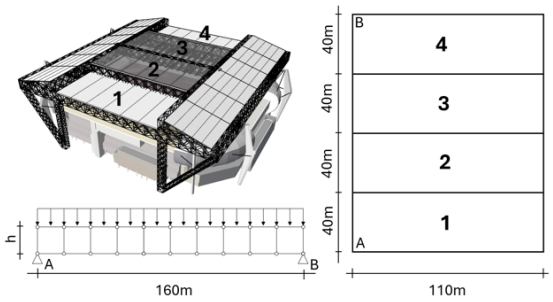


Figure 52: First structural calculations main truss

Steel truss calculation

Steel grade	s235	Unit
Yield strength	355	N/mm²
Tensile strength	360	N/mm²
Weight	7850	kg/m³
Modulus of Elasticity	210000	N/mm²

Cross section	Top chord	Bottom chord	Diagonals/verticals	Unit
Diameter outer circle	780	612	600	mm
Thickness	50	36	30	mm
Diameter inner circle	680	540	540	mm
A	114668,132	65144,06526	53721,23438	mm²
W_y	19677345,4	8863424,88	7292657,567	mm³
I_y	7674164724	2712208013	2187797270	mm⁴
Weight	900,144835	511,3809123	421,7116899	kg/m

Dead load self-weight	Amount	Weight per unit	Total weight	Load over lenght of trus	Unit
Top chord	1			8,830420832	kN/m
Bottom chord	1			5,01664675	kN/m
Verticals	11	66,19186684	728,1105352	4,550690845	kN/m
Diagonals	10	93,60943578	936,0943578	5,850589736	kN/m

Dimensions truss	Amount	Unit
Heigth (h)	16	m
Length	160	m

Dead load roof Johan Cruyff Arena	Amount	Unit	Amount	Unit
Dimensions (Wikipedia)	2x37x120	m	8880	m²
Weight (Wikipedia)	520	ton	5101,2	kN
Dead load per square meter			0,574459459	kN/m²
Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN
Dead load roof per truss per meter			31,59527027	kN/m
			Unit	**Amount**
Total dead load			55,84361843	kN/m
Total variable load (snow)	0,56	kN/m²	15,4	kN/m

Ultimate Limit State (ULS)	Amount	Unit
Safety factor (G)	1,5	
Safety factor (Q)	1	
Moment	317329,3685	kNm
Axial force bottom/top chord	19833,08553	kN
Top chord		
A	111735,6931	mm²
Bottom chord		
A	55867,84656	mm²
Verticals/diagonals		
N	11219,28741	kN
A	47741,64857	mm²

Servicibility Limit State (SLS)	Amount	Unit
Safety factor	1	
Moment	227979,579	kNm
Moment of Inertia		
Top chord	7,34643E+12	mm⁴
Bottom chord	4,17193E+12	mm⁴
Verticals/diagonals		mm⁴
Total	1,15184E+13	mm⁴
Maximum deflection	640	mm
Deflection	251,3358694	mm

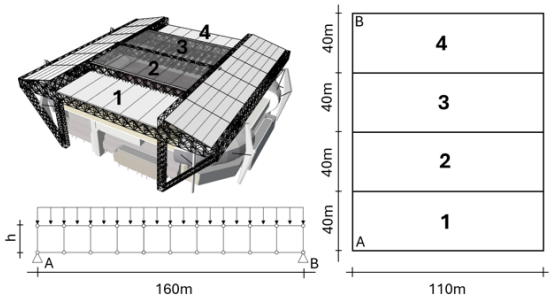


Figure 53: Structural calculation of main truss after modifications

E2. Calculations column truss

Steel column truss calculation											
	Steel grade	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**
	Yield strength	355	N/mm^2					Safety factor (G)	1,5	
	Tensile strength	360	N/mm^2					Safety factor (Q)	1	
	Weight	7850	kg/m^3							
	Modulus of Elasticity	210000	N/mm^2					**Dimensions truss**	**Amount**	**Unit**
								Heigth	45	m
	Cross section	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			Width	5	m
	Diameter outer circle	780	612	600	mm					
	Thickness	50	36	30	mm				**Amount**	**Unit**
	Diameter inner circle	680	540	540	mm			**Reaction force per chord**	3966,617106	kN
	A	114668,1319	65144,06526	53721,23438	mm^2					
	W_y	19677345,45	8863424,88	7292657,567	mm^3			**Area of column**		
	I_y	7674164724	2712208013	2187797270	mm^4			**Cross section**	**Top chord**	
	Weight	900,1448351	511,3809123	421,7116899	kg/m			Diameter outer circle	400	
								Thickness	20	
	Dead load self-weight	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**		Diameter inner circle	360	
	Top chord	1			8,830420832	kN/m		A	23876,10417	
	Bottom chord	1			5,01664675	kN/m				
	Verticals	11	66,19186684	728,1105352	4,550690845	kN/m			**Amount**	**Unit**
	Diagonals	10	93,60943578	936,0943578	5,850589736	kN/m		**Stress in column**	166,1333473	N/mm^2
								Compressive strenght	177,5	N/mm^2
	Dimensions truss	**Amount**	**Unit**							
	Heigth (h)	16	m							
	Length	160	m							
	Dead load roof Johan Cruyff Arena	**Amount**	**Unit**	**Amount**	**Unit**					
	Dimensions (Wikipedia)	2x37x120	m	8880	m^2					
	Weight (Wikipedia)	520	ton	5101,2	kN					
	Dead load per square meter			0,574459459	kN/m^2					
	Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN					
	Dead load roof per truss per meter			31,59527027	kN/m					
				Unit	**Amount**					
	Total dead load			55,84361843	kN/m					
	Total variable load (snow)	0,56	kN/m^2	15,4	kN/m					

Figure 54: Structural calculations column truss

E3. Calculations roof truss 1-4

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Truss roof 1-4**													
2														
3	**Steel grade**	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**		**Servicibility Limit State (SLS)**	**Amount**	**Unit**
4	Yield strength	355	N/mm^2					Safety factor (G)	1,3			Safety factor	1	
5	Tensile strength	360	N/mm^2					Safety factor (Q)	1,65			Moment	11182,2054	kNm
6	Weight	7850	kg/m^3					Moment	16908,46702	kNm				
7	Modulus of Elasticity	210000	N/mm^2					Axial force bottom/top chord	3381,693403	kN		**Moment of Inertia**		
8												Top chord	1,22212E+11	mm^4
9	**Cross section**	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			**Top chord**				Bottom chord	64023293732	mm^4
10	Diameter outer circle	508	273	139,7	mm			A	19051,79382	mm^2		Verticals/diagonals		mm^4
11	Thickness	12,5	12,5	8	mm			**Bottom chord**				Total	1,86235E+11	mm^4
12	Diameter inner circle	483	248	123,7	mm			A	9525,89691	mm^2				
13	A	19458,2395	10229,81108	3309,98202	mm^2			**Verticals/diagonals**						
14	W_y	2352574,89	637175,774	103119,386	mm^3			N	869,5339406	kN		**Maximum deflection**	440	mm
15	I_y	597554022	86974493,15	7202889,111	mm^4			A	3700,144428	mm^2		**Deflection**	360,3805691	mm
16	Weight	152,7471801	80,30401696	25,98335886	kg/m									
17														
18	**Dead load self-weight**	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**								
19	Top chord	1			1,498449836	kN/m								
20	Bottom chord	1			0,787782406	kN/m								
21	Verticals	23	1,274483752	29,31312629	0,266482966	kN/m								
22	Diagonals	22	1,802392206	39,65262854	0,360478441	kN/m								
23														
24	**Dimensions truss**	**Amount**	**Unit**											
25	Heigth (h)	5	m											
26	Length	110	m											
27														
28	**Dead load roof Johan Cruyff Arena**	**Amount**	**Unit**	**Amount**	**Unit**									
29	Dimensions (Wikipedia)	2x37x120	m	8880	m^2									
30	Weight (Wikipedia)	520	ton	5101,2	kN									
31	Dead load per square meter			0,574459459	kN/m^2									
32	Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN									
33	Dead load roof per truss per meter			45,95675676	kN/m									
34														
35				**Unit**	**Amount**									
36	**Total dead load**			2,91319365	kN/m									
37	**Total variable load (snow)**	0,56	kN/m^2	4,48	kN/m									

E4. Calculations retractable roof 2-3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Retractable roof 2-3**													
2														
3	**Steel grade**	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**		**Servicibility Limit State (SLS)**	**Amount**	**Unit**
4	Yield strength	355	N/mm^2					Safety factor (G)	1,3			Safety factor	1	
5	Tensile strength	360	N/mm^2					Safety factor (Q)	1,65			Moment	11493,97826	kNm
6	Weight	7850	kg/m^3					Moment	17313,77174	kNm				
7	Modulus of Elasticity	210000	N/mm^2					Axial force bottom/top chord	3462,754348	kN		**Moment of Inertia**		
8												Top chord	1,22212E+11	mm^4
9	**Cross section**	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			**Top chord**				Bottom chord	64023293732	mm^4
10	Diameter outer circle	508	273	150	mm			A	19508,4752	mm^2		Verticals/diagonals		mm^4
11	Thickness	12,5	12,5	10	mm			**Bottom chord**				Total	1,86235E+11	mm^4
12	Diameter inner circle	483	248	130	mm			A	9754,2376	mm^2				
13	A	19458,2395	10229,81108	4398,229715	mm^2			**Verticals/diagonals**						
14	W_y	2352574,89	637175,774	144408,5423	mm^3			N	890,3771201	kN		**Maximum deflection**	440	mm
15	I_y	597554022	86974493,15	10830640,67	mm^4			A	3788,838809	mm^2		**Deflection**	370,4283976	mm
16	Weight	152,7471801	80,30401696	34,52610326	kg/m									
17														
18	**Dead load self-weight**	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**								
19	Top chord	1			1,498449836	kN/m								
20	Bottom chord	1			0,787782406	kN/m								
21	Verticals	23	1,693505365	38,9506234	0,354096576	kN/m								
22	Diagonals	22	2,394978255	52,6895216	0,478995651	kN/m								
23														
24	**Dimensions truss**	**Amount**	**Unit**											
25	Heigth (h)	5	m											
26	Length	110	m											
27														
28	**Dead load roof Johan Cruyff Arena**	**Amount**	**Unit**	**Amount**	**Unit**									
29	Dimensions (Wikipedia)	2x37x120	m	8880	m^2									
30	Weight (Wikipedia)	520	ton	5101,2	kN									
31	Dead load per square meter			0,574459459	kN/m^2									
32	Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN									
33	Dead load roof per truss per meter			45,95675676	kN/m									
34														
35				**Unit**	**Amount**									
36	**Total dead load**			3,11932447	kN/m									
37	**Total variable load (snow)**	0,56	kN/m^2	4,48	kN/m									

E5. Calculations roof truss 5-6

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Truss roof 5-6**													
2														
3	**Steel grade**	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**		**Servicibility Limit State (SLS)**	**Amount**	**Unit**
4	Yield strength	355	N/mm^2					Safety factor (G)	1,3			Safety factor	1	
5	Tensile strength	360	N/mm^2					Safety factor (Q)	1,65			Moment	1600,711579	kNm
6	Weight	7850	kg/m^3					Moment	2570,925052	kNm				
7	Modulus of Elasticity	210000	N/mm^2					Axial force bottom/top chord	514,1850105	kN		**Moment of Inertia**		
8												Top chord	20694590513	mm^4
9	**Cross section**	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			**Top chord**				Bottom chord	10733071609	mm^4
10	Diameter outer circle	139,7	114,3	88,9	mm			A	2896,81696	mm^2		Verticals/diagonals		mm^4
11	Thickness	8	5	5	mm			**Bottom chord**				Total	31427662122	mm^4
12	Diameter inner circle	123,7	104,3	78,9	mm			A	1448,40848	mm^2				
13	A	3309,98202	1716,880385	1317,898118	mm^2			**Verticals/diagonals**						
14	W_y	103119,386	44955,41637	26180,84664	mm^3			N	290,8669661	kN		**Maximum deflection**	200	mm
15	I_y	7202889,111	2569202,045	1163738,633	mm^4			A	1237,731771	mm^2		**Deflection**	63,16121735	mm
16	Weight	25,98335886	13,47751102	10,34550023	kg/m									
17														
18	**Dead load self-weight**	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**								
19	Top chord	1			0,25489675	kN/m								
20	Bottom chord	1			0,132214383	kN/m								
21	Verticals	11	0,507446786	5,581914648	0,111638293	kN/m								
22	Diagonals	10	0,717638127	7,17638127	0,143527625	kN/m								
23														
24	**Dimensions truss**	**Amount**	**Unit**											
25	Heigth (h)	5	m											
26	Length	50	m											
27														
28	**Dead load roof Johan Cruyff Arena**	**Amount**	**Unit**	**Amount**	**Unit**									
29	Dimensions (Wikipedia)	2x37x120	m	8880	m^2									
30	Weight (Wikipedia)	520	ton	5101,2	kN									
31	Dead load per square meter			0,574459459	kN/m^2									
32	Dead load roof per truss	(4x40x110)/2	m	5055,243243	kN									
33	Dead load roof per truss per meter			101,1048649	kN/m									
34														
35				**Unit**	**Amount**									
36	**Total dead load**			0,642277052	kN/m									
37	**Total variable load (snow)**	0,56	kN/m^2	4,48	kN/m									

E6. Calculations main arch

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Main arch**													
2														
3	**Steel grade**	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**		**Servicibility Limit State (SLS)**	**Amount**	**Unit**
4	Yield strength	355	N/mm^2					Safety factor (G)	1,3			Safety factor	1	
5	Tensile strength	360	N/mm^2					Safety factor (Q)	1,65			Moment	335716,2658	kNm
6	Weight	7850	kg/m^3					Vertical reaction force	12165,17864					
7	Modulus of Elasticity	210000	N/mm^2					Horizontal reaction force	24330,35728	kN		**Moment of Inertia**		
8								Axial total force at the top	24330,35728			Top chord	7,4776E+12	mm^4
9	**Cross section**	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			Axial force at the bottom	27199,81879			Bottom chord	4,00934E+12	mm^4
10	Diameter outer circle	1208	813	457	mm							Verticals/diagonals		mm^4
11	Thickness	20	16	10	mm			**Top chord**				Total	1,14869E+13	mm^4
12	Diameter inner circle	1168	781	437	mm			A	76619,20785	mm^2				
13	A	74644,24145	40061,58952	14042,91916	mm^2			**Bottom chords**						
14	W_y	21808476,82	7828332,713	1535725,343	mm^3			A	38309,60393	mm^2		**Maximum deflection**	640	mm
15	I_y	13172320000	3182217248	350913240,8	mm^4			A	38309,60393	mm^2		**Deflection**	371,1227334	mm
16	Weight	585,9572954	314,4834777	110,2369154	kg/m									
17														
18	**Dead load self-weight**	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**								
19	Top chord	1			5,748241068	kN/m								
20	Bottom chord	4			12,34033167	kN/m								
21	Verticals	51	8,651393122	441,2210492	2,757631558	kN/m								
22	Diagonals	32	15,13993796	484,4780148	3,027987593	kN/m								
23														
24	**Dimensions truss**	**Amount**	**Unit**											
25	Heigth (h)	20	m											
26	Length	160	m											
27														
28	**Dead load roof Johan Cruyff Arena**	**Amount**	**Unit**	**Amount**	**Unit**									
29	Weight roof 1	20x55	m	34,23002539	kN/m									
30	Weight roof 2	20x25	m	2,007115787	kN/m									
31	Total dead weight			36,23714118	kN/m									
32														
33														
34														
35				**Unit**	**Amount**									
36	**Total dead load**			60,11133306	kN/m									
37	**Total variable load (snow)**	0,56	kN/m^2	44,8	kN/m									

E7. Calculations side arch

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Side arch**													
2														
3	**Steel grade**	**s355**	**Unit**					**Ultimate Limit State (ULS)**	**Amount**	**Unit**		**Servicibility Limit State (SLS)**	**Amount**	**Unit**
4	Yield strength	355	N/mm^2					Safety factor (G)	1,3			Safety factor	1	
5	Tensile strength	360	N/mm^2					Safety factor (Q)	1,65			Moment	76972,6454	kNm
6	Weight	7850	kg/m^3					Vertical reaction force	2893,610975	kN				
7	Modulus of Elasticity	210000	N/mm^2					Horizontal reaction force	5787,221951	kN		**Moment of Inertia**		
8								Axial total force at the top	5787,221951			Top chord	1,94642E+12	mm^4
9	**Cross section**	**Top chord**	**Bottom chord**	**Diagonals/verticals**	**Unit**			Axial force at the bottom	6469,75244			Bottom chord	1,02307E+12	mm^4
10	Diameter outer circle	508	273	193,7	mm							Verticals/diagonals		mm^4
11	Thickness	12,5	12,5	10	mm			**Top chord**				Total	2,96949E+12	mm^4
12	Diameter inner circle	483	248	173,7	mm			A	18224,65476	mm^2				
13	A	19458,2395	10229,81108	5771,105705	mm^2			**Bottom chords**						
14	W_y	2352574,89	637175,774	252099,9673	mm^3			A	9112,32738	mm^2		**Maximum deflection**	640	mm
15	I_y	597554022	86974493,15	24415881,83	mm^4			A	9112,32738	mm^2		**Deflection**	329,1577187	mm
16	Weight	152,7471801	80,30401696	45,30317978	kg/m									
17														
18	**Dead load self-weight**	**Amount**	**Weight per unit**	**Total weight**	**Load over lenght of truss**	**Unit**								
19	Top chord	1			1,498449836	kN/m								
20	Bottom chord	4			3,151129626	kN/m								
21	Verticals	51	3,555393549	181,325071	1,133281694	kN/m								
22	Diagonals	32	6,221938711	199,1020388	1,244387742	kN/m								
23														
24	**Dimensions truss**	**Amount**	**Unit**											
25	Heigth (h)	20	m											
26	Length	160	m											
27														
28	**Dead load roof Johan Cruyff Arena**	**Amount**	**Unit**	**Amount**	**Unit**									
29	Weight extra elements roof 2		m	1,019587001	kN/m									
30	Weight roof 2	20x25	m	2,007115787	kN/m									
31	Total dead weight			3,026702789	kN/m									
32														
33														
34														
35				**Unit**	**Amount**									
36	**Total dead load**			10,05395169	kN/m									
37	**Total variable load (snow)**	0,56	kN/m^2	14	kN/m									

Appendix F SCIA Engineer results

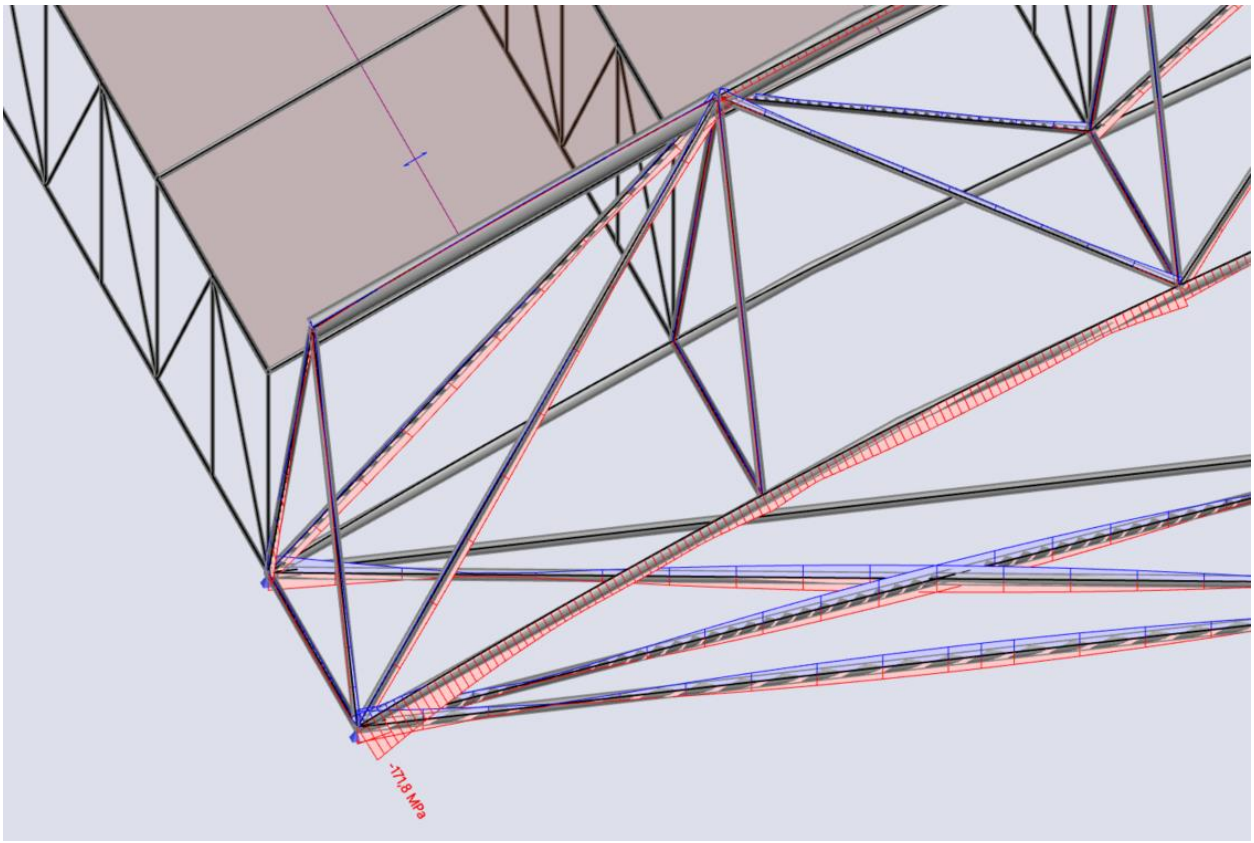


Figure 55: Snow load ULS

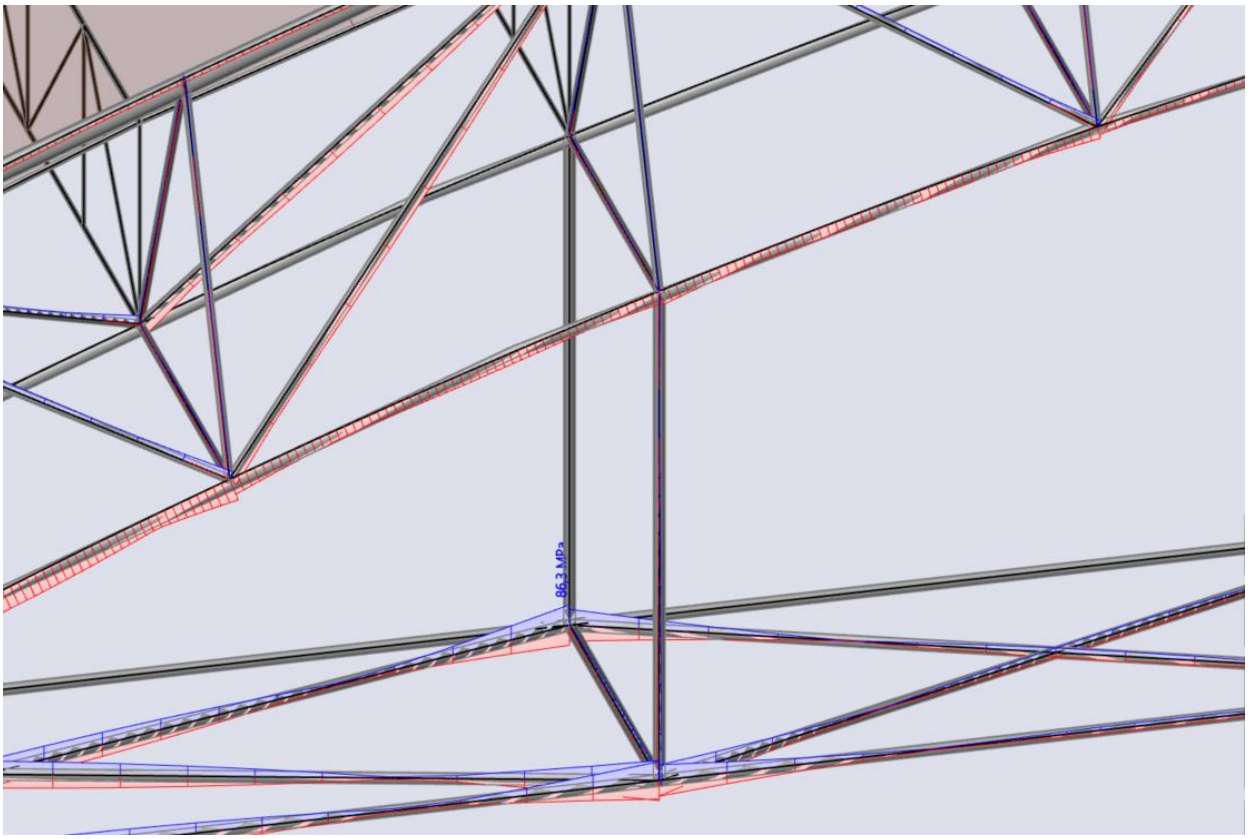


Figure 56: Snow load ULS

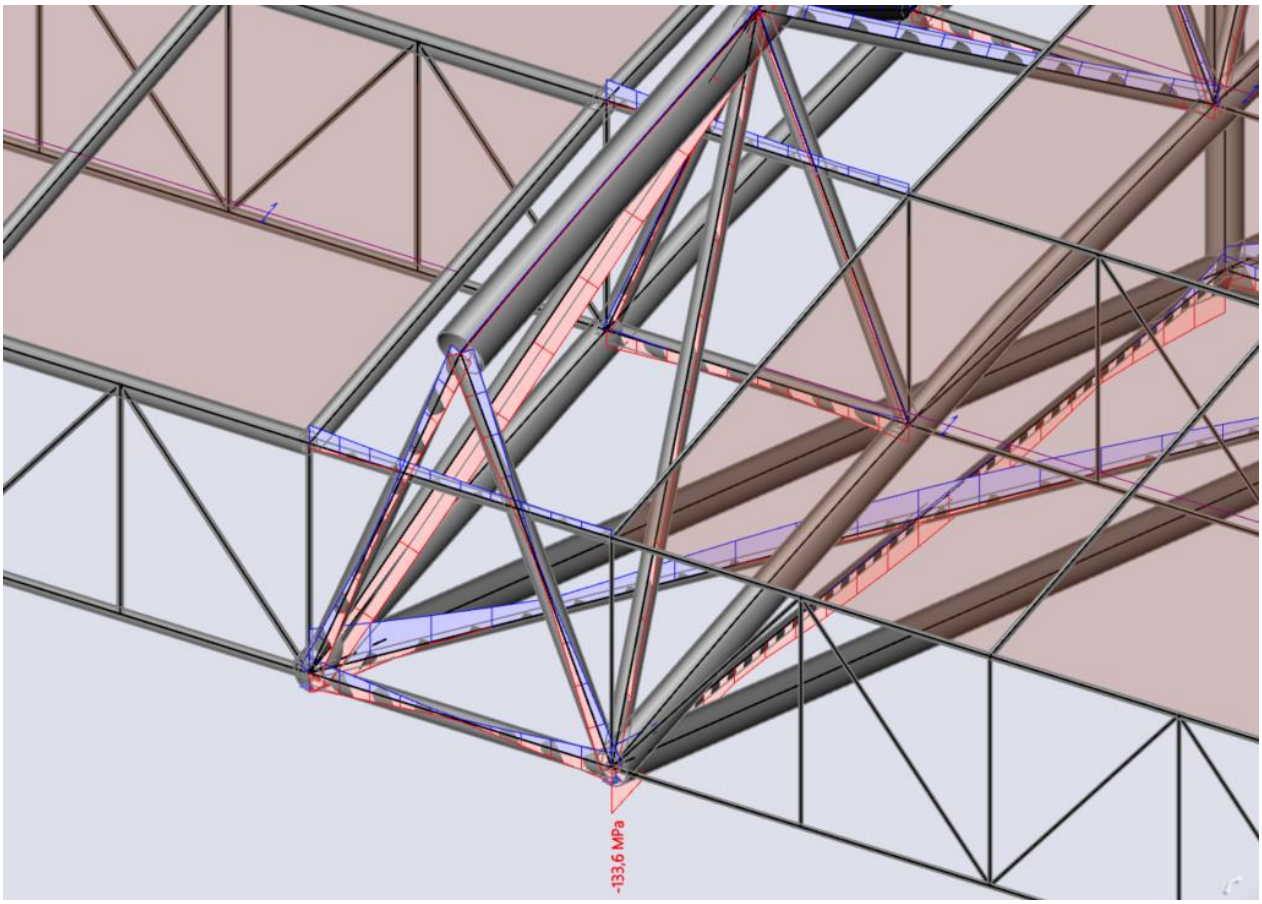


Figure 57: Snow load ULS

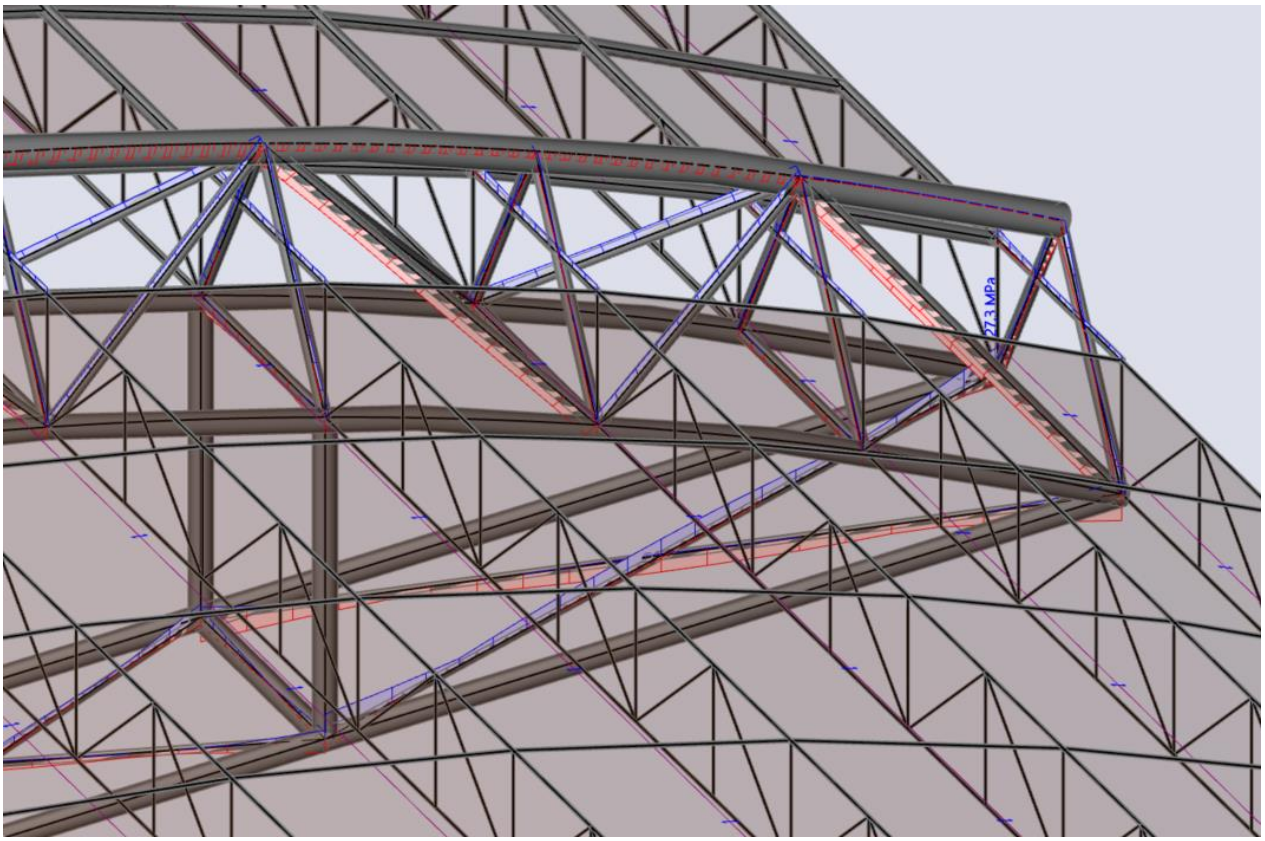


Figure 58: Snow load ULS

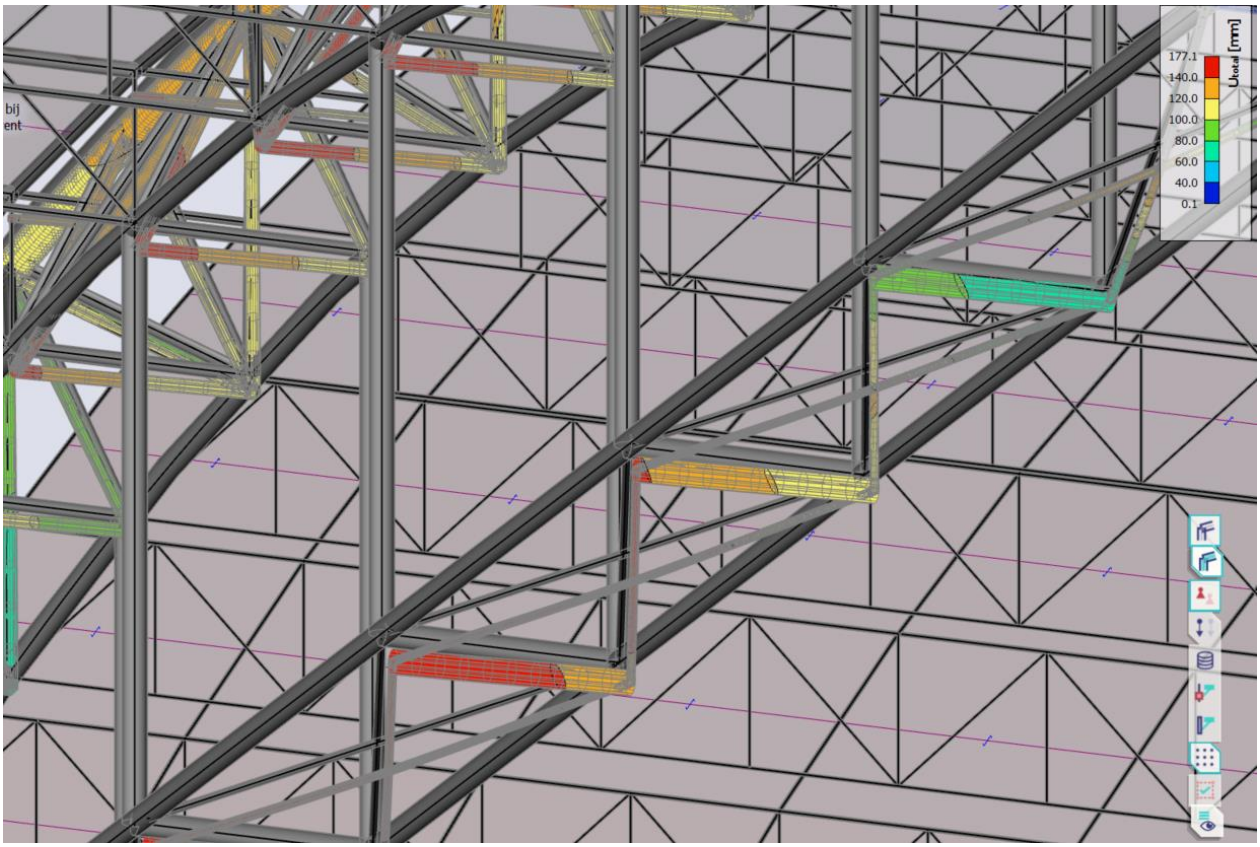


Figure 59: Snow load ULS

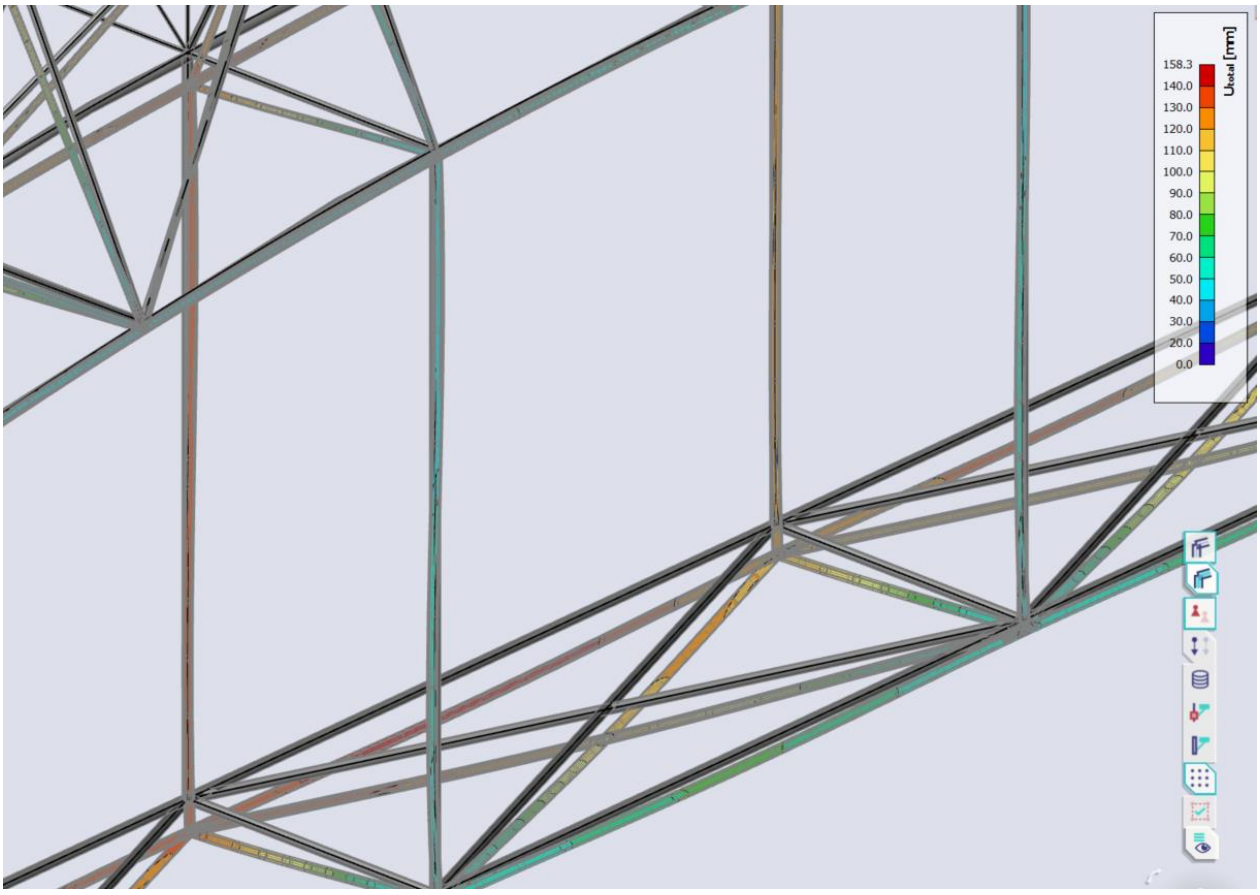


Figure 60: Snow load ULS

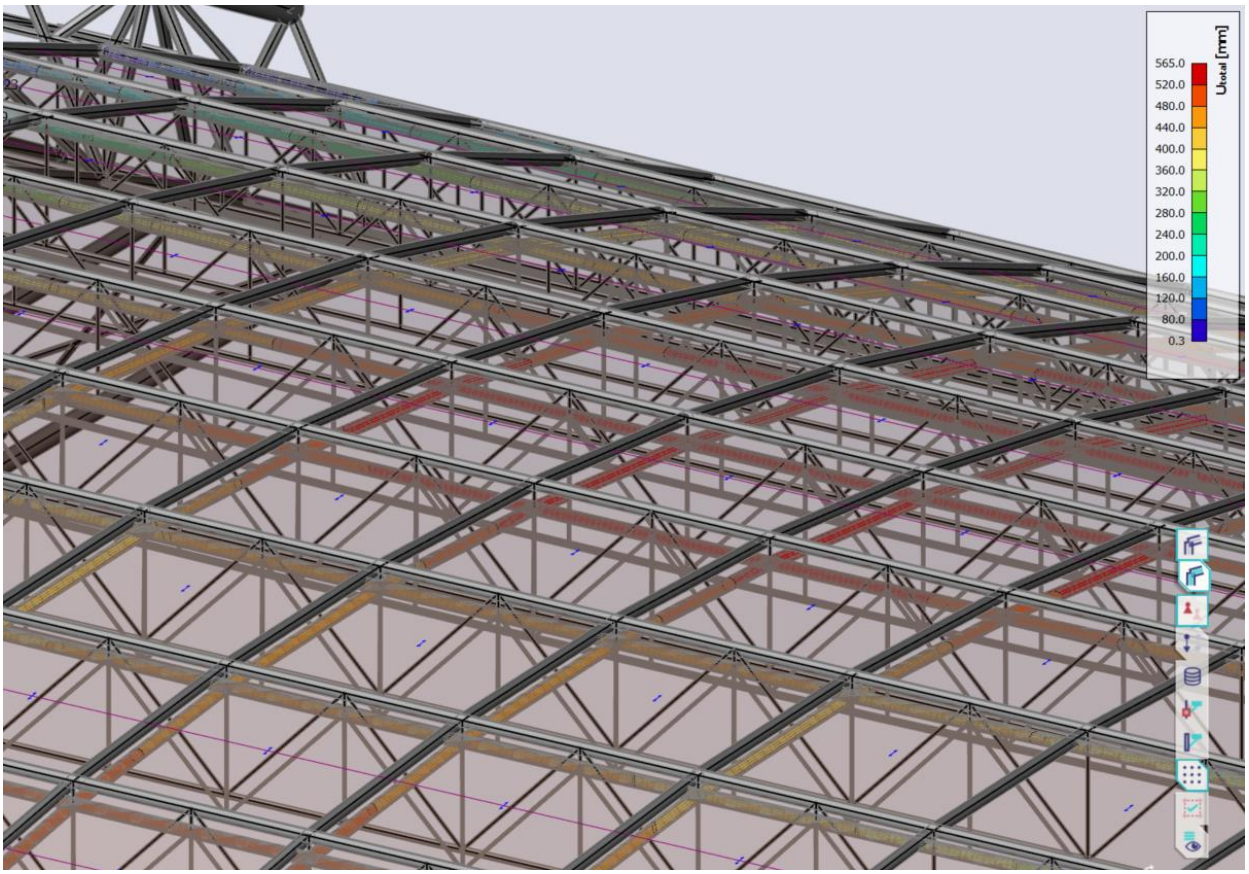


Figure 61: Snow load ULS

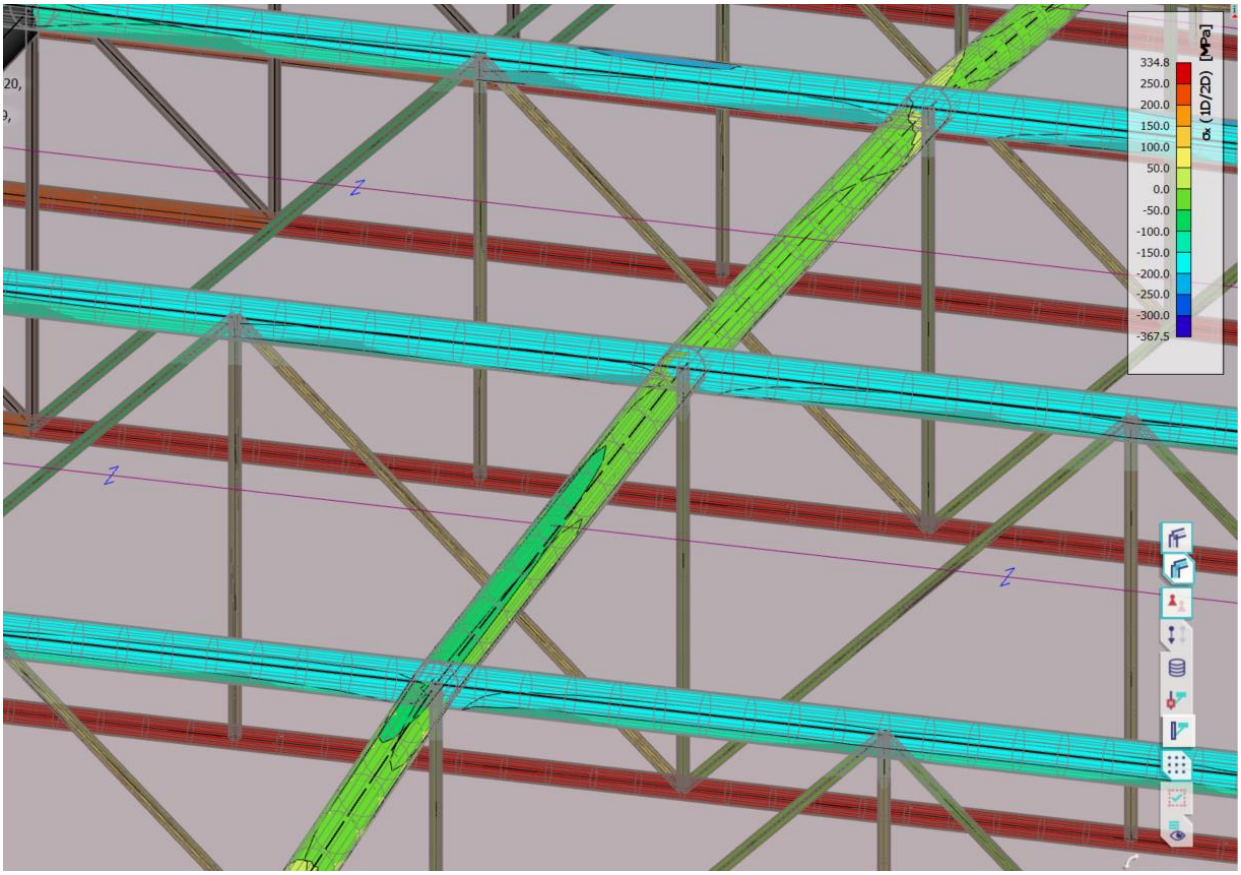


Figure 62: Snow load ULS

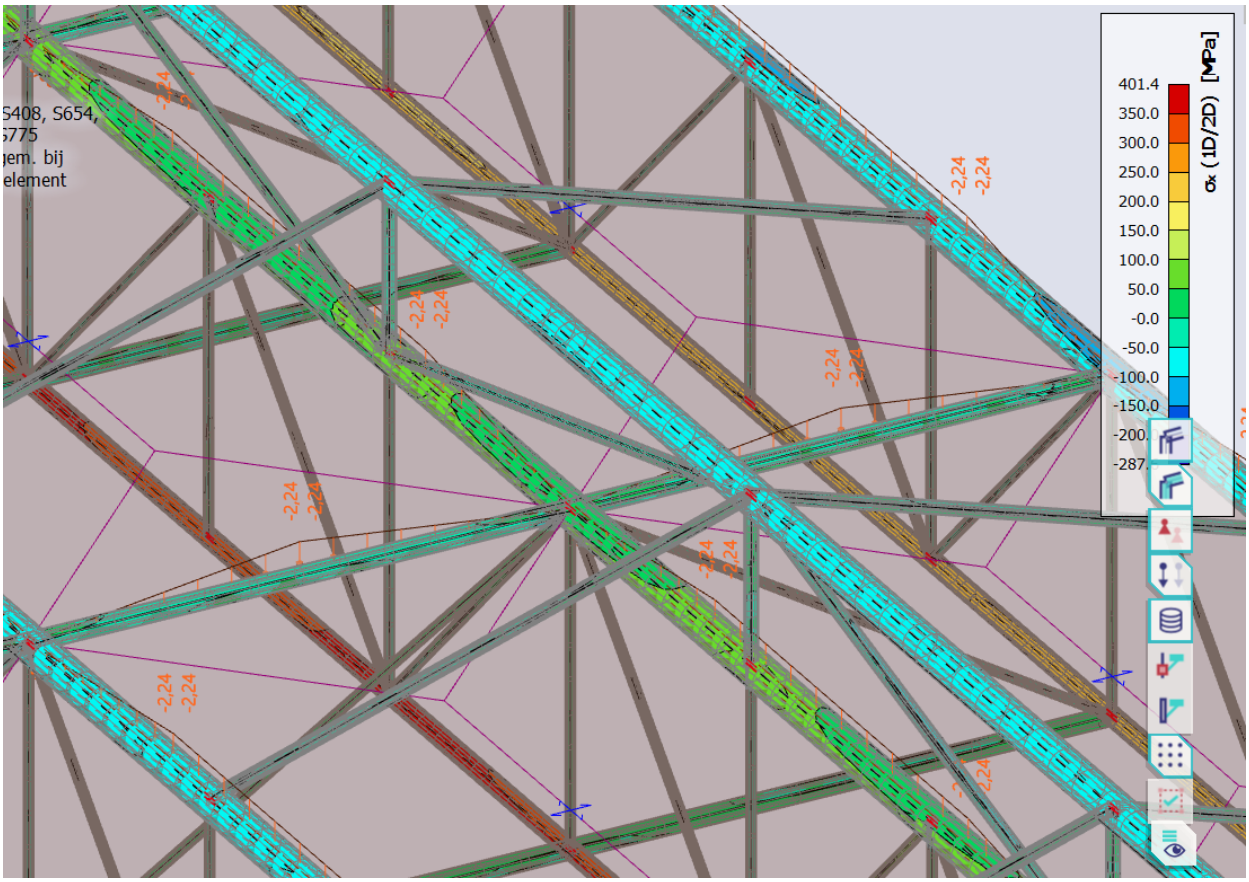


Figure 63: Retractable roof

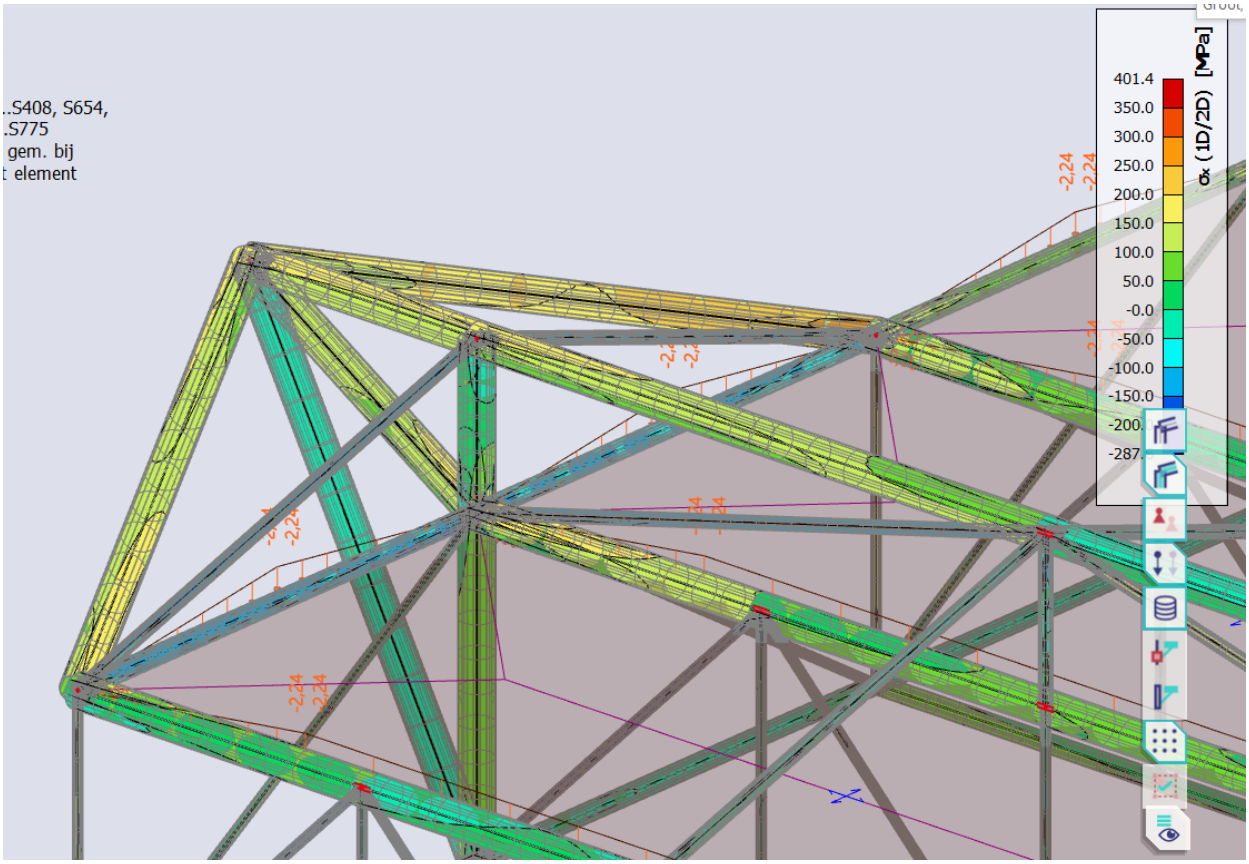


Figure 64: Retractable roof

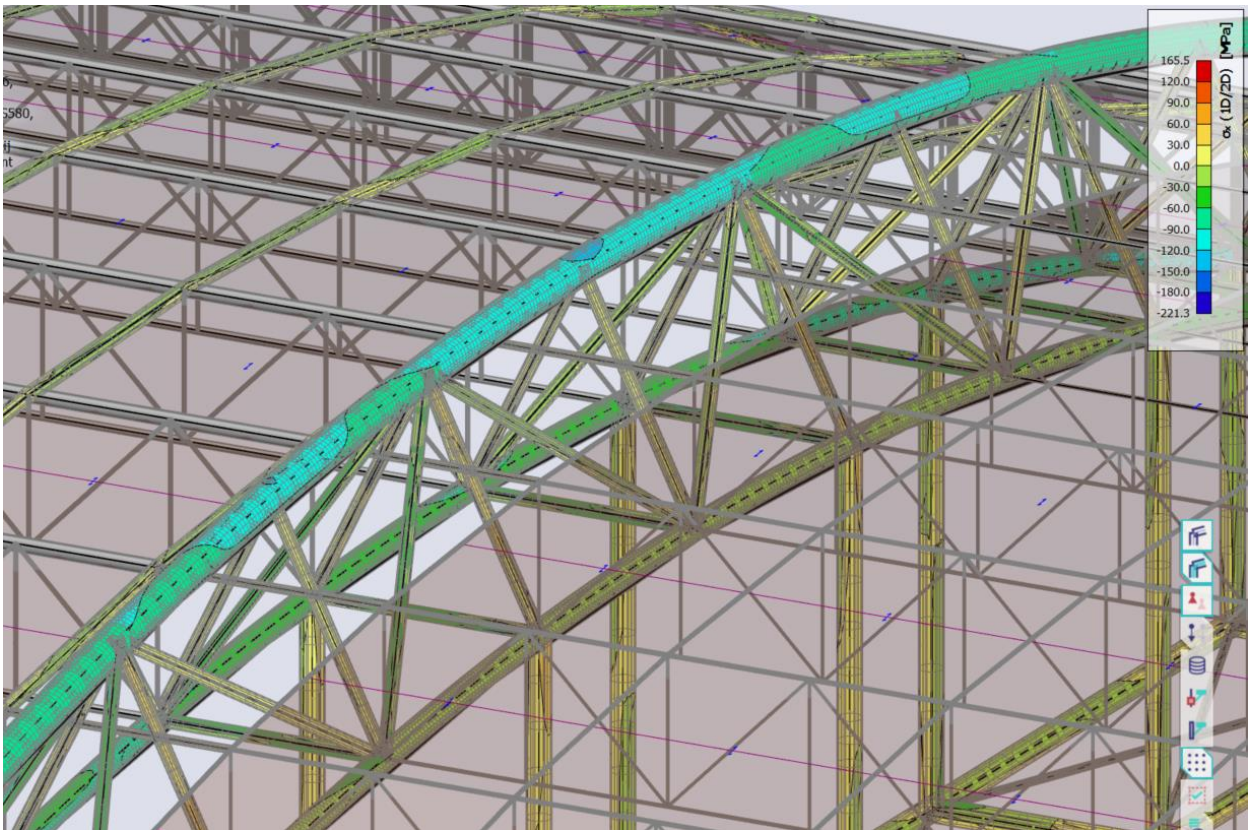


Figure 65: Wind load ULS

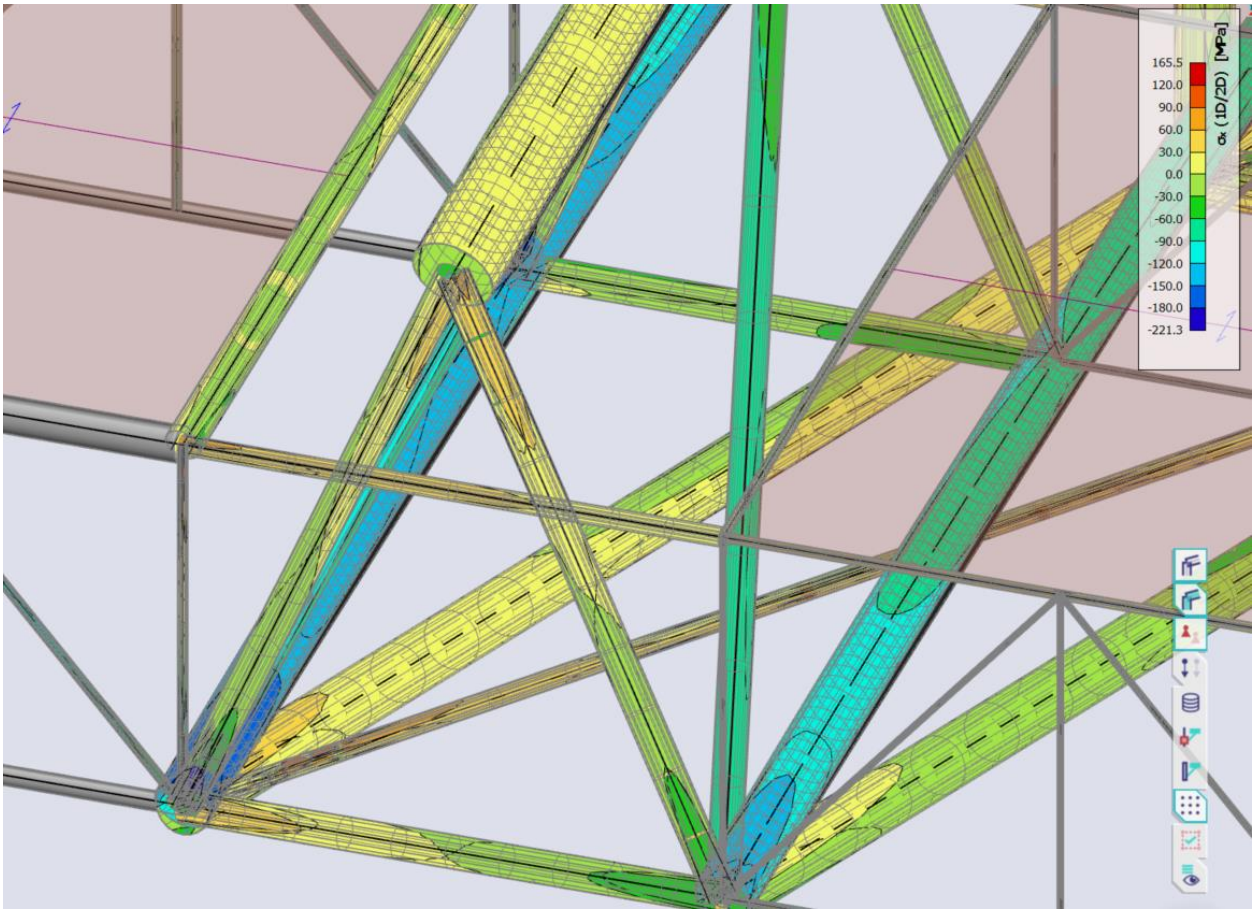


Figure 66: SLS wind load

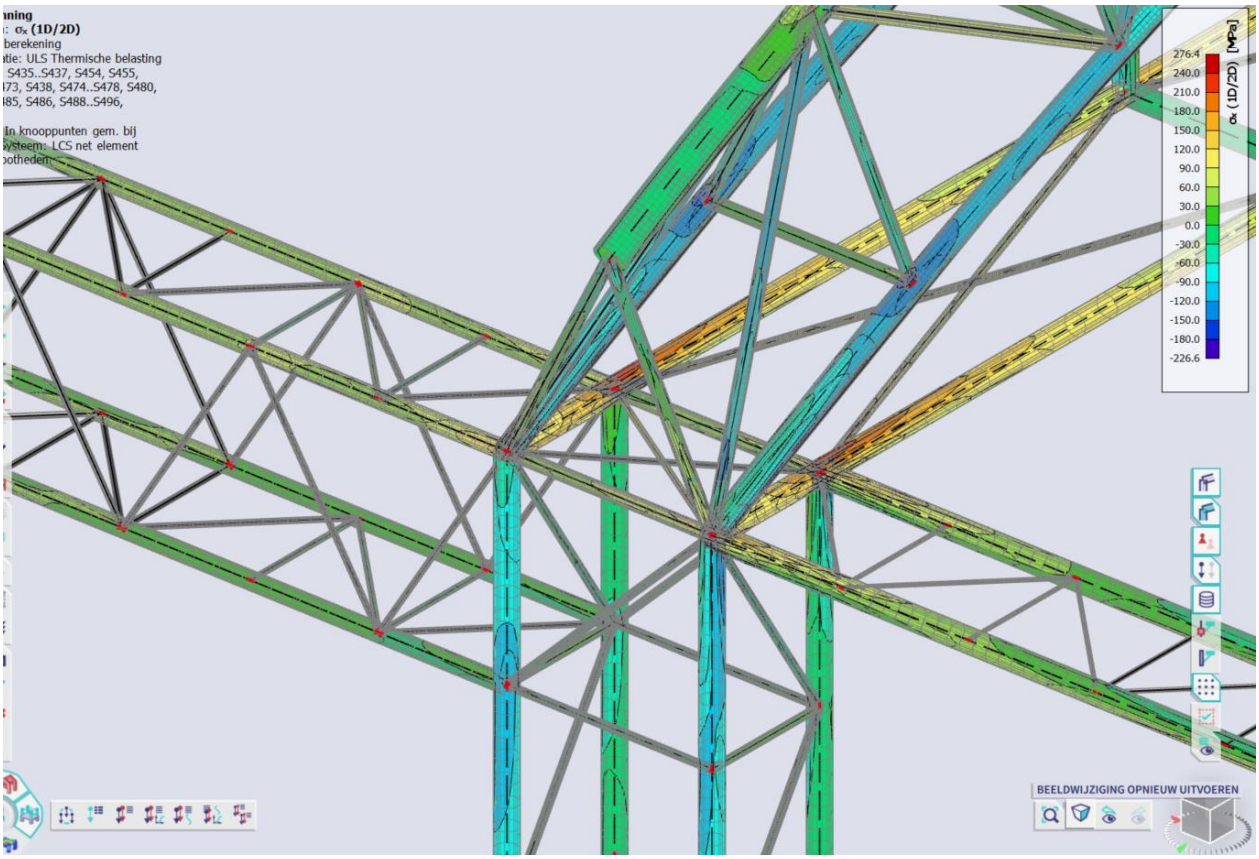


Figure 67: Thermal load ULS

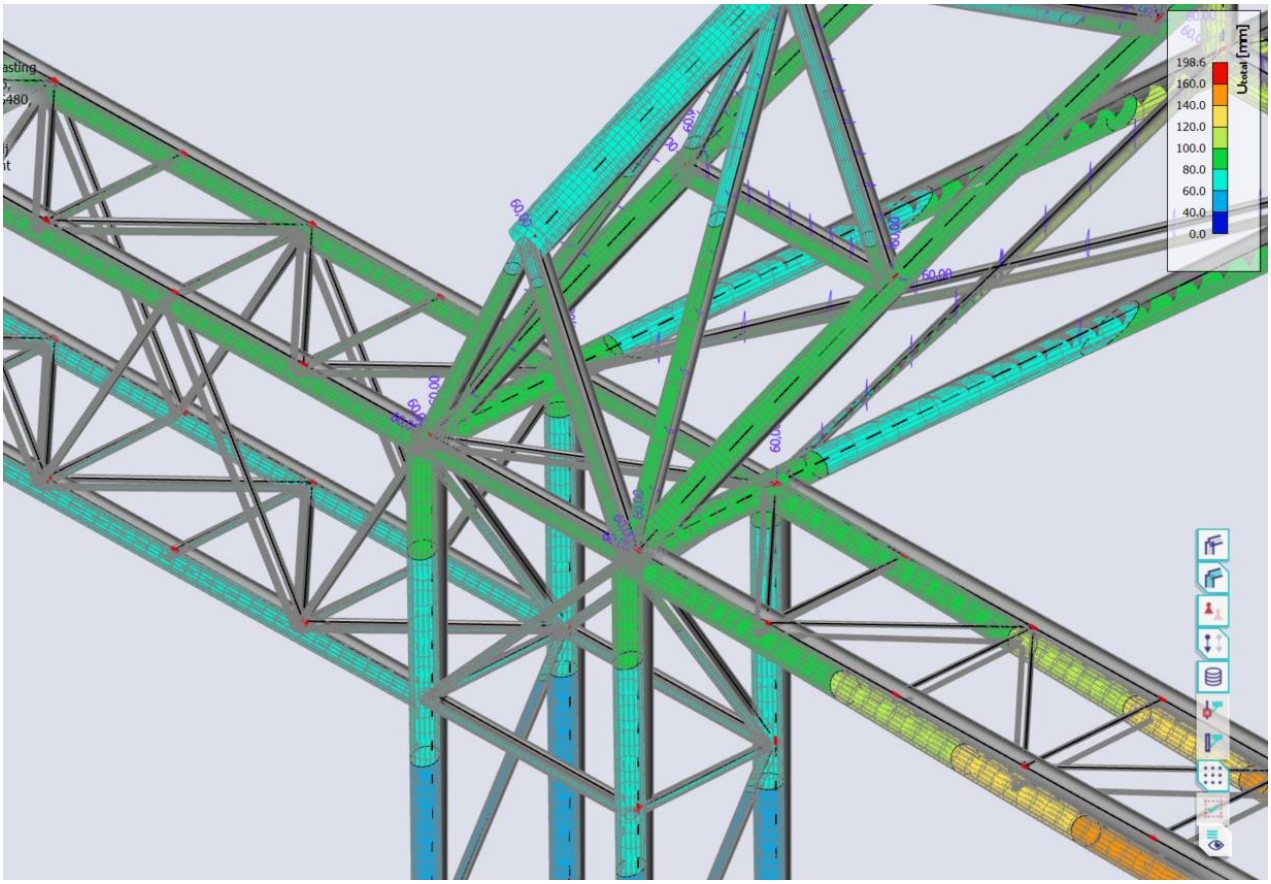
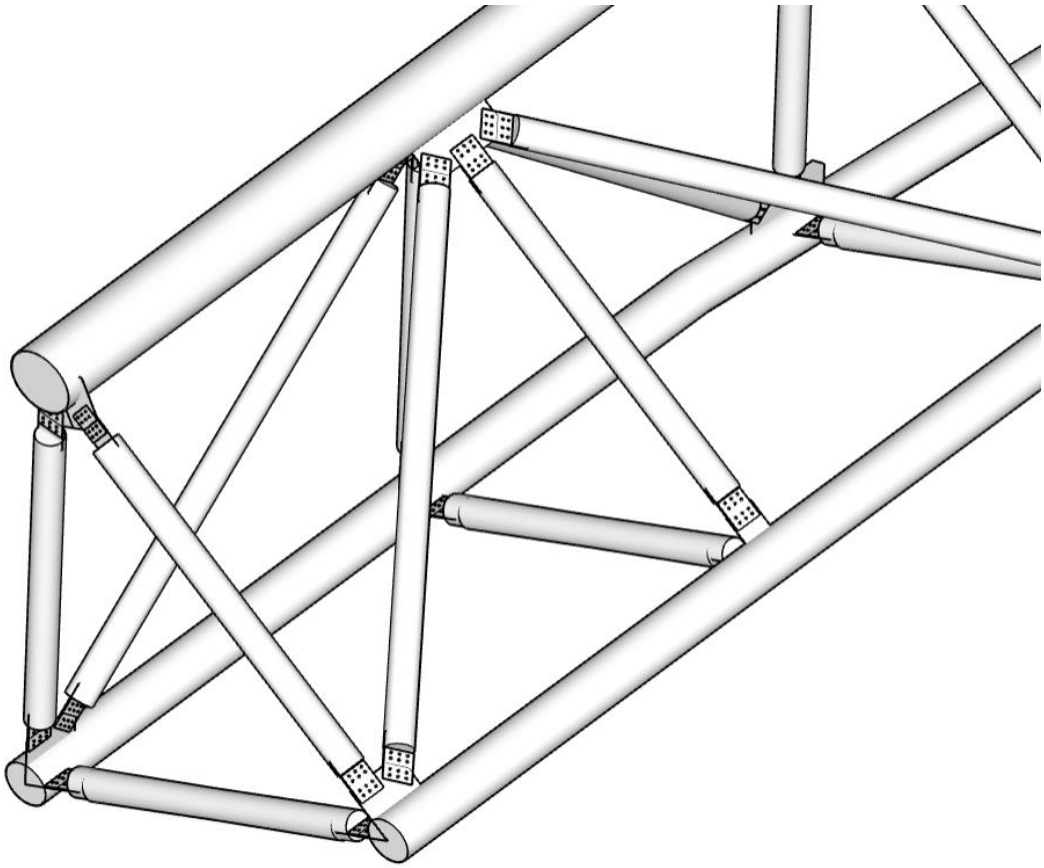
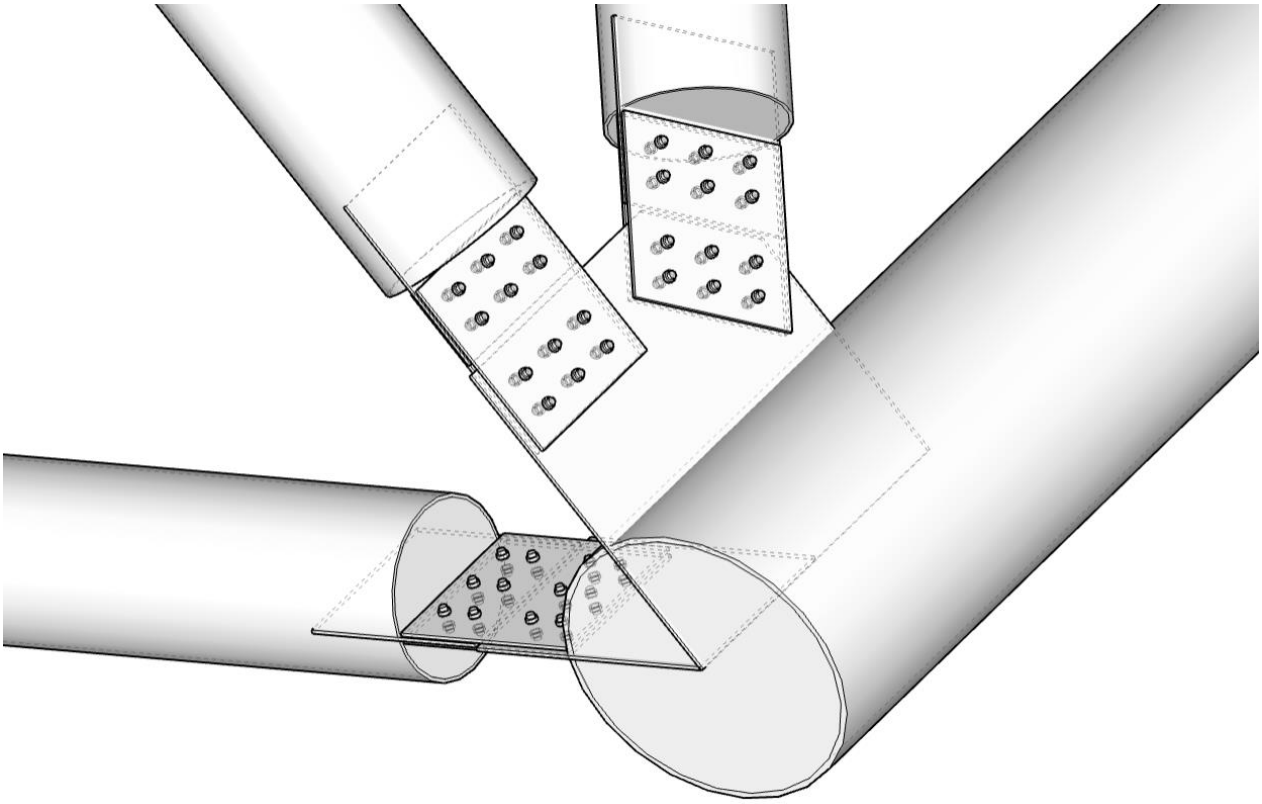
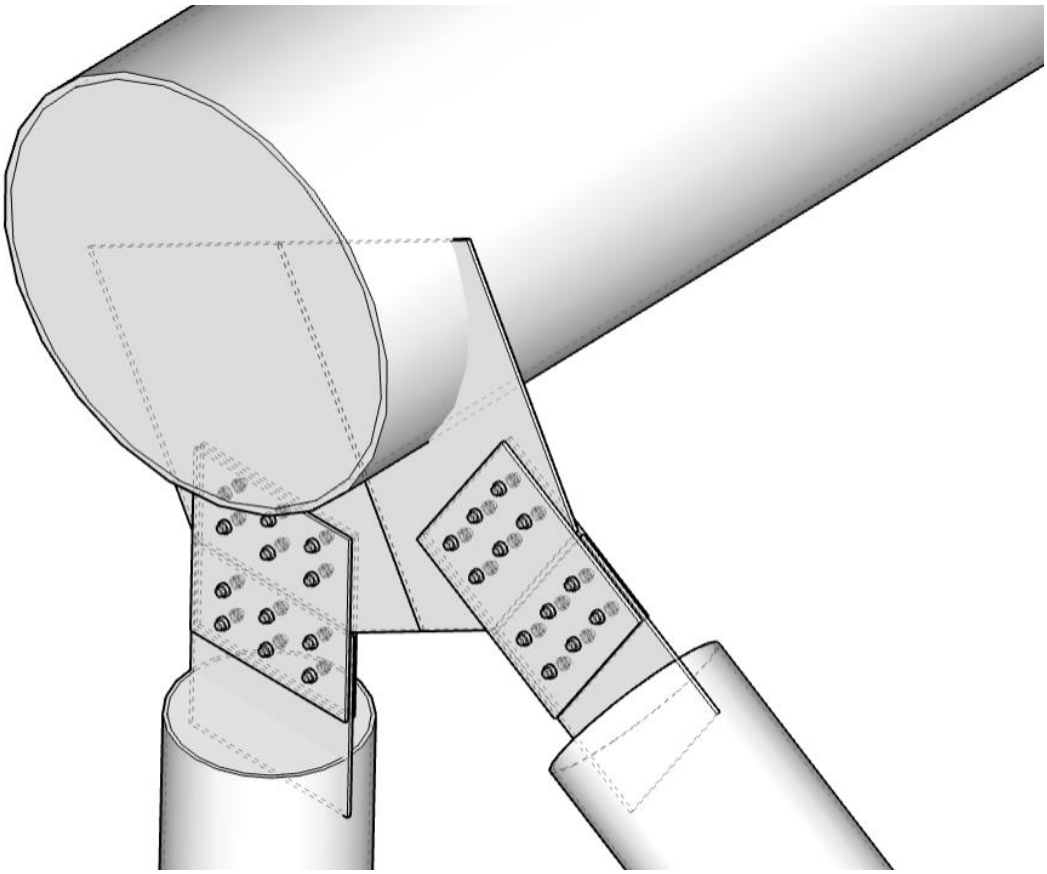


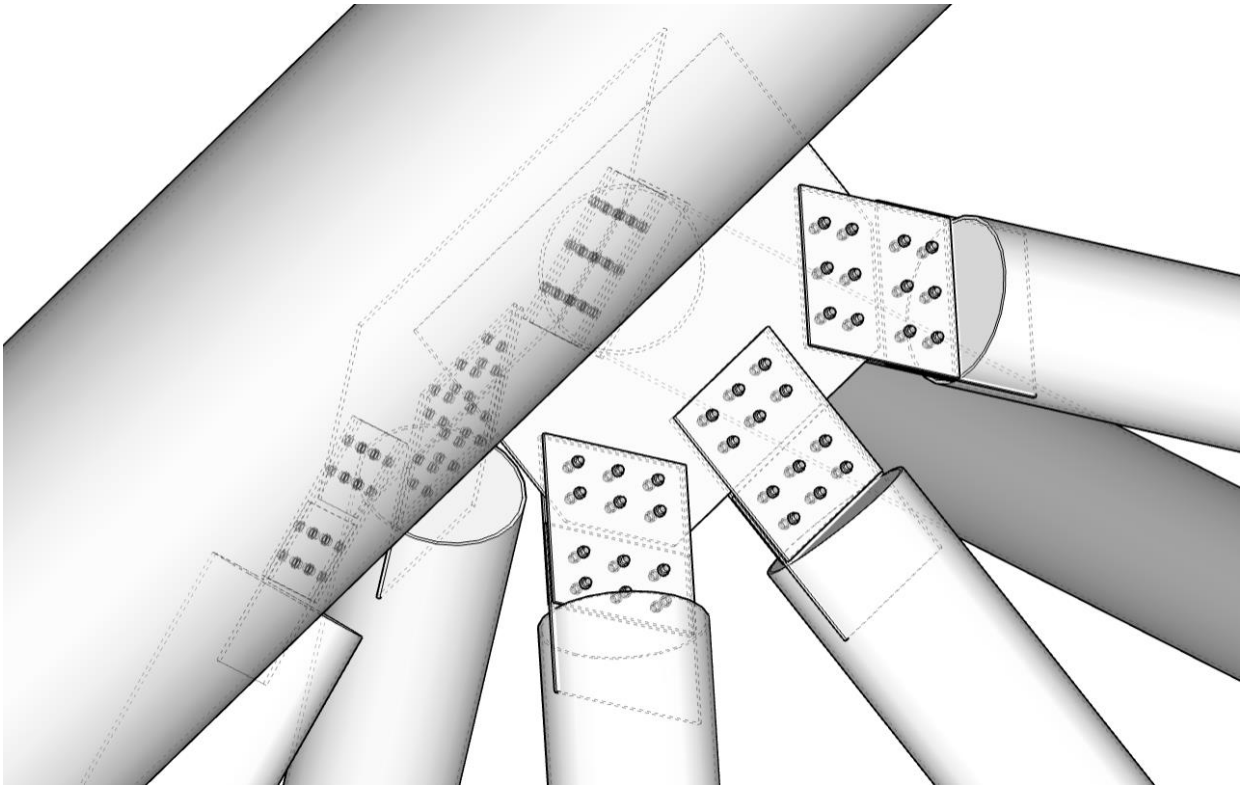
Figure 68: Thermal load SLS

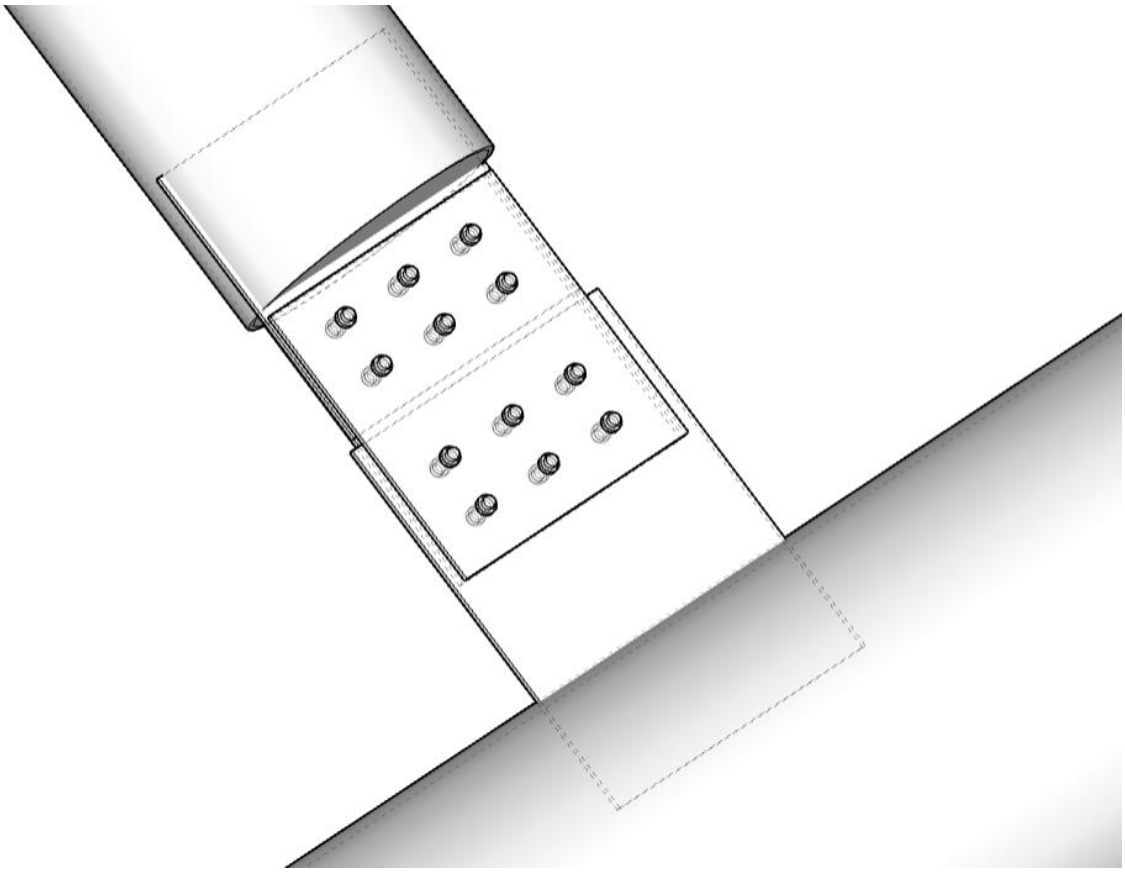
Appendix G Details

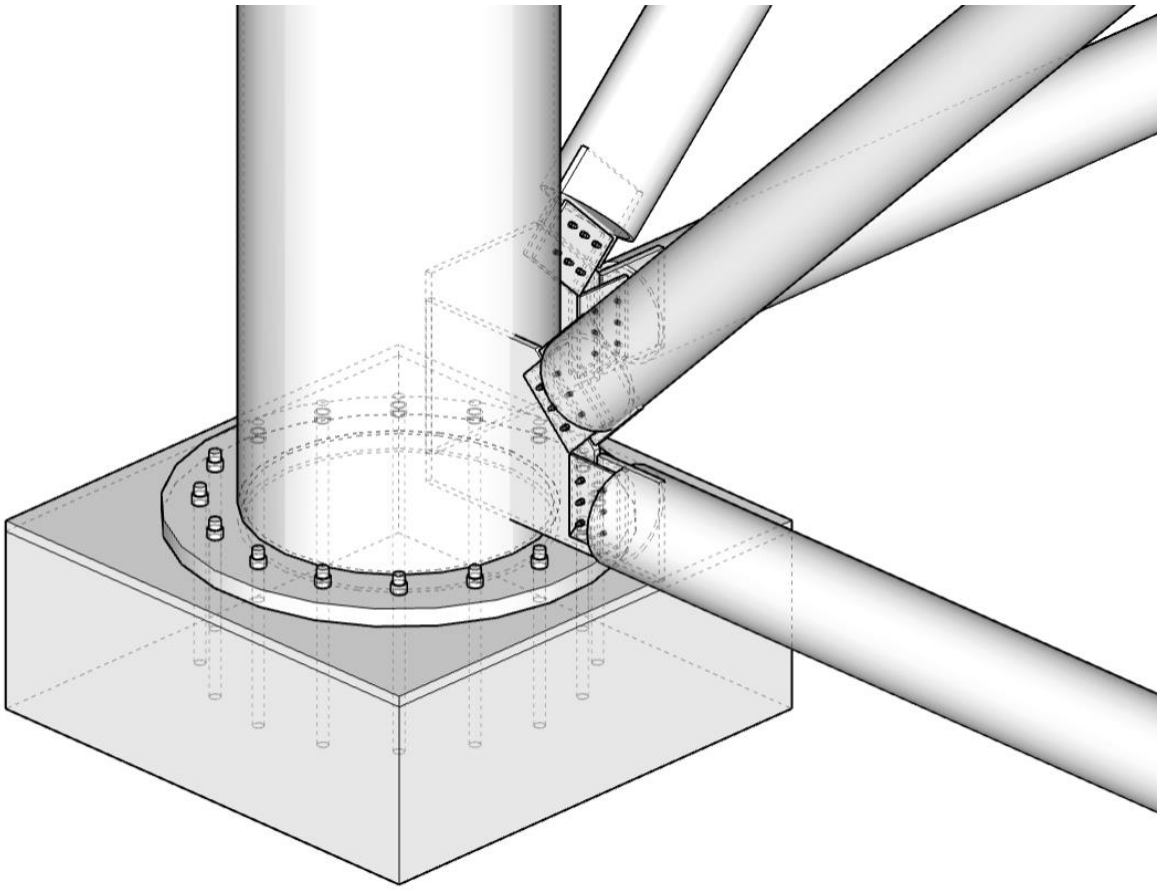




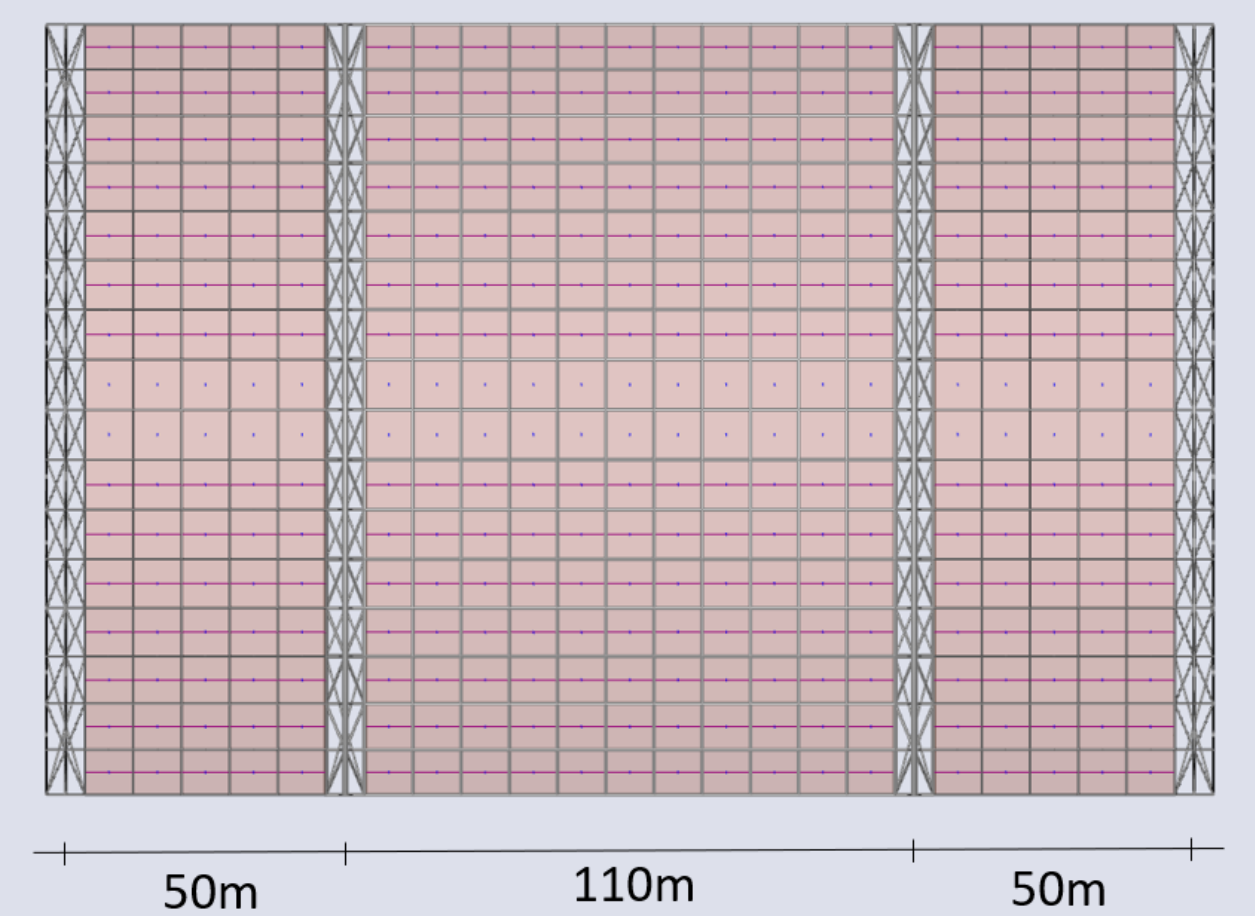


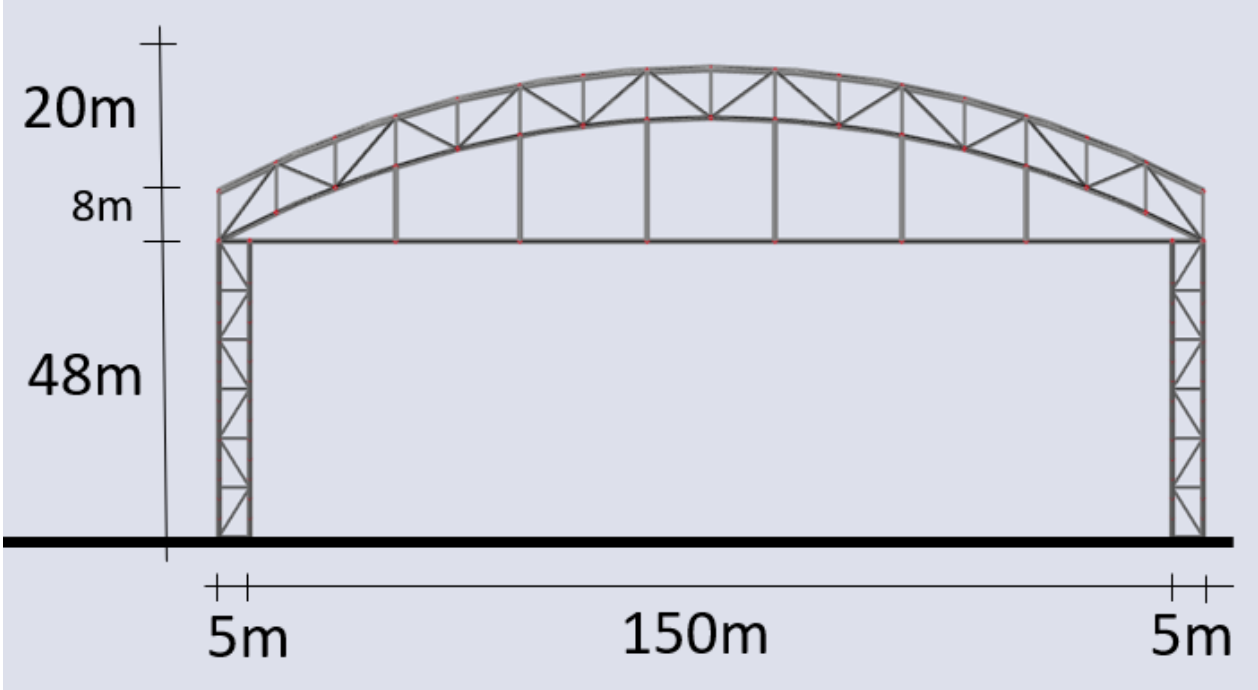


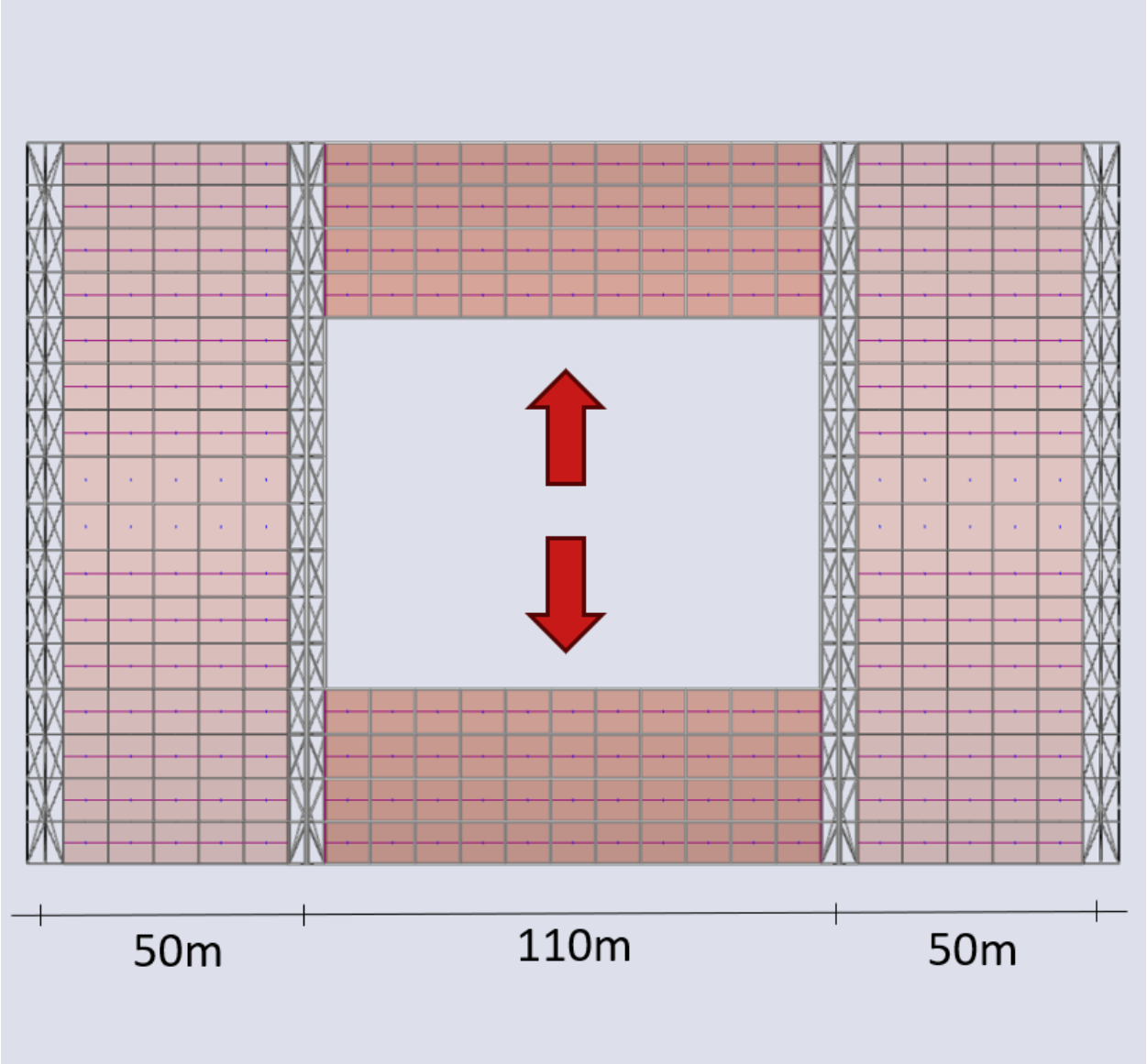


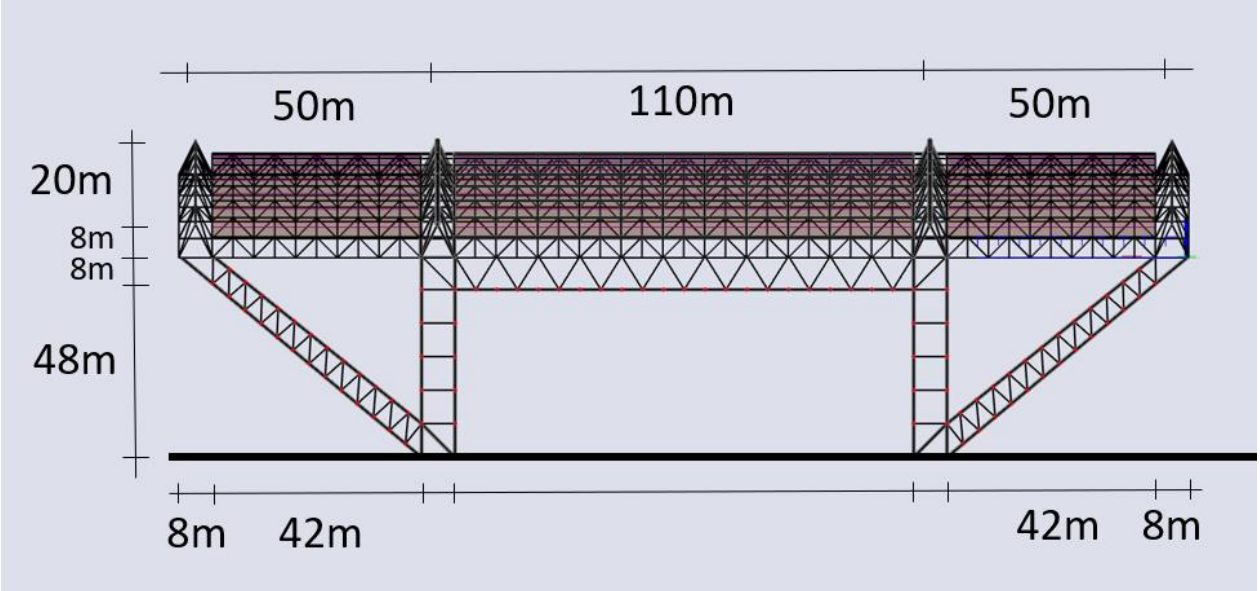


Appendix H Drawings






50m
110m
50m


50m
110m
50m
20m
8m
8m
48m
8m
42m
42m
8m